拓扑学作业

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Chapter 2 Continuity

Problem 1

Verify each of the following for arbitrary subsets A, B of a space X:

(a) $\overline{A \cup B} = \overline{A} \cup \overline{B}$; (b) $\overline{A \cap B} \subset \overline{A} \cap \overline{B}$; (c) $\overline{\overline{A}} = \overline{A}$;

(d) $(A \cup B)^\circ \supset \mathring{A} \cup \mathring{B}$; (e) $(A \cap B)^\circ = \mathring{A} \cap \mathring{B}$; (f) $\left(\mathring{A}\right)^\circ = \mathring{A}$

Show that equality need not hold in (b) and (c).

Proof: (a) $\subset$: $x \in \overline{A \cup B} \iff \forall U_x \ni x, U_x \cap (A \cup B) \neq \emptyset$. 反设 $x \notin \overline{A} \cup \overline{B}$, 则 $\exists x$ 的邻域 U_1, U_2 使得 $U_1 \cap A = \emptyset, U_2 \cap B = \emptyset$, 从而 $(U_1 \cap U_2) \cap (A \cup B) = \emptyset$ 且 $U_1 \cap U_2$ 为 x 的邻域, 矛盾!

$\supset$: $x \in \overline{A} \cup \overline{B}$, 设 $x \in \overline{A}$, 对任意 x 的邻域 U_x, $U_1 \cap A \neq \emptyset \Longrightarrow U_1 \cap (A \cup B) \neq \emptyset$, 即 $x \in \overline{A \cup B}$.

(b) $x \in \overline{A \cap B} \iff \forall U_x \ni x, U_x \cap A \cap B \neq \emptyset \Longrightarrow \forall U_x \ni x, U_x \cap A \neq \emptyset, U_x \cap B \neq \emptyset \iff x \in \overline{A} \cap \overline{B}$, 即 $\overline{A \cap B} \subset \overline{A} \cap \overline{B}$.

真包含的例子: 取 $X = \mathbb{R}, A = (-1, 0), B = (0, 1)$, 则 $\overline{A} \cap \overline{B} = \{0\}, \overline{A \cap B} = \emptyset$.

(c) $\overline{A}$ 为闭集, 闭集的闭包为自身.

对于 (d), (e), (f), 我们先证明 $\overline{A^c} = \left(\mathring{A}\right)^c$, 这是因为 $x \in \left(\mathring{A}\right)^c \iff x \notin \mathring{A} \iff \forall U_x \ni x, U_x \cap A^c \neq \emptyset \iff x \in \overline{A^c}$.

(d) 由 (b) 知 $\overline{A^c \cap B^c} \subset \overline{A^c} \cap \overline{B^c} \iff \overline{(A \cup B)^c} \subset \left(\mathring{A}\right)^c \cap \left(\mathring{B}\right)^c \iff \left((A \cup B)^\circ\right)^c \subset \left(\mathring{A} \cup \mathring{B}\right)^c \iff (A \cup B)^\circ \supset \mathring{A} \cup \mathring{B}$.

真包含: 取 $A = [-1, 0], B = [0, 1]$, 则 $(A \cup B)^\circ = (-1, 1), \mathring{A} \cup \mathring{B} = (-1, 1) \setminus \{0\}$.

(e) 将 (a) 中 A, B 替换为 A^c, B^c, 利用 $\overline{A^c} = \left(\mathring{A}\right)^c$ 和 De-Morgan 律即证.

(f) $\mathring{A}$ 为开集, 开集的内点集为自身.

Problem 2

Find a family of closed subsets of the real line whose union is not closed.

解: 取 $A_n = \left[\frac{1}{n}, 1\right], n \geq 1$ 为 $\mathbb{R}$ 上的闭集族, 则 $\bigcup_{n \geq 1} A_n = (0, 1]$ 不是闭集.

Problem 3

Specify the interior, closure and frontier of each of the following subsets of the plane:

(a) $\left\{(x, y) | 1 < x^2 + y^2 \leq 2\right\}$; (b) $\mathbb{R}^2$ with both axes removed;

(c) $\mathbb{R}^2 - \left\{\left(x, \sin \frac{1}{x}\right) | x > 0\right\}$.

Solution: (a) 内点集 $\left\{(x, y) | 1 < x^2 + y^2 < 2\right\}$, 闭包 $\left\{(x, y) | 1 \leq x^2 + y^2 \leq 2\right\}$, 边界点 $\left\{(x, y) | x^2 + y^2 = 1 \text{ 或 } 2\right\}$.

(b) 内点集 $\{(x, y) | x \neq 0, y \neq 0\}$, 闭包 $\mathbb{R}^2$, 边界点集 $\{(x, y) | x = 0 \text{ 或 } y = 0\}$.

(c) 内点集 $\mathbb{R}^2-\left(\left\{\left(x,\sin\frac{1}{x}\right)|x>0\right\}\cup\{(0,0)\}\right)$, 闭包 $\mathbb{R}^2$, 边界点集 $\left\{\left(x,\sin\frac{1}{x}\right)|x>0\right\}\cup\{(0,0)\}$.

Problem 4

Find all the limit points of the following subsets of the real line:

(a) $\left\{\frac{1}{m}+\frac{1}{n}|m,n=1,2,\cdots\right\}$; (b) $\left\{\frac{1}{n}\sin n|n=1,2,\cdots\right\}$.

Solution: (a) 令 $m=n\to\infty$ 可得 0 为极限点, 固定 n, 令 $m\to\infty$ 可得每个 $\frac{1}{n}$ 为极限点. 记 $A=\left\{\frac{1}{m}+\frac{1}{n}|n,m=1,2,\cdots\right\}, B=\{0\}\cup\left\{\frac{1}{n}|n=1,2,\cdots\right\}$. 下证明如下命题: 若区间 $I=[\alpha,\beta]\cap B=\emptyset$, 则 $I\cap A$ 为有限集.

$\beta<0$ 及 $\beta>1$ 时比较显然, 由 $\beta\notin B$, 考虑 $\beta\in(0,1)$ 的情况. 设 $\frac{1}{k+1}<\beta<\frac{1}{k}$, 则 $\frac{1}{k+1}<\alpha<\beta<\frac{1}{k}$. 取 $x=\frac{1}{m}+\frac{1}{n}\in I\cap A$, 于是

$$\frac{1}{k+1}<\alpha<\frac{1}{m}+\frac{1}{n}<\beta<\frac{1}{k}$$

对固定的 m, 可以解得 n 应满足

$$\frac{km}{m-k}<n<\frac{m(k+1)}{m-k-1}$$

注意到上述区间在 $m\to\infty$ 时趋于 $(k,k+1)$, 于是当 m 充分大时, n 只可能取 $k,k+1$. 若 $n=k$, 代回可知 $\frac{1}{m}<0$ 矛盾, $n=k+1$ 时有

$$m<\frac{1}{\alpha-\frac{1}{k+1}}$$

故 m 充分大时上述方程无解, 进而满足条件的 (m,n) 只有有限对. 上述命题得证.

对任意 $x\notin B$, 总能找到 $(x-\varepsilon,x+\varepsilon)\cap B=\emptyset$, 进而由命题知 $(x-\varepsilon,x+\varepsilon)$ 中仅含有限个 A 中的点, 存在更小的 $(x-\delta,x+\delta)$ 与 $A\setminus\{x\}$ 交集为空, 即 x 不是 A 的极限点. 综上 A 的极限点为 $B=\{0\}\cup\left\{\frac{1}{n}|n=1,2,\cdots\right\}$.

(b) 由 $\lim_n\frac{1}{n}\sin n=0$ 可知极限点只有 0.

Problem 5

If A is a dense subset of a space X, and if O is open in X, show that $O\subset\overline{A\cap O}$.

Proof: 取 $x\in O$, 若 $x\in A$ 则显然, 现设 $x\in O\setminus A$. 对任意 x 的邻域 U, $U\cap O$ 仍为包含 x 的邻域, 故由 $x\in X=\overline{A}$ 知 $U\cap O\cap A\neq\emptyset$, 即 $x\in\overline{A\cap O}\Longrightarrow O\subset\overline{A\cap O}$.

Problem 6

If Y is a subspace of X, and Z is a subspace of Y, prove that Z is a subspace of X.

Proof: 设 U 为 Z 中的开集, 即存在 $V\subset Y$ 为开集, 使得 $U=Z\cap V$, 于 V 而言, 存在 $W\subset X$ 为开集使得 $V=Y\cap W$, 于是 $U=Z\cap Y\cap W=Z\cap W$, 可见 Z 上的拓扑为 X 的子空间拓扑, 即 Z 是 X 的子空间.

Problem 9

Let Y be a subspace of X. If A is open(closed) in Y, and if Y is open(closed) in X, show that A is open(closed) in X.

Proof: 存在 X 中的开集 U 使得 $A = U \cap Y$, 由 Y 为开集即知 A 为 X 中开集. 对于闭集的情况, 可知 $Y \setminus A$ 为 Y 中开集, 存在 X 中开集 W 使得 $Y \setminus A = W \cap Y$, 于是 $A = Y \setminus W = Y \cap W^c$ 为 X 中闭集之交, 从而 A 为 X 中闭集.

Problem 16

What topology must X have if every real-valued function defined on X is continuous?

Solution: X 上的拓扑应为: 任意 X 的子集均为开集. 任取 $A \subset X$, 定义 $f_A : A \to \mathbb{R}$ 满足 $f|_A \equiv 0, f|_{X\setminus A} \equiv 1$, 于是由连续性可知 $f^{-1}\left(\left(-\frac{1}{2}, \frac{1}{2}\right)\right) = A$ 为开集.

Problem 14

Prove that the function $h(x) = \frac{e^x}{1+e^x}$ is a homeomorphism from the real line to the open interval $(0, 1)$.

Solution: 易见

$$h(x) = 1 - \frac{1}{1+e^x}$$

为单调递增函数, $h(-\infty) = 0, h(\infty) = 1$, 从而 h 连续且为 $\mathbb{R} \to (0,1)$ 的双射. 易求

$$h^{-1}(y) = \ln\left(\frac{y}{1-y}\right)$$

为 $(0,1) \to X$ 的连续映射, 从而 h 为同胚.

Problem 17

Let X denote the set of all real numbers with the finite-complement topology, and define $f : \mathbb{R} \to X$ by $f(x) = x$. Show that f is continuous, but is not a homeomorphism.

Proof: 设 X 为 finite-complement topology 意义下的开集, 从而 X 也是通常意义下的开集, 故 $f^{-1}(X) = X$ 为开集, 即 f 连续. 下说明 $g = f^{-1} : X \to \mathbb{R}$ 不连续. 设 X 为 $\mathbb{R}$ 中开集, 则 $g^{-1}(X) = X$ 在 finite-complement topology 意义下未必构成开集, 例如 $\mathbb{R}$ 中开集 $(0,1)$ 在 finite-complement topology 下不是开集. 从而 f 不是同胚.

Problem 27

Show that $d(x, A) = 0$ iff $x \in \overline{A}$.

Proof: 根据定义即知

$$\begin{aligned} d(x,A) = 0 &\iff \exists \{a_n\} \subset A \text{ s.t. } d(x, a_n) \to 0 \\ &\iff \forall x \text{ 的邻域 } U,\ \exists a \in A \cap U \\ &\iff x \in \overline{A} \end{aligned}$$

Problem 28

If A, B are disjoint closed subsets of a metric space, find disjoint open sets U, V s.t. $A \subset U, B \subset V$.

Solution: 考虑函数

$$f(x) = \frac{d(x,A)}{d(x,A) + d(x,B)}$$

课上已证, 对于闭集 A, $d(x,A) = 0 \iff x \in A$, 于是由 $A \cap B = \emptyset$ 可知 $d(x,A) + d(x,B) > 0$, 从而 f 为良定的连续函数, 故如下构造的 U, V 为不相交的开集.

$$U = \left\{x : f(x) < \frac{1}{2}\right\}, \quad V = \left\{x : f(x) > \frac{1}{2}\right\}$$

并且若 $x \in A$, 则 $f(x) = 0 \Longrightarrow x \in U$, 若 $x \in B$, 则 $f(x) = 1 \Longrightarrow x \in V$. 于是 $A \subset U, B \subset V$ 且 $U \cap V = \emptyset$. U, V 即为所求.

Problem 29

Show one can define a distance fucntion on an arbitary set X by $d(x,y)=1$ if $x\neq y$ and $d(x,x)=0$. What topology does d give to X?

Proof: 正定性和对称性显然, 对于三角不等式, 取 $x,y,z\in X$, 若 $x=y$ 则显然

$$d(x,y)\leq d(x,z)+d(z,y)$$

若 $x\neq y$, 则 $d(x,z),d(z,y)$ 中至少有一个为 1, 从而三角不等式仍然成立, 即 d 为距离. 易见该距离空间中的开球为单点集 $\{x\}$ 或全空间 X. 故 $\forall A\subset X$ 有

$$A=\bigcup_{x\in A}\{x\}$$

为开集, 从而 X 的任意子集均为开集.

Problem 30

Show that every closed subset of a metric space is the intersection of a countable number of open sets.

Proof: 由 $d(x,A)$ 为连续函数知

$$U_n=\left\{x:d(x,A)<\frac{1}{n}\right\},\quad n=1,2,\cdots$$

为一列包含 A 的开集, 于是

$$A\subset\bigcap_{n=1}^{\infty}U_n$$

设 $x\in\bigcap_{n=1}^{\infty}U_n$, 则

$$d(x,A)<\frac{1}{n},\forall n\Longrightarrow d(x,A)=0\Longleftrightarrow x\in A$$

从而 $A=\bigcap_{n=1}^{\infty}U_n$.

Problem 31

If A,B are subsets of a metric space, their *distance apart* $d(A,B)$ is the infimum of the numbers $d(x,y)$ where $x\in A$ and $y\in B$. Find two disjoint closed subsets of the plane which are zero distance apart. The *diameter* of A is the supremum of the numbers $d(x,y)$ where $x,y\in A$. Check that both of the closed sets which you have just found have infinite diameter.

Solution: 取 $X=\mathbb{R}^2$, 再取

$$A=\left\{(x,y)|\ y\geq\frac{1}{x},x>0\right\},\quad B=\left\{(x,y)|y\leq-\frac{1}{x},x>0\right\}$$

则 A,B 为闭集, 取 $x_n=\left(n,\frac{1}{n}\right)\in A,y_n=\left(n,-\frac{1}{n}\right)\in B$, 则

$$d(x_n,y_n)=\frac{2}{n}\to0\Longrightarrow d(A,B)=0$$

易见此时 $\operatorname{diam}(A)=\operatorname{diam}(B)=\infty$.

Problem 35

Given a map $f : X \to \mathbb{R}^{n+1} \setminus \{0\}$, find a map $g : X \to S^n$ which agrees with f on the set of $f^{-1}(S^n)$.

Solution: 取

$$g(x) = \frac{f(x)}{|f(x)|}$$

则 $g(X) \subset S^n$, 且若 $f(x) \in S^n$, 则 $g(x) = f(x)$.

Chapter 3 Compactness and Connectedness

Prob 3.6

Show that the Hausdorff condition cannot be relaxed in theorem (3.7).

theorem 3.7: A one-one (injective), onto (surjective), and continuous function from a compact space X to a Hausdorff space Y is a homeomorphism.

Solution: Consider $X = [0,1] \subset \mathbb{R}$, which is compact under the standard topology of $\mathbb{R}$. Take $Y = [0,1]$ with the finite complement topology, and let $f : X \to Y$ be the identity map.

For any open set $A \subset Y$, we have that $[0,1] \setminus A$ is a finite set in $[0,1]$, or $A = [0,1]$. Hence

$$A = [0, x_1) \cup (x_1, x_2) \cup \cdots \cup \left(x_{\{n-1\}}, x_n\right) \cup (x_n, 1], \quad 0 \le x_1 < \cdots < x_n \le 1.$$

It follows that $f^{-1}(A) = A$ is an open set in X. Thus $f : X \to Y$ is a continuous bijection.

To show that f^{-1} is not continuous, it suffices to find an open set $B \subset X$ such that $f(B) = B \subset Y$ is not open. Taking $B = \left(\frac{1}{3}, \frac{2}{3}\right)$ gives the desired counterexample.

Prob 3.8

(*Linderlof's theorem*) If X has a countable base for its topology, prove that any open set cover of X contains a countable subcover.

Proof: Let β be a basis of X, and let $X \subset \bigcup_\alpha U_\alpha$ be an open cover of X. For each $x \in X$, choose U_x from the cover such that $x \in U_x$. Since U_x is open, we can write it as $U_x = \bigcup_{i=1}^{\infty} B_i$ with $B_i \in \beta$. Then there exists some B_k such that $x \in B_k$, and we denote is as B_x.

Define

$$\mathcal{B} = \{B_x \mid x \in X\} \subset \beta$$

Since β is countable, $\mathcal{B}$ is countable as well. Now write $\mathcal{B} = \left\{B_{x_1}, B_{x_2}, \cdots\right\}$. For each i, we know $B_{x_i} \subset U_{x_i}$. Hence we have

$$X \subset \bigcup_{i=1}^{\infty} B_{x_i} \Longrightarrow X \subset \bigcup_{i=1}^{\infty} U_{x_i}$$

which is a countable subcover.

Prob 3.9

Prove that two disjoint compact subsets of a Hausdorff space always possess disjoint neighborhoods.

Proof: Let X be a Hausdorff space. Firstly, we prove that if $x \in X$ and $B \subset X$ be a compact set. Then exists a neighborhood U_x of x and B_x of B s.t. $B_x \cap U_x = \emptyset$.

$\forall y \in B, \exists$ neighborhood V_y of y and U_y of x s.t. $U_y \cap V_y = \emptyset$. From $B \subset \bigcup_{y \in B} V_y$ and compactness, $\exists y_1, \cdots, y_k$ s.t. $B \subset \bigcup_{i=1}^{k} V_{y_i}$. Let $U_x = \bigcap_{i=1}^{k} U_{y_i}$ be a neighborhood of x, $B_x = \bigcup_{i=1}^{k} V_{y_i}$ be a neighborhood of B, then $U_x \cap B_x = \emptyset$.

Consider $A, B \subset X$ be compact sets. For each $x \in A, \exists$ neighborhoods $U_x \ni x, B_x \supset B$ such that $U_x \cap B_x = \emptyset$. From $A \subset \bigcup_{x \in A} U_x$, $\exists x_1, \cdots, x_k$ s.t. $A \subset \bigcup_{i=1}^{k} U_{x_i}$. Denote $C = \bigcup_{i=1}^{k} U_{x_i}$ and $D = \bigcap_{i=1}^{k} B_{x_i}$, then C, D are neighborhoods of A, B and $C \cap D = \emptyset$.

Prob 3.10

Let A be a compact susbet of a metric space X. Show that the diameter of A is equal to $d(x, y)$ for some pair of points $x, y \in A$. Given $x \in X$, show that $d(x, A) = d(x, y)$ for some $y \in A$. Given a closed subset B, disjoint from A, show that $d(A, B) > 0$.

Proof: We have proved that: If X is a compact set and $f : X \to \mathbb{R}$ is continuous, then f is bounded and attains its bounds.

Consider the map $f : A \times A \to \mathbb{R}$, $f(a, b) = d(a, b)$. Then f is continuous and $A \times A$ is compact since A is compact. Hence $\operatorname{diam}(A) = \sup_{a,b\in A} f(a, b)$ can be attained.

Next, consider $g : A \to \mathbb{R}$, $g(a) = d(x, a)$. Then g is continuous and attains its infimum, i.e. $d(x, A)$.

For a compact set A and a closed set B with $A \cap B = \emptyset$, we have proved that $d(x, B) > 0, \forall x \notin B$. Thus $d(a, B) > 0, \forall a \in A$. Consider $h : A \to \mathbb{R}$, $h(a) = d(a, B)$. We have

$$\forall b \in B, a_1, a_2 \in A : d(a_1, B) \le d(a_1, b) \le d(a_1, a_2) + d(a_2, b)$$
$$\implies d(a_1, B) \le d(a_1, a_2) + d(a_2, B) \implies |h(a_1) - h(a_2)| \le d(a_1, a_2)$$

i.e. h is continuous and attains its infimum. Thus we have

$$d(A, B) = \inf_{a\in A} \inf_{b\in B} d(a, b) = \inf_{a\in A} h(a) = h(a') > 0.$$

Prob 3.11

Find a topological space and a compact subset whose closure is not compact.

Solution: Fixed a point $p \in X$, given X the topology as followed: $A \subset X$ is a open set if and only if $p \in A$ or $A = \emptyset$. Let $B = \{p\}$. For any open sets covering $B \subset \bigcup_\alpha U_\alpha$, there must exists $U_k \ni p \implies B \subset U_k$. Therefore B is a compact set.

For any $x \in X$ and a neighborhood U of x, by definition we know $p \in U \implies B \cap U \neq \emptyset \implies x \in \overline{B}$. Thus $\overline{B} = X$.

Now let $X = \mathbb{R}$ and $p = 0, B = \{0\}$, then $X \subset \bigcup_{i=1}^{\infty}(-i, i)$ is a open sets covering and don't exists a finite covering, i.e. $X = \overline{B}$ is not compact.

Prob 3.13

Let $f : X \to Y$ be a closed map with the property that the inverse image of each point of Y ia a compact subset of X. Show that $f^{-1}(K)$ is compact whenever K is compact in Y. Can the condition that f is closed be removed? (*closed map: the image of a closed set is closed*)

Proof: Consider an open sets covering of $f^{-1}(K) \subset \bigcup_\alpha U_\alpha$. For each $y \in K$, $\{U_\alpha\}$ is also an open sets covering of $f^{-1}(y)$. Since $f^{-1}(y)$ is compact, we can find $U_1, \cdots, U_k$ s.t. $f^{-1}(y) \subset \bigcup_{i=1}^{k} U_i$. Consider $X \setminus \bigcup_{i=1}^{k} U_i$ be a closed of X, then $f\left(X \setminus \bigcup_{i=1}^{k} U_i\right)$ is closed in Y, denote

$$V_y = Y \setminus f\left(X \setminus \bigcup_{i=1}^{k} U_i\right)$$

Then V_y is open in Y. Since $f^{-1}(y) \subset \bigcup_{i=1}^{k} U_i$, we know that $y \in V_y$ i.e. V_y is a neighborhood of y. Therefore we can write

$$K \subset \bigcup_{y\in K} V_y \overset{\text{compact}}{\Longrightarrow} K \subset \bigcup_{i=1}^{n} V_{y_i} \Longrightarrow f^{-1}(K) \subset \bigcup_{i=1}^{n} f^{-1}\left(V_{y_i}\right)$$

For each i, if $a \in f^{-1}\left(V_{y_i}\right)$, then $f(a) \notin f\left(X \setminus \bigcup_{j=1}^{k_i} U_{ij}\right) \Longrightarrow a \notin X \setminus \bigcup_{j=1}^{k_i} U_{ij} \Longrightarrow a \in \bigcup_{j=1}^{k_i} U_{ij}$. Thus we have $f^{-1}\left(V_{y_i}\right) \subset \bigcup_{j=1}^{k_i} U_{ij}$. Therefore

$$f^{-1}(K) \subset \bigcup_{i=1}^{n} \bigcup_{j=1}^{k_i} U_{ij}$$

is a countable covering of $f^{-1}(K) \Longrightarrow f^{-1}(K)$ is compact.

Prob 3.14

If $f : X \to Y$ is a one-one map, and if $f : X \to f(X)$ is a homeomorphism when we give $f(X)$ the induced topology from Y, we call f an *embedding* of X in Y. Show that a one-one map from a compact space to a Hausdorff space must be an embedding. (f *should be continuous*)

Proof: Since $f : X \to f(X)$ is a continuous bijective, we only need to show that $f^{-1} : f(X) \to X$ is continuous. That is: foy any closed set $A \subset X$, $\left(f^{-1}\right)^{-1}(A) = f(A)$ is a closed set in $f(X)$. As a closed set in a compact space, A is compact as well. From the continuity of f we know $f(A)$ is compact in a Hausdorff space. Therefore $f(A)$ is a closed set in $f(X)$. So f is an embedding.

M-Prob 7.4

Show that the metric space (X, d) is complete if and only if for any nested sequence $A_1 \supset A_2 \supset \cdots$ of nonempty closed sets of X such that $\operatorname{diam}(A_n) \to 0$, the intersection of the sets A_n is nonempty.

Proof: $\Longrightarrow$: Choose a sequence $\{x_n\}$ s.t. $x_n \in A_n$. $\forall \varepsilon > 0, \exists N$ s.t. $\forall n \geq N, \operatorname{diam}(A_n) < \varepsilon$. Thus $\forall m, n > N, x_n, x_m \in A_N \Longrightarrow d(x_n, x_m) < \operatorname{diam}(A_N) < \varepsilon \Longrightarrow \{x_n\}$ is a Cauchy sequence. Since X is complete, $\{x_n\}$ converge to un unique point $x \in X$ (uniqueness is from the metric on X). We prove that $x \in A_n, \forall n$. Suppose that $\exists N$ s.t. $x \notin A_N$. From A_N is closed, we know $d(x, A_N) > 0$. When n is large enough, x_n satisfies $d(x, x_n) < \frac{1}{2} d(x, A_N) \Longrightarrow x_n \notin A_N$. But when $n > N$, we have $x_n \in A_n \subset A_N$. It's a contradiction. Therefore $x \in A_n, \forall n \Longrightarrow \bigcup_{n=1}^{\infty} A_n \neq \emptyset$.

$\Longleftarrow$: Let $\{x_n\} \subset X$ be a Cauchy sequence. Suppose that $\{x_n\}$ isn't converge to any point in X, then $\overline{\{x_n\}} = \{x_n\}$. Let

$$A_k = \overline{\{x_n \mid n \geq k\}} = \{x_n \mid n \geq k\}$$

Then $\{A_n\}$ is a nested nonempty closed sets with $\operatorname{diam}(A_N) = \sup_{m,n\geq N} d(x_n, x_m) \to 0$. By condition we know that $\bigcap_{i=1}^{\infty} A_i \neq \emptyset$. But by difinition of A_n we have $\bigcap_{i=1}^{\infty} A_i = \emptyset$ which is a contradiction. Therefore $\{x_n\}$ converge to some $x \in X$ and thus X is complete.

Prob 3.21

If A and B are compact, and if W is a neighbourhood of $A \times B$ in $X \times Y$, find a neighbourhood U of A in X and a neighbourhood V of B in Y such that $U \times V \subset W$.

Proof: We can write W as

$$W = \bigcup_{\alpha} U_\alpha \times V_\alpha, \quad U_\alpha \subset A \text{ and } V_\alpha \subset B \text{ are open}$$

Then $\forall a \in A, b \in B$, we can find $U_{a,b} \times V_{a,b}$ s.t. $(a,b) \in U_{a,b} \times V_{a,b}$. Fix a, then $\left\{V_{a,b}\right\}_{b \in B}$ is an open covering of B. Since B is compact, there exists V_{a,b_i} s.t. $B \subset \bigcup_{i=1}^{n_a} V_{a,b_i}$. Denote that

$$V_a := \bigcup_{i=1}^{n_a} V_{a,b_i}, \quad U_a := \bigcap_{i=1}^{n_a} U_{a,b_i}$$

Then U_a is a neighborhood of a, $B \subset V_a$ and $U_a \times V_a \subset W$. Since $\bigcup_{a \in A} U_a$ is an open covering of A, we have finite open covering U_{a_i} i.e. $A \subset \bigcup_{i=1}^{n} U_{a_i}$. Denote

$$U = \bigcup_{i=1}^{n} U_{a_i}, \quad V = \bigcap_{i=1}^{n} V_{a_i}$$

Then $A \subset U, B \subset V$ i.e. U, V are neighborhoods of A, B. Let $(x,y) \in U \times V$, assume $x \in U_{a_k}$, then $(x,y) \in U_{a_k} \times V_{a_k} \in W$. Therefore $U \times V \subset W$.

Prob 3.25

Show that the diagonal map $\Delta : X \to X \times X$ defined by $\Delta(x) = (x,x)$ is indeed a map, and chek that X is Hausdorff iff $\Delta(X)$ is closed in $X \times X$.

Proof: Consider projections $p : X \times X \to X$, then $p \circ \Delta = \mathrm{id}$ is continuous, thus Δ is continuous.

$\Longrightarrow$: $\forall x, y \in X$, find neighborhood U_x of x and neighborhood U_y of y s.t. $U_x \cap U_y = \emptyset$. Consider $U_x \times U_y \subset X \times X$. We claim that $\left(U_x \times U_y\right) \cap \Delta(X) = \emptyset$ since $U_x \cap U_y = \emptyset$. Thus $U_x \times U_y$ is an open subset of $(\Delta(X))^c$. Therefore

$$(\Delta(X))^c = \bigcup_{x \neq y \in X} U_x \times U_y \text{ is open in } X \times X$$

Thus $\Delta(X)$ is closed.

$\Longleftarrow$: From $(\Delta(X))^c$ is open, we can write

$$(\Delta(X))^c = \bigcup_{\alpha} U_\alpha \times V_\alpha$$

$\forall x \neq y \in X, (x,y) \in (\Delta(X))^c \Longrightarrow \exists U_1 \times V_1$ s.t. $(x,y) \in U_1 \times V_1 \Longrightarrow x \in U_1, y \in V_1$. Thus

$$(U_1 \times V_1) \cap \Delta(X) = \emptyset \Longrightarrow U_1 \cap V_1 = \emptyset \Longrightarrow X \text{ is Hausdorff.}$$

Prob 3.26

We know that the projections $p_1 : X \times Y \to X$, $p_2 : X \times Y \to Y$ are open maps. Are they always closed?

Solution: NO. Consider $X = Y = \mathbb{R}$ with the standard topology of $\mathbb{R}$. Let

$$A = \left\{(x, y) \mid y \geq \frac{1}{x}, x > 0\right\} \subset \mathbb{R}^2$$

be a closed set. Then $p_1(A) = (0, +\infty)$ is open in $\mathbb{R}$. Similarly p_2 is not closed map.

Prob 3.30

Let X be the set of all points in the plane which have at least one rational coordinate. Show that X, with the induced topology, is a connected space.

Solution: We prove X is path-connected. Let $(x_1, y_1), (x_2, y_2) \in X$.

Case 1: $x_1, x_2 \in \mathbb{Q}$ or $y_1, y_2 \in \mathbb{Q}$. WLOG, assume $x_1, x_2, \in \mathbb{Q}$. Choose $z \neq y_1, y_2, z \in \mathbb{Q}$, then

$$\gamma(t) = \begin{cases} (x_1, y_1 + 3t(z - y_1)), & 0 \leq t \leq \frac{1}{3} \\ \big(x_1 + 3\big(t - \frac{1}{3}\big)(x_2 - x_1), z\big), & \frac{1}{3} \leq t \leq \frac{2}{3} \\ \big(x_2, z + 3\big(t - \frac{2}{3}\big)(y_2 - z)\big), & \frac{2}{3} \leq t \leq 1 \end{cases}$$

is a continous map from $[0, 1]$ to X and $\gamma(0) = (x_1, y_1), \gamma(1) = (x_2, y_2)$.

Case 2: $x_1, y_2 \in \mathbb{Q}$ or $x_2, y_1 \in \mathbb{Q}$. Assume $x_1, y_2 \in \mathbb{Q}$, then

$$\gamma(t) = \begin{cases} (x_1, y_1 + 2t(y_2 - y_1)), & 0 \leq t \leq \frac{1}{2} \\ \big(x_1 + 2\big(t - \frac{1}{2}\big)(x_2 - x_1), y_2\big), & \frac{1}{2} \leq t \leq 1 \end{cases}$$

is a continous map from $[0, 1]$ to X and $\gamma(0) = (x_1, y_1), \gamma(1) = (x_2, y_2)$.

Therefore X is path-connected and thus is connected.

Prob 3.32

If X has only a finite number of components, show that each component is both open and closed. Find a space none of whose components are open sets.

Proof: We have proved that each connected componet is closed in X. If X has finite connected components, assume $X = \bigcup_{i=1}^{n} A_i$, from $\{A_i\}$ is seperated, we have

$$A_i^c = \bigcup_{1 \leq j \leq n, j \neq i} A_j \text{ is closed}$$

Thus A_i is open and closed.

Prob 3.33

(*Intermediate value theorem*). If $f : [a, b] \to \mathbb{R}$ is a map such that $f(a) < 0$ and $f(b) > 0$, use the connectedness of $[a, b]$ to establish the existence of a point c for which $f(c) = 0$.

Proof: Suppose $\forall c \in [a, b], f(c) \neq 0$. Consider

$$A = \{x \in [a, b] \mid f(x) > 0\}, \quad B = \{x \in [a, b] \mid f(x) < 0\}$$

then $a \in A, b \in B$ and $[a, b] = A \cup B$. Let $x \in \overline{A}$, then $\exists \{x_n\} \subset A \text{ s.t. } x_n \to x$. We have

$$f(x) = \lim_n f(x_n) \geq 0 \Longrightarrow f(x) > 0 \Longrightarrow x \in A \Longrightarrow \overline{A} \subset A \Longrightarrow A = \overline{A}$$

Similarly $B = \overline{B}$. Then $\overline{A} \cap B = A \cap \overline{B} = A \cap B = \emptyset$, which is contradict with the connectedness of $[a, b]$. Therefore $\exists c \in [a, b] \text{ s.t. } f(c) = 0$.

Prob 3.34

A space X is *locally connected* if for each $x \in X$, and each neighbourhood U of x, there is a connected neighbourhood V of x which is contained in U. Show that any euclidean space, and therefore any space which is locally euclidean (like a surface), is locally connected. If

$$X = \{0\} \cup \left\{ \frac{1}{n} \mid n = 1, 2, \cdots \right\}$$

with the subspace topology from the real line, show that X is not locally connected.

Proof: Let X be euclidean. $\forall x \in X$ and neighborhood U of x, there exists $B(x, \varepsilon) \text{ s.t. } B(x, \varepsilon) \subset U$. $B(x, \varepsilon)$ is a connected neighborhood of x. Thus X is locally connected.

Let $X = \{0\} \cup \left\{ \frac{1}{n} \mid n = 1, 2, \cdots \right\}$. We first prove that if $C \subset X$ be connected $\iff C$ is a one-point set. It's trivial that $\{0\}, \left\{ \frac{1}{n} \right\}, n \geq 1$ are connected sets in X. Now let C be a connected subset of X. Consider $\iota : X \to \mathbb{R}, \iota(x) = x$ which embed X in $\mathbb{R}$. Then ι is continuous and thus $\iota(C) = C$ is connected in $\mathbb{R} \implies C$ must be an interval or a one-point set $\implies |C| = 1$.

Now we show that X isn't locally connected: Choose $0 \in X$ and $U \subset X$ be a neighborhood of 0. Any neighborhood of 0 in $\mathbb{R}$ must contain $(-\delta, \delta)$ for small enough δ. Thus U contains $(-\delta, \delta) \cap X \implies \{0\} \cup \left\{ \frac{1}{n} \mid n > \frac{1}{\delta}, n \in \mathbb{Z}^+ \right\} \subset U \implies U$ can't be connected. Therefore 0 doesn't have a connected neighborhood. Thus X is not locally connected.

Prob 3.35

Show that local connectedness is preserved by a homeomorphism, but need not be preserved by a continuous function.

Proof: Let $f : X \to Y$ be a homeomorphism, and suppose that X is locally connected. $\forall y \in Y$ and any neighborhood V of y, there exists a unique $x \in X$ with $f(x) = y$. Since f is continuous, $f^{-1}(V)$ is a neighborhood of x in X. By local connectedness of X, there exists a connected neighborhood U of x s.t. $U \subset f^{-1}(V)$. Because f and f^{-1} are continuous, $f(U)$ is a connected neighborhood of y, and $f(U) \subset V$. Thus Y is locally connected.

Consider

$$X = \{0, 1, 2, \cdots\}, \quad Y = \{0\} \cup \left\{ \frac{1}{n} \mid n = 1, 2, \cdots \right\}$$

$$f : X \to Y, \quad f(0) = 0, \ f(n) = n^{-1}, \forall n \geq 1$$

X, Y both equipped with the subspace topology. Then $\{n\}_{n \geq 0}$ is a basis for X and $\left\{0, \frac{1}{n}\right\}_{n \geq 1}$ is a basis for Y. Thus f is continuous. For $\forall x \in X$ and neighborhood V of x, $\{x\}$ is a connected neighborhood of x satisfying $\{x\} \subset V$. Therefore X is locally connected. From 3.34 we know Y is not locally connected. Thus only continuous map can not preserve the locally connectedness.

Prob 3.37

Show that the continuous image of a path-connected space is path-connected.

Proof: Let X be path-connected and $f : X \to Y$ be continuous. $\forall x' \neq y' \in f(X)$, we can find $x \neq y \in X \text{ s.t. } f(x) = x', f(y) = y'$. Choose $\gamma : [0, 1] \to X \text{ s.t. } \gamma(0) = x, \gamma(1) = y$, then $f \circ \gamma$ is a continuous cap from $[0, 1]$ to $f(X)$ and $f \circ \gamma(0) = x', f \circ \gamma(1) = y'$. Thus $f(X)$ is path-connected.

Prob 3.39

Prove that the product of two path-connected spaces is path-connected.

Proof: Let X, Y be path-connected. Let $(x_1, y_1) \neq (x_2, y_2) \in X \times Y$. Choose continuous maps

$$\gamma_x : [0,1] \to X \text{ s.t. } \gamma_x(0) = x_1, \gamma_x(1) = x_2$$
$$\gamma_y : [0,1] \to Y \text{ s.t. } \gamma_y(0) = y_1, \gamma_y(1) = y_2$$

Let $\gamma : [0,1] \to X \times Y, t \mapsto \big(\gamma_x(t), \gamma_y(t)\big)$. From $p_1 \circ \gamma = \gamma_x, p_2 \circ \gamma = \gamma_y$ is continuous we know γ is continuous and $\gamma(0) = (x_1, y_1), \gamma(1) = (x_2, y_2)$. Therefore $X \times Y$ is path-connected.

Prob 3.40

If A and B are path-connected subsets of a space, and if $A \cap B$ is nonempty, prove that $A \cup B$ is path-connected.

Proof: Let $x, y \in A \cup B$ and assume $x \in A \setminus B, y \in B \setminus A$. Choose $z \in A \cap B$ and γ_1 join x and z in A, γ_2 join z and y in B. Let

$$\gamma(t) = \begin{cases} \gamma_1(2t), & 0 \leq t \leq \frac{1}{2} \\ \gamma_2(2t-1), & \frac{1}{2} \leq t \leq 1 \end{cases}$$

then γ is a continuous map that join x and y in $A \cup B$. Therefore $A \cup B$ is path-connected.

Indentification Spaces

Prob 4.2

Which space do we obtain if we take a Möbius strip and identify its boundary circle to a point?

Solution: 考虑一个矩形, 莫比乌斯带可以认为是将矩形的左右两边按相反顺序识别, 因此将莫比乌斯带的边界圆周识别为一点, 相当于再将矩形的上下两边按相反顺序识别. 根据课上所讲, 这会得到 $\mathbb{R}P^2$ 空间.

Prob 4.3

Let $f : X \to Y$ be an identification map, let A be a subspace of X, and give $f(A)$ the induced topology from Y. Show that the restriction $f|_A : A \to f(A)$ need not be an identification map.

Proof: 取 $X = [0, 1], Y$ 为将 $0, 1$ 识别为一点 p 得到的空间, 其同胚于 S^1. 取 $A = [0, 1)$, 则 $f(A) = Y$. 取 $E = \left[0, \frac{1}{2}\right) \subset A$ 为 A 中的开集, 于是 $f|_A\ (E) = f(E)$ 为以 p 为端点的半开弧, 易见 $f(A) = Y$ 中任何包含 p 的开集都必须包含一段含 p 的开弧, 因此 $f|_A\ (E)$ 不是 $f(A)$ 中的开集, 而 $f|_A^{-1}\ (E) = E$ 是 A 中的开集. 由此可见 $f|_A$ 不是识别映射.

Prob 4.4

With the terminology of Problem 3, show that if A is open in X and if f takes open sets to open sets, or if A is closed in X and f takes closed sets to closed sets, then $f|_A$ is an identification map.

Proof: Let $B \subset f(A)$ be an open set. Then $\exists U \subset Y$ be open s.t. $B = f(A) \cap U$. Since A is open and f is an open map, we know $f(A)$ is open. Thus B is open and $f^{-1}(B)$ is open due to f is an identification map.

Let $B \subset f(A)$ and $f^{-1}(B)$ is open. Since f is an identification map, B is open in $Y \Longrightarrow B$ is open in $f(A)$. Above all we know $f|_A$ is an identification map.

Prob 4.5

Let X denote the union of the circles $\left(x - \frac{1}{n}\right)^2 + y^2 = \left(\frac{1}{n}\right)^2, n \geq 1$ with the subspace topology from the plane, and let Y denote the identification space obtained from the real line by identifying all the integers to a single point. Prove X and Y are not homeomorphic. (X is called the *Hawaiian earring*.)

Proof: 我们定义拓扑 X 在 $x \in X$ 处为局部紧的当且仅当存在 x 的邻域 U s.t. $\overline{U}$ 为紧集. 易见同胚保持局部紧性. 由 X 为 $\mathbb{R}^2$ 中的闭集, 因此 $\forall x \in X$ 以及 x 的邻域 $U, \overline{U}$ 为 X 中的闭集, 故 $\overline{U}$ 为 $\mathbb{R}^2$ 中的有界闭集, 从而 $\overline{U}$ 在 $\mathbb{R}^2$ 中紧, 进而在 X 中紧. 因此 X 在每一点处局部紧.

考虑 Y 中将所有整数识别在一起得到的 $q \in Y$. 下证对任意 q 的邻域 $V, \overline{V}$ 在 Y 中不紧. 事实上, 设 $f : \mathbb{R} \to Y$ 为识别映射, V 是 q 的一个邻域, 则 $f^{-1}(V) \subset \mathbb{R}$ 为开集且 $\mathbb{Z} \subset f^{-1}(V)$. $\mathbb{R}$ 中开集可以写作可列个开区间的并, 因此不妨设

$$f^{-1}(V) = \bigcup_{n \in \mathbb{Z}} (n - \varepsilon_n, n + \delta_n), \quad 0 < \varepsilon_n, \delta_n < \frac{1}{4}$$

$$\Longrightarrow V = \{q\} \bigcup \left(\bigcup_{n \in \mathbb{Z}} (n - \varepsilon_n, n + \delta_n) \setminus \{n\} \right) \Longrightarrow \overline{V} = \{q\} \bigcup \left(\bigcup_{n \in \mathbb{Z}} [n - \varepsilon_n, n + \delta_n] \setminus \{n\} \right)$$

考虑 $\overline{V}$ 的开覆盖

$$\overline{V} \subset \bigcup_{n\in\mathbb{Z}} V_n, \quad V_n := f([n-2\varepsilon_n, n+2\delta_n])$$

由 $\overline{V}$ 以及 V_i 的形式可知上述开覆盖不存在有限子覆盖, 因此 $\overline{V}$ 对任意 V 均不紧, 从而 $q \in Y$ 不具备局部紧性. 综上 X 与 Y 不同胚.

Prob 4.11

Show that the fucntion $f : [0, 2\pi] \times [0, \pi] \to \mathbb{R}^5$ defined by

$$f(x, y) = (\cos x, \cos 2y, \sin 2y, \sin x \cos y, \sin x \sin y)$$

induces an embedding of the Klein bottle in $\mathbb{R}^5$.

Proof: 考虑对矩形 $[0, 2\pi] \times [0, \pi]$ 的边界进行如下识别: 将上下两边按相反方向识别, 将左右两边按相同方向识别, 即将 $(x, 0)$ 与 $(2\pi - x, \pi)$ 识别为一点, 将 $(0, y)$ 与 $(2\pi, y)$ 识别为一点, 由此得到克莱因瓶. 考虑将 f 作用在识别空间上:

$$\begin{aligned} i : X \to Y, \quad \overline{f} : Y &\to \mathbb{R}^5 \\ [(x, y)] &\mapsto f(x, y) \end{aligned}$$

容易验证 $f(x, 0) = f(2\pi - x, \pi), f(0, y) = f(2\pi, y)$, 因此 $\overline{f}$ 定义良好且连续. 下证 $\overline{f}$ 为单射: 设 $\overline{f}([(x, y)]) = \overline{f} = ([(x, y)])$, 则 $f(x, y) = f(x', y') \Longrightarrow \cos x = \cos x', \cos 2y = \cos 2y', \sin 2y = \sin 2y'$, 解得 $x = 2\pi - x', y = y'$ 或 $y = \pi - y'$. 不妨设 $(x, y) \neq (x', y')$.

若 $y = y'$, 则由 $\cos x = \cos x', \sin x = \sin x'$ 可得 $x = x'$ 或 $\{x, x'\} = \{0, 2\pi\}$, 故只能是后者, 此时有 $[(x, y)] = [(x', y')]$.

若 $y = \pi - y'$, 代入 $\sin x \cos y = \sin x' \cos y', \sin x \sin y = \sin x' \sin y'$ 得 $x = 2\pi - x', \{y, y'\} = \{0, \pi\}$, 此时亦有 $[(x, y)] = [(x', y')]$. 因此 $\overline{f}$ 为单射.

又由于 $[0, 2\pi] \times [0, \pi]$ 紧, 故克莱因瓶紧, 而 $\mathbb{R}^5$ 是 Hausdorff 空间, 因此 Prob 3.14 可知 $\overline{f}$ 是嵌入.

Exercise 1

(A subset in $\mathbb{R}^n$ is compact if and only if it is closed and bounded)

(1) Give an example to show that a closed and bounded subset in a metric space is not necessarily compact in general.

(2) Let X be a metric space and let $\varepsilon > 0$. A subset $E \subset X$ is called an *ε-grid* if for any $x \in X$ there exists $e \in E$ such that $d(e, x) < \varepsilon$. A metric space is called *totally bounded* if for any $\varepsilon > 0$ there exists a finite ε-grid. Prove that a subset in a complete metric space is compact if and only if it is closed and totally bounded.

Proof: (1) Let X be a set with the distance:

$$d(x, y) = \begin{cases} 1, x \neq y \\ 0, x = y \end{cases}$$

Then (X, d) is a metric space. $\{x\}$ and X are open balls in $X \Longrightarrow \forall A \subset X$, A is open $\Longrightarrow \forall A \subset X$, A is open and closed. Let $|X| = \infty$, then X is a closed and bounded set (since $X \subset X$ be open ball). But for open covering

$$X = \bigcup_{x \in X} \{x\}$$

we can not find any finite sub-covering. Therefore X is not compact.

(2) compact $\Longrightarrow$ closed and totally bounded:

Let (X, d) be a complete metric space and $A \subset X$ be compact. $\forall \varepsilon > 0$, we have an open covering

$$A \subset \bigcup_{x \in A} B(x, \varepsilon).$$

By compactness, there exist finitely many points $x_1, \cdots, x_n$ such that

$$A \subset \bigcup_{i=1}^{n} B(x_i, \varepsilon).$$

Thus $\{x_1, \cdots, x_n\}$ is an ε-grid for A.

Since X is a Hausdorff space and A is compact, A must be closed.

closed and totally bounded $\Longrightarrow$ compact:

Let $A \subset X$ be closed and totally bounded. First we prove that A is *sequentially compact*. Let $\{x_n\} \subset A$ and set $\varepsilon_1 = \frac{1}{4}$. Then there exist points $\left\{a_1, \cdots, a_{n_1}\right\}$ s.t.

$$A \subset \bigcup_{i=1}^{n_1} B(a_i, \varepsilon_1).$$

Hence there exists some $1 \leq k \leq n_1$ such that $B(a_k, \varepsilon_1)$ contains infinitely many x_j. That is, there exists a subsequence $\{x_k^1\} \subset \{x_n\}$ such that $\left|x_k^1 - x_{k'}^1\right| < \frac{1}{2}$.

Repeat this construction with $\varepsilon_n = 2^{-n-1}$, obtaining nested subsequences $\{x_k^n\}_{k=1}^{\infty} \subset \{x_k^{n-1}\}_{k=1}^{\infty}$ satisfying $|x_k^n - x_{k'}^n| < \frac{1}{2^n}$. Define the diagonal subsequence $\{x_n^n\}_{n=1}^{\infty}$. Then for all $m, n > k$,

$$|x_n^n - x_m^m| < \frac{1}{2^k}.$$

Thus $\{x_n^n\}$ is a Cauchy sequence. By completeness, there exists $x \in X$ with $x_n^n \to x$. Since A is closed, we have $x \in A$. Therefore every sequence in A has a convergent subsequence.

Now suppose A is not compact. Let $A \subset \bigcup_{\alpha} U_\alpha$ be an open covering. By Lindelöf's theorem, there exists a countable subcover $A \subset \bigcup_{i=1}^{\infty} U_i$. Since A does not have a finite subcover, we can choose

$$x_1 \in A \setminus U_1, \quad x_2 \in A \setminus (U_1 \cup U_2), \cdots, \quad x_n \in A \setminus \left(\bigcup_{i=1}^{n} U_i\right), \cdots$$

This produces $\{x_n\} \subset A$. By sequential compactness, $\exists x \in A$ s.t. $x_n \to x$. Suppose $x \in U_M$. Then there exists $\varepsilon > 0$ s.t. $B(x, \varepsilon) \subset U_M$. Since $x_n \to x$, $\exists N$ s.t. $\forall n > N$, $x_n \in B(x, \varepsilon) \subset U_M$. But this contradicts the construction of $\{x_n\}$ when $n > \max\{N, M\}$, since $x_n \notin \bigcup_{i=1}^{n} U_i$.

Therefore A is compact.

Prob 4.10

Let S^2 be the unit sphere in $\mathbb{R}^2$ and define $f \to \mathbb{R}^4$ by

$$f(x,y,z) = (x^2 - y^2, xy, xz, yz)$$

Show that f induces an embedding of the projective plane in $\mathbb{R}^4$.

Proof: 考虑将 S^2 的对径点识别得到的空间, 即 $\mathbb{R}P^2$, 易见

$$f(x,y,z) = f(-x,-y,-z)$$

因此 f 可以作用在 $\mathbb{R}P^2$ 上. 设 $U \subset \mathbb{R}^4$ 为开集, 将 f 延拓为 $\overline{f} : \mathbb{R}^3 \to \mathbb{R}^4$, 于是 $\overline{f}$ 为多项式, 进而 $\overline{f}^{-1}(U)$ 为 $\mathbb{R}^3$ 中的开集. 从而 $f^{-1}(U) = S^2 \cap \overline{f}^{-1}(U)$ 为 S^2 中的开集, 进而为 $\mathbb{R}P^2$ 中的开集, 因此 f 连续. 下验证 f 为单射: 将 $(x,y,z) \in S^2$ 写作 极坐标形式

$$(x,y,z) = (\cos\theta\sin\varphi, \sin\theta\sin\varphi, \cos\varphi), \quad \theta \in [0, 2\pi), \varphi \in [0, \pi]$$

于是 $f(x,y,z) = f(x',y',z')$ 等价于

$$\begin{cases} \cos 2\theta \sin^2\varphi = \cos 2\theta' \sin^2\varphi' \\ \sin 2\theta \sin^2\varphi = \sin 2\theta' \sin^2\varphi' \\ \cos\theta \sin 2\varphi = \cos\theta' \sin 2\varphi' \\ \sin\theta \sin 2\varphi = \sin\theta' \sin 2\varphi' \end{cases}$$

解得 $(\theta', \varphi') = (\theta, \varphi)$ 或 $(\theta + \pi, \pi - \varphi)$, 即 $(x',y',z') = (x,y,z)$ 或 $(x',y',z') = (-x,-y,-z)$. 因此 f 作用在 $\mathbb{R}P^2$ 为单射. 故由 $\mathbb{R}P^2$ 为紧空间以及 $\mathbb{R}^4$ 为 Hausdorff 空间可知 f 为嵌入.

Prob 4.13

Show that the product of two topological groups is a topological group.

Proof: 令 G, H 为拓扑群, 则 $G \times H$ 作为拓扑空间是拓扑积空间. 定义乘法

$$m : (g_1, h_1) \cdot (g_2, h_2) = (g_1 \cdot g_2, h_1 \cdot h_2)$$

于是

$$(g,h)^{-1} = (g^{-1}, h^{-1})$$

下验证乘法与取逆映射的连续性: 设 $K \subset G \times H$ 为开集, 于是 $K = \bigcup_\alpha G_\alpha \times H_\alpha$, 其中 G_α, H_α 分别为 G, H 中的开集. 记 G, H 中的乘法为 m_G, m_H, 则

$$m^{-1}(K) = \bigcup_\alpha m^{-1}(G_\alpha \times H_\alpha)$$

下证 $m^{-1}(G_\alpha \times H_\alpha)$ 为 $(G \times H) \times (G \times H)$ 中的开集. 我们有

$$\begin{aligned} ((g_1,h_1),(g_2,h_2)) \in m^{-1}(G_\alpha \times H_\alpha) &\iff (g_1 \cdot g_2, h_1 \cdot h_2) \in G_\alpha \times H_\alpha \\ &\iff g_1 \cdot g_2 \in G_\alpha, h_1 \cdot h_2 \in H_\alpha \\ &\iff (g_1, g_2) \in m_G^{-1}(G_\alpha), (h_1, h_2) \in m_H^{-1}(H_\alpha) \end{aligned}$$

通过换序可知 $(G \times H) \times (G \times H) \cong (G \times G) \times (H \times H)$. 而 $m_G^{-1}(G_\alpha) \times m_H^{-1}(H_\alpha)$ 为后者中的开集, 因此由上述等价于关系即知 $m^{-1}(G_\alpha \times H_\alpha)$ 为前者中的开集, 故 m 连续. 同理可证取逆映射连续. 因此 $G \times H$ 为拓扑群.

Prob 4.14

Let G be a topological group. If H is a subgroup of G, show that its closure $\overline{H}$ is also a subgroup, and that if H is normal then so is $\overline{H}$.

Proof: 我们已证, 对连续映射 f, 有 $f\left(\overline{H}\right) \subset \overline{f(H)}$. 对取逆映射 i, 我们有

$$i\left(\overline{H}\right) \subset \overline{i(H)} \overset{H \text{ is a subgroup}}{=\!=\!=} \overline{H} \Longrightarrow i\left(\overline{H}\right) = \overline{H}$$

对于乘法映射 m, 我们有

$$m\left(\overline{H} \times \overline{H}\right) \subset m\left(\overline{H \times H}\right) \subset \overline{m(H \times H)} \overset{H \text{ is a subgroup}}{=\!=\!=} \overline{H}$$

从而 $\overline{H}$ 对乘法和取逆均封闭, 且 $e \in H \subset \overline{H}$, 故 $\overline{H}$ 为子群.

若 H 为正规子群, 则 $\forall g \in G$, 有 $gHg^{-1} = H$. 考虑映射 $f_g : G \to G, f_g(h) = ghg^{-1}$, 则 f_g 连续且 $f_g(H) = H$. 因此

$$f_g\left(\overline{H}\right) \subset \overline{f_g(H)} = \overline{H}$$

由群论的知识知 $g\overline{H}g^{-1} \subset \overline{H}, \forall g \iff g\overline{H}g^{-1} = \overline{H}, \forall g$, 故 $\overline{H}$ 亦为正规子群.

Prob 4.15

Let G be a compact Hausdorff space which has the structure of a group. Show that G is a topological group if the multiplication function $m : G \times G \to G$ is continuous.

Proof: 只需证明取逆映射 i 连续. 设 $K \subset G$ 为闭集, 下证 $i^{-1}(K)$ 连续. 注意到 $\{e\} \subset G$ 为闭集, 由 m 的连续性, 可知

$$m^{-1}(\{e\}) = \{(x, y) \in G \times G \mid x \cdot y = e\} = \{(x, i(x)) \mid x \in G\} \subset G \times G$$

为闭集. 上述集合即为 i 的图像, 即为 $\Gamma(i)$. 考虑 $G \times K \subset G \times G$, 由定义知这是 $G \times G$ 中的闭集. 由 G 为紧的 Hausdorff 空间知 $G \times G$ 亦为紧的 Hausdorff 空间, 于是 $G \times F, \Gamma(i)$ 均为紧集. 考虑投影映射 $p : G \times G \to G, p(x, y) = x$, 则

$$p(G \times K \cap \Gamma(i)) = i^{-1}(K)$$

从而由 p 的连续性知 $i^{-1}(K)$ 为紧集, 进而由 G 为 Hausdorff 空间知 $i^{-1}(K)$ 为闭集. 因此 i 连续, 故 G 为拓扑群.

Prob 4.16

Prove that $\mathrm{O}(n)$ is homeomorphic to $\mathrm{SO}(n) \times \mathbb{Z}^2$. Are these two isomorphic as topological groups?

Solution: 先证明 $\mathrm{O}(n)$ 与 $\mathrm{SO}(n) \times \mathbb{Z}_2$ 同胚: 取 $R = \mathrm{diag}(-1, 1, \cdots, 1)$ 为对角阵, 记

$$\varphi : \mathrm{SO}(n) \times \mathbb{Z}_2 \to \mathrm{O}(n)$$

$$(A, \varepsilon) \mapsto \begin{cases} A, & \varepsilon = 1 \\ AR, & \varepsilon = -1 \end{cases}$$

单射: 若 $\varphi(A_1, \varepsilon_1) = \varphi(A_2, \varepsilon_2)$, 若 $\varepsilon_1 = \varepsilon_2$, 自然 $A_1 = A_2$, 若 $\varepsilon_1 = -\varepsilon_2$, 不妨设 $\varepsilon_1 = 1$, 则 $A_1 = A_2R \Longrightarrow |A_1| = -|A_2|$ 矛盾. 故 $(A_1, \varepsilon_1) = (A_2, \varepsilon_2)$.

满射: 设 $A \in \mathrm{O}(n)$, 若 $|A| = 1$, 则 $(A, 1) \in \mathrm{SO}(n) \times \mathbb{Z}_2$ 且 $\varphi(A, 1) = A$; 若 $|A| = -1$, 则 $(AR, -1) \in \mathrm{SO}(n) \times \mathbb{Z}_2$, 且 $\varphi(AR, -1) = AR^2 = A$. 故 φ 满, 从而为双射.

连续: 易见 $\mathrm{SO}(n)\times\mathbb{Z}_2$ 的连通分支为 $\mathrm{SO}(n)\times\{1\}$ 与 $\mathrm{SO}(n)\times\{-1\}$, 二者均为既开又闭的子集, φ 在二者上分别为 A 与 AR, 分别连续, 故 φ 连续. 同理, φ^{-1} 有表达式

$$\varphi^{-1}(B)=\begin{cases}(B,1), & |B|=1\\ (BR,-1), & |B|=-1\end{cases}$$

在 $\mathrm{O}(n)$ 的两个连通分支 $\mathrm{SO}(n)$ 与 $\mathrm{O}(n)\setminus\mathrm{SO}(n)$ 上分别连续, 故 φ^{-1} 连续.

综上 φ 为同胚, 故 $\mathrm{SO}(n)\times\mathbb{Z}_2$ 与 $\mathrm{O}(n)$ 同胚.

下证明当 n 为奇数时有 $\mathrm{SO}(n)\times\mathbb{Z}_2\cong\mathrm{O}(n)$, 偶数时二者不同构.

当 n 为奇数时, 容易验证

$$\Psi:\mathrm{O}(n)\to\mathrm{SO}(n)\times\mathbb{Z}_2,\quad A\mapsto(|A|A,|A|)$$

为双射 (注意 $||A|A|=|A|^{n+1}=1$), 连续性类似同胚时的论述即得, 故 Ψ 为同构.

当 n 为偶数时, 假设存在 $\phi:\mathrm{SO}(n)\times\mathbb{Z}_2\to\mathrm{O}(n)$ 为同构. 考虑群的中心化子

$$Z(G):=\{g\in G\mid\forall h\in G,gh=hg\}$$

易见同构保持群的中心化子. 而 $Z(\mathrm{SO}(n)\times\mathbb{Z}_2)=\{(I,1),(I,-1)\}$, 两个元素属于 $\mathrm{SO}(n)\times\mathbb{Z}_2$ 的不同连通分支, 故 $\phi(I,1)$ 与 $\phi(I,-1)$ 在 $\mathrm{O}(n)$ 中属于不同连通分支, 且均为 $\mathrm{O}(n)$ 的中心化子. 而 $\phi((I,1))=I$, 于是必有 $|\phi(I,-1)|=-1$. 但此时 $Z(\mathrm{O}(n))=\{I,-I\}$, 其中 $|-I|=(-1)^n=1$, 矛盾! 故同构 ϕ 不存在. 此时两个空间作为拓扑群不同构.

Prob 4.24

If T is an automorphism of the topological group $\mathbb{R}$ (i.e., $T:\mathbb{R}\to\mathbb{R}$ is a isomorphism of topological groups) show that $T(r)=rT(1)$ for any rational number r. Deduce that $T(x)=xT(1)$ for any real number x, and hence that the automorphism group of $\mathbb{R}$ is isomorphic to $\mathbb{R}\times\mathbb{Z}_2$.

Proof: 设 T 为 $\mathbb{R}$ 的一个拓扑群自同构, 则 $\forall r=\frac{p}{q}\in\mathbb{Q}$, 有

$$T(r)=T\left(\frac{1}{q}+\frac{1}{q}+\cdots+\frac{1}{q}\right)=pT\left(\frac{1}{q}\right),$$

$$T(1)=T\left(\frac{1}{q}+\cdots+\frac{1}{q}\right)=qT\left(\frac{1}{q}\right)\Longrightarrow T\left(\frac{1}{q}\right)=\frac{1}{q}T(1)\Longrightarrow T(r)=rT(1)$$

对任意实数 $x\in\mathbb{R}$, 取有理数列 $\{r_n\}\subset\mathbb{Q}$ 使得 $r_n\to x$, 则

$$T(x)=T\Big(\lim_n r_n\Big)=\lim_n T(r_n)=\lim_n r_nT(1)=xT(1)$$

因此 T 由 $T(1)$ 唯一确定.

下考察 $\mathrm{Aut}(\mathbb{R})$ 的结构. $\forall T\in\mathrm{Aut}(\mathbb{R}),T(1)\neq0$, 且任意 $T(1)\neq0$决定一个 $T\in\mathrm{Aut}(\mathbb{R})$, 故定义

$$\phi:\mathrm{Aut}(\mathbb{R})\to\mathbb{R}\times\mathbb{Z}_2,\quad\phi(T)=(\log|T(1)|,\mathrm{sgn}(T(1)))$$

容易验证 $\phi^{-1}((0,1))=\mathrm{id}_{\mathbb{R}}$, 故 ϕ 为单射, 满射也显然, 因此 ϕ 为双射. 任取 $T_1,T_2\in\mathrm{Aut}(\mathbb{R})$, 则

$$\begin{aligned}\phi(T_1\circ T_2)&=(\log|T_1(1)T_2(1)|,\mathrm{sgn}(T_1(1)T_2(1)))\\&=(\log|T_1(1)|+\log|T_2(1)|,\mathrm{sgn}(T_1(1))\,\mathrm{sgn}(T_2(1)))=\phi(T_1)\cdot\phi(T_2)\end{aligned}$$

因此 ϕ 为同构, $\mathrm{Aut}(\mathbb{R})\cong\mathbb{R}\times\mathbb{Z}_2$.

Prob 4.25

Show that the automorphism group of the circle is isomorphic to $\mathbb{Z}_2$.

Proof: 设 $S^1=\left\{e^{i\theta}\mid\theta\in[0,2\pi)\right\}$, 类似 4.24, 可以证明若 $T:S^1\to S^1$ 为同态, 则存在 $n\in\mathbb{Z}$ 使得 $T(z)=z^n$. 当 $|n|\geq 2$ 时, T 并非单射, 当 $n=0$ 时, T 亦非满射. 容易验证 $n=\pm1$ 时 T 为同构, 因此 $\mathrm{Aut}(S^1)=\left\{z,z^{-1}\right\}$, 由此可见 $\mathrm{Aut}(S^1)\cong\mathbb{Z}_2$.

Prob 4.26

Give an action of $\mathbb{Z}$ on $\mathbb{R}\times[0,1]$ which has the Möbius strip as orbit space.

Solution: 定义 $\mathbb{Z}\curvearrowright\mathbb{R}\times[0,1]$ 为

$$n\cdot(x,y)=\begin{cases}(x+n,y), & n\text{ 为偶数}\\(x+n,1-y), & n\text{ 为奇数}\end{cases}$$

先验证这确实是群作用: 易见 $0\cdot(x,y)=(x,y)$, 任取 $m,n\in\mathbb{Z}$, 有

$$m\cdot(n\cdot(x,y))=\begin{cases}(x+m+n,y), & m,n\text{ 奇偶性相同}\\(x+m+n,1-y), & m,n\text{ 奇偶性不同}\end{cases}=(m+n)\cdot(x,y)$$

可见这确实是群作用. 由该作用的定义, 在 $(\mathbb{R}\times[0,1])/\mathbb{Z}$ 中可以选取 $[0,1)\times[0,1]$ 中的元素作为代表元, 并且 $(0,y)$ 与 $(1,1-y)$ 在同一个轨道中. 因此 $(\mathbb{R}\times[0,1])/\mathbb{Z}$ 同胚于将矩形的左右两边按相反方向识别得到的 Möbius 带.

Prob 4.27

Find an action of $\mathbb{Z}_2$ on the torus with orbit space the cylinder.

Solution: 设 $T=S^1\times S^1$ 为二维环面, 定义 $\mathbb{Z}_2\curvearrowright T$ 为

$$0\cdot(z_1,z_2)=(z_1,z_2),\quad 1\cdot(z_1,z_2)=(z_1,\overline{z_2})$$

容易验证这是群作用, 并且 $[(z_1,z_2)]=\{(z_1,z_2),(z_1,\overline{z_2})\}$. 于是商空间 $T/\mathbb{Z}_2\cong S^1\times S'^1$, 其中 S'^1 为上半圆周, 即 $S'^1\cong[0,\pi]$, 从而 $T/\mathbb{Z}_2\cong S^1\times[0,\pi]=\text{cylinder}$.

Prob 4.28

Describe the orbits of the natural action of $\mathrm{SO}(n)$ on $\mathbb{R}^n$ as a group of linear transformations, and identify the orbit space.

Solution: 任取 $x\in\mathbb{R}^n$, 则

$$[x]=\{Ax\mid A\in\mathrm{SO}(n)\}$$

对 $y=Ax, A\in\mathrm{SO}(n)$, 有 $\|y\|=\|Ax\|=\|x\|$, 于是 $[x]\subset\|x\|S^{n-1}$. 反之, 为证明 $\|x\|S^{n-1}\subset[x]$, 不妨设 $\|x\|=1$, 取 $y\in S^{n-1}$, 分别将 x,y 扩充为 $\mathbb{R}^n$ 的标准正交基 B_x,B_y 使得 $|B_x|=\left|B_y\right|=1$, x,y 分别为 B_x,B_y 的第一列. 定义线性映射 $A:\mathbb{R}^n\to\mathbb{R}^n$ 使得 A 将 B_x 映为 B_y, 则 $Ax=y$, 且

$$I_n=B_y^TB_y=(AB_x)^T(AB_x)=B_x^TA^TAB_x\Longrightarrow A^TA=I_n$$

$$1=\left|B_y\right|=|AB_x|=\texttt{abs}\ (A)\Longrightarrow A\in\mathrm{SO}(n)$$

综上, 我们证明了 $[x]=\|x\|S^{n-1}$.

因此当 $n\geq2$ 时, 轨道空间即为 $[0,\infty)$; 当 $n=1$ 时, $\mathrm{SO}(1)=\{1\}$, 轨道空间为 $\mathbb{R}$.

Prob 4.31

The *stabilizer* of a point $x \in X$ consists of those elements $g \in G$ for which $g \cdot x = x$. Show that the stabilizer of any point is a closed subgroup of G when X is Hausdorff, and that points in the same orbit have conjugate stabilizers for any X.

Proof: 将 $x \in X$ 的稳定化子记作 $\mathrm{Stab}(x)$. 记群作用映射为 $f(g,x) = g \cdot x$, 于是 $f^{-1}(\{x\})$ 为 $G \times X$ 中的闭集. 注意 $G \times \{x\}$ 为 $G \times X$ 中的闭集, 且

$$f^{-1}(\{x\}) \cap (G \times \{x\}) = \{(g,x) \mid g \cdot x = x\} = \mathrm{Stab}(x) \times \{x\}$$

因此 $\mathrm{Stab}(x) \times \{x\}$ 为 $G \times X$ 中的闭集, 进而为 $G \times \{x\}$ 中的闭集, 定义

$$p : G \to G \times \{x\}, \quad g \mapsto (g,x)$$

则 p 为连续映射, 且 $\mathrm{Stab}(x) = p^{-1}(\mathrm{Stab}(x) \times \{x\})$, 因此 $\mathrm{Stab}(x)$ 为 G 中的闭集. 容易验证 $\mathrm{Stab}(x)$ 为子群, 故 $\mathrm{Stab}(x)$ 为闭子群.

设 $y = g \cdot x$, 取 $h \in \mathrm{Stab}(y)$, 则

$$hy = y \Longrightarrow h(gx) = gx \Longrightarrow g^{-1}hg \cdot x = x \Longrightarrow g^{-1}hg \in \mathrm{Stab}(x) \Longrightarrow g^{-1}\,\mathrm{Stab}(y)g \subset \mathrm{Stab}(x)$$

任取 $k \in \mathrm{Stab}(x)$, 则

$$kx = x \Longrightarrow gkg^{-1}y = y \Longrightarrow gkg^{-1} \in \mathrm{Stab}(y) \Longrightarrow k \in g^{-1}\,\mathrm{Stab}(y)g \Longrightarrow \mathrm{Stab}(x) \subset g^{-1}\,\mathrm{Stab}(y)g$$

综上 $\mathrm{Stab}(x) = g^{-1}\,\mathrm{Stab}(y)g$, 故 x, y 在同一轨道上时其稳定化子共轭.

Prob 4.32

If G is compact, X Hausdorff, and if G acts transitively on X, show that X is homeomorphic to the orbit space $G/\,\mathrm{Stab}(x)$ for any $x \in X$.

Proof: 固定 $x \in X$, 记 $H = \mathrm{Stab}(x)$, 定义 x 的右移映射为

$$p : G \to X, \quad p(g) = g \cdot x, \forall g \in G$$

由 $G \curvearrowright X$ 为转移的, 可知 p 为满射. 商映射 $\pi : G \to G/H, g \mapsto gH$, 考虑将 p 作用在 G/H 上:

$$p : G/H \to X, \quad p(gH) = p(g) = g \cdot x$$

先验证 p 定义良好: 任取 $g' \in gH$, 则 $g' = gh, h \in H$, 因此

$$p(g'H) = g' \cdot x = (gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = p(gH)$$

再验证 p 为双射: 满射已证明, 下证单射. 设 $p(g_1H) = p(g_2H)$, 则

$$g_1 \cdot x = g_2 \cdot x \Longrightarrow g_2^{-1}g_1 \cdot x = x \Longrightarrow g_2^{-1}g_1 \in H \Longrightarrow g_1H = g_2H$$

最后验证 p 为同胚: 由群作用映射的连续性知 $p : G \to X$ 连续, 进而由商空间的泛性质知 $p : G/H \to X$ 连续. 由 G 紧可知 G/H 紧, 则 p 为紧空间到 Hausdorff 空间的连续双射, 故为同胚.

The Fundamental Group

Prob 5.1

Let C denote the unit circle in the plane. Suppose $f : C \to C$ is a map which is not homotopic to the identity. Prove that $f(x) = -x$ for some point x of C.

Proof: 反设 $\forall x \in C, f(x) \neq -x$, 定义映射

$$F(x,t) : C \times [0,1] \to C, \quad F(x,t) = \frac{(1-t)f(x) + tx}{\|(1-t)f(x) + tx\|}, \quad \forall x \in C, t \in [0,1]$$

其中 $(1-t)f(x) + tx, t \in [0,1]$ 表示连接 $f(x)$ 与 x 的线段, 其经过原点当且仅当 $f(x) = -x$, 故其对 $\forall x \in C$ 均不经过原点, 从而 F 良定义. 显然 $F(x,0) = f(x), F(x,1) = x$, 故 $f \underset{F}{\simeq} \mathrm{id}_C$, 矛盾! 因此存在 $x \in C$ 使得 $f(x) = -x$.

Prob 5.2

With C as above, show that the map which takes each point of C to the point diametrically opposite is homotopic to the identity.

Proof: 将 C 写作 $C = \left\{e^{2\pi i\theta} \middle| \ 0 \leq \theta < 1\right\}$, 设 $f : C \to C, f(x) = -x$, 定义

$$F(x,t) : C \times [0,1] \to C, \quad F(x,t) = (-x) \cdot e^{\pi i t}, \quad \forall x \in C, t \in [0,1]$$

于是 $F(x,0) = f(x), F(x,1) = x$, 故 $f \underset{F}{\simeq} \mathrm{id}_C$.

Prob 5.3

Let D be the disc bounded by C, parametrize D using polar coordinates, and let $h : D \to D$ be the homeomorphism defined by

$$h(0) = 0, h(r,\theta) = (r, \theta + 2\pi r)$$

Find a homotopy F from h to the identity map such that the functions

$$F|_{D\times\{t\}} : D \times \{t\} \to D, \forall 0 \leq t \leq 1$$

are all homeomorphisms.

Solution: 定义

$$F : D \times [0,1] \to D, \quad F(r,\theta,t) = (r, \theta + 2\pi r(1-t)), \quad \forall (r,\theta) \in D, t \in [0,1]$$

于是 $F(r,\theta,0) = h(r,\theta), F(r,\theta,1) = (r,\theta)$, 故 $h \underset{F}{\simeq} \mathrm{id}_D$. 对任意固定 $t \in [0,1]$, 映射

$$F|_{D\times\{t\}} : D \times \{t\} \to D, \quad F|_{D\times\{t\}}\ (r,\theta) = (r, \theta + 2\pi r(1-t))$$

表示将半径为 r 的圆绕中心旋转 $2\pi r(1-t)$ 角度, 显然为双射且连续, 其逆映射为

$$\left(F|_{D\times\{t\}}\right)^{-1} : D \to D \times \{t\}, \quad \left(F|_{D\times\{t\}}\right)^{-1}(r,\theta) = (r, \theta - 2\pi r(1-t))$$

亦连续, 因此 $F|_{D\times\{t\}}$ 为同胚.

Prob 5.4

With the terminology of Problem 3, show that h is homotopic to the identity map relative to C.

Proof: 将 5.3 中的 F 改为

$$F : D \times [0,1] \to D, \quad F(r,\theta,t) = (r, \theta + 2\pi r^{1-t}), \quad \forall (r,\theta) \in D, t \in [0,1]$$

于是 $F(r,\theta,0) = h(r,\theta), F(r,\theta,1) = (r,\theta)$, 故 $h \underset{F}{\simeq} \mathrm{id}_D$. 且 $\forall (1,\theta) \in C$ 有 $F(1,\theta,t) = (1,\theta)$. 即 $h \underset{F}{\simeq} \mathrm{id}_D \text{ rel } C$.

Prob 5.5

Let $f : X \to S^n$ be a map which is not onto. Prove that f is *null homotopic*, that is to say f is homotopic to a map which takes all of X to a single point of S^n.

Proof: 设 $y \in S^n \setminus f(X)$, 考虑

$$F(x,t) = \frac{(1-t)f(x) - ty}{\|(1-t)f(x) - ty\|}, \quad \forall x \in X, t \in [0,1]$$

注意 $(1-t)f(x) + t(-y)$ 表示连接 $f(x)$ 与 $-y$ 的线段, 其经过原点当且仅当 $f(x) = -(-y) = y$, 故其对任意 $x \in X$ 均不经过原点, 从而 F 良定义. 显然 $F(x,0) = f(x), F(x,1) = -\frac{y}{\|y\|} \in S^n$, 故 F 为 f 与常值映射 $g(x) = -\frac{y}{\|y\|}$ 之间的一个同伦, 因此 f 为零同伦映射.

Prob 5.6

As usual, CY denotes the cone on Y. Show that any two maps $f, g : X \to CY$ are homotopic.

Proof: 定义 $h : X \to CY$ 为 $h(x) = q, \forall x \in X$, 其中 q 为 $X \times \{1\}$ 被识别得到的锥点. 定义

$$F(x,t) = (1-t)f(x) + th(x), \quad \forall x \in X, t \in [0,1]$$

注意 CY 为凸集, 因此 F 确为 $X \times [0,1] \to CY$ 的连续映射, 且 $F(x,0) = f(x), F(x,1) = h(x)$, 因此 $f \underset{F}{\simeq} h$, 同理可知 $g \simeq h$, 故 $f \simeq g$.

Prob 5.7

Show that a map from X to Y is null homotopic iff it extends to a map from the cone on X to Y.

Proof: $\Longrightarrow$ 设 $g : X \to Y$ 为常值映射, $f \underset{F}{\simeq} g$, 于是 $F(x,0) = f(x), F(x,1) = g(x) = y \in Y$. 可见 F 在 $X \times \{1\}$ 上恒为 y. 因此 F 可自然地延拓为 $\overline{F} : CX \to Y$, 且 $\overline{F}|_{X \times \{0\}} = f$.

$\Longleftarrow$ 设 $f : X \to Y$ 可延拓为 $\overline{f} : CX \to Y$, 满足 $\overline{f}|_{X \times \{0\}} = f$. 设 $\overline{f}(X \times \{1\}) = y \in Y$, 定义

$$F(x,t) = \overline{f} \circ \pi(x,t), \quad \forall x \in X, t \in [0,1]$$

其中 $\pi : X \to CX$ 为识别映射. 进而 F 连续且 $F(x,0) = f(x), F(x,1) = y$, 故 $f \underset{F}{\simeq} g \equiv y$, 即 f 为零同伦映射.

Prob 5.10

Let γ, σ be two paths in the space X which begin at the point p and end at q. As in the proof of theorem (5.6) these paths induce isomorphisms γ_*, σ_* of $\pi_1(X,p)$ with $\pi_1(X,q)$. Show that σ_* is the composition of γ_* and the inner automorphism of $\pi_1(X,q)$ induced by the element $\langle \sigma^{-1} \cdot \gamma \rangle$.

Proof: 由定义, 我们有

$$\gamma_* : \pi_1(X,p) \to \pi_1(X,q), \quad \gamma_*(\langle \alpha \rangle) = \langle \gamma^{-1} \cdot \alpha \cdot \gamma \rangle,$$
$$\sigma_* : \pi_1(X,p) \to \pi_1(X,q), \quad \sigma_*(\langle \alpha \rangle) = \langle \sigma^{-1} \cdot \alpha \cdot \sigma \rangle$$

$\langle\sigma^{-1}\cdot\gamma\rangle\in\pi_1(X,q)$ 诱导的内自同构为

$$\begin{aligned}\varphi:\pi_1(X,q)\to\pi_1(X,q),\quad \varphi(\langle\beta\rangle)&=\langle\sigma^{-1}\cdot\gamma\rangle\cdot\beta\cdot\langle\sigma^{-1}\cdot\gamma\rangle^{-1}\\&=\langle\sigma^{-1}\cdot\gamma\cdot\beta\cdot\gamma^{-1}\cdot\sigma\rangle\end{aligned}$$

因而

$$\begin{aligned}\varphi\circ\gamma_*(\langle\alpha\rangle)=\varphi(\langle\gamma^{-1}\cdot\alpha\cdot\gamma\rangle)&=\langle\sigma^{-1}\cdot\gamma\cdot(\gamma^{-1}\cdot\alpha\cdot\gamma)\cdot\gamma^{-1}\cdot\sigma\rangle\\&=\langle\sigma^{-1}\cdot\alpha\cdot\sigma\rangle=\sigma_*(\langle\alpha\rangle)\end{aligned}$$

即 $\sigma_*=\varphi\circ\gamma_*$.

Prob 5.11

Let X be a path-connected space. When is it true that for any two points $p,q\in X$ and all paths from p to q induce the same isomorphism between $\pi_1(X,p)$ and $\pi_1(X,q)$?

Proof: 由 X 道路连通知此时 $\pi_1(X,p)\simeq\pi_1(X,q)$. 题中命题成立当且仅当 $\pi_1(X,p)$ 为 Abel 群:

$\Longrightarrow$: $\forall p,q\in X$ 以及任意两条 γ,σ 为从 p 到 q 的路径, 由 5.10 可知 $\sigma_*=\varphi\circ\gamma_*$, 其中 φ 为由 $\langle\sigma^{-1}\cdot\gamma\rangle$ 诱导的内自同构. 由 $\sigma_*=\gamma_*$ 知 $\varphi=\mathrm{id}_{\pi_1(X,q)}$, 即:

$$\forall\langle\alpha\rangle\in\pi_1(X,q).\quad\langle\sigma^{-1}\cdot\gamma\rangle\cdot\langle\alpha\rangle=\langle\alpha\rangle\cdot\langle\sigma^{-1}\cdot\gamma\rangle$$

下说明: $\langle\sigma^{-1}\cdot\gamma\rangle$ 可以取遍 $\pi_1(X,q)$ 中的任意元素. 设 $\langle\beta\rangle\in\pi_1(X,q)$, 取 γ 为从 p 到 q 的任意路径, 定义 $\sigma=\gamma\cdot\beta^{-1}$, 则

$$\langle\sigma^{-1}\cdot\gamma\rangle=\langle\beta\cdot\gamma^{-1}\cdot\gamma\rangle=\langle\beta\rangle$$

因此 $\forall\langle\alpha\rangle,\langle\beta\rangle\in\pi_1(X,q),\langle\beta\rangle\cdot\langle\alpha\rangle=\langle\alpha\rangle\cdot\langle\beta\rangle$, 故 $\pi_1(X,q)$ 为 Abel 群.

$\Longleftarrow$: 由上述推理知只需对任意从 p 到 q 的路径 γ,σ 说明 $\varphi=\mathrm{id}_{\pi_1(X,q)}$. 我们有

$$\varphi(\langle\alpha\rangle)=\langle\sigma^{-1}\cdot\gamma\cdot\alpha\cdot\gamma^{-1}\cdot\sigma\rangle=\langle\gamma^{-1}\cdot\sigma\cdot\sigma^{-1}\cdot\gamma\cdot\alpha\rangle=\langle\alpha\rangle$$

因此 $\gamma_*=\sigma_*$.

Prob 5.13

Let G be a path-connected topological group. Given two loops α,β based at e in G, define a map

$$F:[0,1]\times[0,1]\to G,\quad F(s,t)=\alpha(s)\cdot\beta(t)$$

where the dot denotes multiplication in G. Draw a diagram to show the effect of this map on the square, and prove that the fundamental group of G is abelian.

Proof: 由定义知

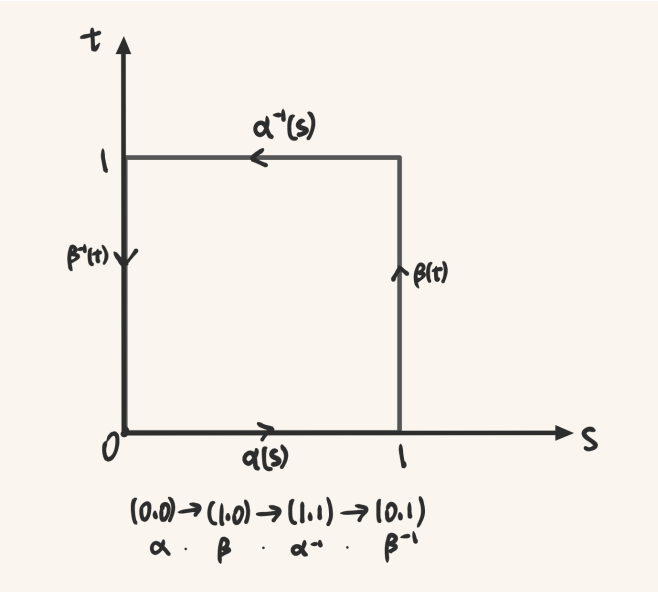
$$\begin{aligned}F(0,t)=\alpha(0)\cdot\beta(t)=e\cdot\beta(t)=\beta(t),&\quad F(1,t)=\alpha(1)\cdot\beta(t)=e\cdot\beta(t)=\beta(t),\\F(s,0)=\alpha(s)\cdot\beta(0)=\alpha(s)\cdot e=\alpha(s),&\quad F(s,1)=\alpha(s)\cdot\beta(1)=\alpha(s)\cdot e=\alpha(s)\end{aligned}$$

因此 F 将正方形的左右边分别映为 β, 上下边分别映为 α.

由图可知, 在正方形所示边界走一圈, 有 $F|_{\partial I^2}=\alpha\cdot\beta\cdot\alpha^{-1}\cdot\beta^{-1}$, 又由 $F:I^2\to G$ 连续, 故由下面的习题 Hatcher 1.1.5 的证明可知 $\alpha\cdot\beta\cdot\alpha^{-1}\cdot\beta^{-1}$ 零同伦, 进而由 G 道路连通知

$$\alpha\cdot\beta\cdot\alpha^{-1}\cdot\beta^{-1}\simeq e_{\pi_1(G)}\Longrightarrow\langle\alpha\rangle\cdot\langle\beta\rangle=\langle\beta\rangle\cdot\langle\alpha\rangle$$

因此 $\pi_1(G)$ 为 Abel 群.



H-Prob 1.1.3

For a path-connected space X, show that $\pi_1(X)$ is abelian iff all basepoint-change homomorphisms β_h depend only on the endpoints of the path h.

Proof: 设题中 h 为从 p 到 q 的路径, 则 $\beta_h = h^* : \pi_1(X,p) \to \pi_1(X,q)$. β_h 只与端点有关当且仅当对任意从 p 到 q 的路径 h_1, h_2 有 $h_1^* = h_2^*$. 可见这就是 Prob 5.11 的结论.

H-Prob 1.1.4

A subspace $X \subset \mathbb{R}^n$ is said to be *star-shaped* if there is a point $x_0 \in X$ such that, for each $x \in X$, the line segment from x_0 to x lies in X.

Show that if a subspace $X \subset \mathbb{R}^n$ is *locally star-shaped*, in the sense that every point of X has a star-shaped **neighborhood** in X, then every path in X is homotopic in X to a piecewise linear path, that is, a path consisting of a finite number of straight line segments traversed at constant speed.

Show this applies when X is open or when X is a union of finitely many closed conveax sets.

Proof: 设 $\gamma : [0,1] \to X$ 为 X 中的一条路径. 由 X 局部星形可知对任意 $t \in [0,1]$, 存在 $\gamma(t)$ 的一个星形邻域 $U_t \subset X$. 由 γ 的连续性和 $[0,1]$ 的紧性可知 $\gamma([0,1])$ 在 X 中紧, 进而 $\{U_t\}$ 存在有限子覆盖. 存在有限个点 $0 = t_0 < t_1 < .... < t_n = 1$ 使得 $\forall i = 0,1,...,n-1, \gamma([t_i, t_{i+1}]) \subset U_{t_i}$. 记 $x_i = \gamma(t_i), \forall i = 0,1,...,n$. 由于 U_{t_i} 为星形域, 因此存在 $y_i \in U_{t_i}$ 使得连接 y_i 与任意 $x \in U_{t_i}$ 的线段均在 U_{t_i} 内. 定义

$$F_i : [t_i, t_{i+1}] \times [0,1] \to U_{t_i}, \quad F_i(s,t) = (1-t)\gamma(s) + t y_i$$

于是 F_i 为 $\gamma|_{[t_i,t_{i+1}]}$ 与常值映射 $g_i(s) = y_i$ 之间的同伦. 考虑连接 x_i, y_i, x_{i+1} 得到的折线, 由星形性质可知该折线在 U_{t_i} 内, 进而 g_i 与该折线对应的线性路径 l_i 同伦, 故 $\gamma|_{[t_i,t_{i+1}]} \simeq l_i$. 上述定义的 $\gamma|_{[t_i,t_{i+1}]}$ 与 l_i 的首尾均为 $x_i, x_{i+1}, i = 0,1,\cdots,n-1$, 因此将各段同伦拼接可知 $\gamma \simeq l$, 其中 l 为连接 $x_0, x_1, \cdots, x_n$ 的折线, 故 l 为分段线性路径.

特别地, 当 $X \subset \mathbb{R}^n$ 为开集时, $\forall x \in X, \exists r > 0$ s.t. $B_r(x) \subset X$, 显然开球是星形集, 故 X 为局部星形集. 当 $X = \bigcup_{i=1}^n X_i$, 其中 X_i 均为闭凸集时, $\forall x \in X, \exists i$ s.t. $x \in X_i$, 由于闭凸集为星形集, 故 X 为局部星形集.

H-Prob 1.1.5

Show that for a space X, the following three conditions are equivalent:

(a) Every map $S^1 \to X$ is homotopic to a constant map, with image a point.

(b) Every map $S^1 \to X$ extends to a map $D^2 \to X$

(c) $\pi_1(X, x_0) = 0$ for $\forall x_0 \in X$.

Deduce that a space X is simply-connected iff all maps $S^1 \to X$ are homotopic. [In this problem, 'homotopic' means 'homotopic without regard to basepoints.']

Proof: (a) $\Longrightarrow$ (b): 设 $f : S^1 \to X$ 为任意映射, 由 (a) 可知 $f \simeq g$, 其中 $g : S^1 \to X$ 为常值映射. 设 $g(S^1) = x_0 \in X$, 对应的同伦映射为 $F : S^1 \times [0,1] \to X$. 将 D^2 中点用极坐标表示, 定义

$$\overline{F} : D^2 \to X, \quad \overline{F}(0,0) = x_0,\ \overline{F}(r,\theta) = F\big(e^{i\theta}, 1-r\big), \quad \forall (r,\theta) \in D^2 \setminus \{(0,0)\}$$

于是 $\overline{F}(1,\theta) = F\big(e^{i\theta}, 0\big) = f\big(e^{i\theta}\big)$, $\overline{F}$ 为 f 在 D^2 上的延拓.

(b) $\Longrightarrow$ (c): 设 $x_0 \in X$, 取任意 $\langle \alpha \rangle \in \pi_1(X, x_0)$, 由 (b) 可知存在 $\overline{\alpha} : D^2 \to X$ 使得 $\overline{\alpha}|_{S^1} = \alpha$. 记 $y = \alpha(0,0) \in X$, 定义

$$F : S^1 \times [0,1] \to X, \quad F\big(e^{i\theta}, t\big) = \overline{\alpha}(1-t, \theta), \quad \forall e^{i\theta} \in S^1, t \in [0,1]$$

于是 $F\big(e^{i\theta}, 0\big) = \overline{\alpha}(1,\theta) = \alpha\big(e^{i\theta}\big)$, $F\big(e^{i\theta}, 1\big) = \overline{\alpha}(0,0) = y$, 故 $\alpha \underset{F}{\simeq} g \equiv y$. 注意本题中 $\simeq$ 不要求保持基点, 因此 $g \simeq e \equiv x_0$, 进而 $\alpha \simeq e$, 因此 $\langle \alpha \rangle = e_{\pi_1(X, x_0)}$, 故 $\pi_1(X, x_0) = 0$.

(c) $\Longrightarrow$ (a): 设 $f : S^1 \to X$, 由 $\pi_1(X, x_0) = 0$ 即得 $f \simeq e_{\pi_1(X, x_0)}$, 即 f 同伦于常值映射.

simply-connected 的定义是: X 道路连通且 $\pi_1(X, x_0) = 0$ 对任意 $x_0 \in X$ 成立.

当 X 是 simply-connected 时, 由上述等价关系可知任意映射 $S^1 \to X$ 均同伦于常值映射, 再由道路连通性即知任意常值映射同伦, 因此任意两个映射 $S^1 \to X$ 均同伦.

若任意映射 $S^1 \to X$ 均同伦, 则任意常值映射均同伦, 可知 X 道路连通, 且任意映射均同伦于常值映射, 由上述等价关系可知 $\pi_1(X, x_0) = 0$ 对任意 $x_0 \in X$ 成立, 因此 X simply-connected.

H-Prob 1.1.6

We can regard $\pi_1(X, x_0)$ as the set of basepoint-preserving homotopy classes of maps $(S^1, s_0) \to (X, x_0)$. Let $[S^1, X]$ be the set of homotopy classes of maps $S^1 \to X$, with no conditions on basepoints. Thus there is a natural map

$$\phi : \pi_1(X, x_0) \to [S^1, X]$$

obtained by ignoring basepoints.

Show that ϕ is onto if X is path-connected, and that $\phi(\langle f \rangle) = \phi(\langle g \rangle)$ iff $\langle f \rangle$ and $\langle g \rangle$ are conjugate in $\pi_1(X, x_0)$. Hence induces a one-to-one correspondence between $[S^1, X]$ and the set of conjugacy classes in $\pi_1(X)$, when X is path-connected.

Proof:

(a) 若 X 道路连通, 则 ϕ 满射: 取任意 $\langle h \rangle \in [S^1, X]$, 选代表 $h : S^1 \to X$. 由于 X 道路连通, 存在路径 $a : [0,1] \to X$ 使 $a(0) = h(s_0), a(1) = x_0$. 定义以 x_0 为基点的环路

$$l = a^{-1} \cdot h \cdot a : (S^1, s_0) \to (X, x_0).$$

于是 h 与 l 自由同伦 (即不要求保持基点), 于是 $\phi(\langle l\rangle)=[h]$, 故 ϕ 满射.

(b) $\phi(\langle f\rangle)=\phi(\langle g\rangle)$ 当且仅当 $\langle f\rangle$ 与 $\langle g\rangle$ 在 $\pi_1(X,x_0)$ 中共轭:

"$\Longrightarrow$": 若 $\phi(\langle f\rangle)=\phi(\langle g\rangle)$, 则存在自由同伦

$$H:S^1\times[0,1]\to X\ \text{ s.t. }\ H(x,0)=f(x),H(x,1)=g(x).$$

注意 $f,g\in\pi_1(X,x_0)$, 于是 $x_0=f(s_0),g(s_0)$. 设 $a(t)=H(s_0,t)$, 则 $a(0)=a(1)=x_0$, 故 $a\in\pi_1(X,x_0)$, 且 $a(t)$ 表示基点 x_0 的移动轨迹.

考虑圆柱 $S^1\times[0,1]$ 的"边界": 从 $(s_0,0)$ 出发, 沿 $S^1\times\{0\}$ 逆时针走一圈到达 s_0, 再沿 $\{s_0\}\times[0,1]$ 从底部走到顶部, 然后沿 $S^1\times\{1\}$ 顺时针走一圈回到 s_0, 最后沿 $\{s_0\}\times[0,1]$ 从顶部走到底部回到起点. 由 H 可知, 在该边界上走一圈的映射为

$$H|_{\partial(S^1\times[0,1])\simeq([0,1]\times[0,1])}=f\cdot a\cdot g^{-1}\cdot a^{-1}.$$

从而由 1.1.5 的证明可知 $f\cdot a\cdot g^{-1}\cdot a^{-1}$ 零同伦, 故

$$\langle f\rangle=\langle a^{-1}\cdot g\cdot a\rangle$$

即 $\langle f\rangle$ 与 $\langle g\rangle$ 共轭。

"$\Longleftarrow$": 若存在 a 使 $\langle g\rangle=\langle a^{-1}\cdot f\cdot a\rangle$, 则把基点沿 a 推移的过程给出 f 与 g 的自由同伦, 故 $\phi(\langle f\rangle)=\phi(\langle g\rangle)$.

Prob 5.21

Describe the homomorphism $f_*:\pi_1(S,1)\to\pi_1(S,f(1))$ induced by each of the following maps:

(a) The antipodal map $f\left(e^{i\theta}\right)=e^{i(\theta+\pi)},0\le\theta\le 2\pi$;

(b) The map $f\left(e^{i\theta}\right)=e^{in\theta},0\le\theta\le 2\pi$, where $n\in\mathbb{Z}$;

(c) The map

$$f\left(e^{i\theta}\right)=\begin{cases}e^{i\theta}, & 0\le\theta\le\pi\\ e^{i(2\pi-\theta)}, & \pi\le\theta\le 2\pi\end{cases}.$$

Solution: 我们有 $\pi_1(S^1,1)\simeq\mathbb{Z}$, 取生成元为 $\langle\alpha\rangle$, 其中 $\alpha:[0,1]\to S^1,\alpha(t)=e^{2\pi it}$, 于是

$$\pi_1(S^1,1)=\left\{\langle\alpha\rangle^k\mid k\in\mathbb{Z}\right\}$$

(a) $\pi_1(S^1,f(1))=\pi_1(S^1,-1)$, 取生成元为 $\langle\beta\rangle$, 其中 $\beta:[0,1]\to S^1,\beta(t)=e^{\pi i+2\pi it}$, 则

$$\pi_1(S^1,-1)=\left\{\langle\beta\rangle^k\mid k\in\mathbb{Z}\right\}$$

易见

$$f_*(\langle\alpha\rangle)=\langle f\circ\alpha\rangle=\langle\beta\rangle$$

因此 $f_*:\pi_1(S^1,1)\to\pi_1(S^1,-1)$ 为同构, 且 $f_*\left(\langle\alpha\rangle^k\right)=\langle\beta\rangle^k$.

(b) $f(1)=1$, 因此 $\pi_1(S^1,f(1))=\pi_1(S^1,1)$. 我们有

$$f_*(\langle\alpha\rangle)=\langle f\circ\alpha\rangle=\langle\gamma\rangle$$

其中 $\gamma:[0,1]\to S^1,\gamma(t)=e^{2\pi int}$, 于是 $\gamma=\alpha^n$, 故

$$f_*(\langle\alpha\rangle)=\langle\alpha\rangle^n$$

因此 $f_*\left(\langle\alpha\rangle^k\right)=\langle\alpha\rangle^{nk}$.

(c) $f(1)=1$, 因此 $\pi_1(S^1,f(1))=\pi_1(S^1,1)$. 我们有

$$f_*(\langle\alpha\rangle)=\langle f\circ\alpha\rangle=\langle\delta\rangle$$

其中 $\delta:[0,1]\to S^1$ 为

$$\delta(t)=\begin{cases}e^{2\pi it}, & 0\le t\le\frac{1}{2}\\ e^{2\pi i(1-t)}, & \frac{1}{2}\le t\le 1\end{cases}.$$

易见 $\delta\simeq e_{\pi_1(S^1,1)}$, 故 $f_*(\langle\alpha\rangle)=e_{\pi_1(S^1,1)}$. 因此 $f_*\left(\langle\alpha\rangle^k\right)=e_{\pi_1(S^1,1)}$.

Prob 5.24

If $X\simeq Y$ and $X'\simeq Y'$, show that $X\times X'\simeq Y\times Y'$. Show also that CX is contractible for any space X.

Proof: 设 $f:X\to Y,g:Y\to X$ s.t.

$$f\circ g\underset{F}{\simeq}\mathrm{id}_Y,g\circ f\underset{G}{\simeq}\mathrm{id}_X$$

$f':X'\to Y',g':Y'\to X'$ s.t.

$$f' \circ g' \underset{F'}{\simeq} \mathrm{id}_{Y'}, g' \circ f' \underset{G'}{\simeq} \mathrm{id}(X')$$

定义

$$\begin{aligned} \varphi : X \times X' \to Y \times Y', \qquad & \psi : Y \times Y' \to X \times X' \\ (x, x') \mapsto (f(x), f(x')) \qquad & (y, y') \mapsto (g(y), g(y')) \end{aligned}$$

于是 $\varphi \circ \psi = (f \circ g, f' \circ g')$, 定义

$$\begin{aligned} \mathcal{F}((y, y'), t) = (F(y, t), F'(y', t)) \Longrightarrow \mathcal{F}((y, y'), 0) &= (f \circ g(y), f' \circ g'(y')) = \varphi \circ \psi(y, y') \\ \mathcal{F}((y, y'), 1) &= (y, y') = \mathrm{id}_{Y \times Y'}(y, y') \end{aligned}$$

因此 $\varphi \circ \psi \underset{\mathcal{F}}{\simeq} \mathrm{id}_{Y \times Y'}$, 同理知 $\psi \circ \varphi \simeq \mathrm{id}_{X \times X'}$, 因此 $X \times X' \simeq Y \times Y'$.

对于 CX, 记 $X \times \{1\}$ 被粘合为点 p, 定义

$$F : CX \times I \to CX, \quad ((x, s), t) \mapsto (x, (1 - t)s + t)$$

于是 $F((x, s), 0) = (x, s), F((x, s), 1) = (x, 1) = p$, 因此 $\mathrm{id}_{CX} \underset{F}{\simeq} c_p$, 即 CX 可缩.

Prob 5.26

Consider the following examples of a circle C embedded in a surface S:

(a) $S =$ Möbius strip, $C =$ boundary circle;

(b) $S =$ torus, $C =$ diagonal circle $= \{(x, y) \in S^1 \times S^1 \mid x = y\}$;

(c) $S =$ cylinder, $C =$ one of boundary circles.

In each case, choose a base point in C, describe generators for the fundamental groups of C and S, and write down in terms of these generators the homomorphism of fundamental groups induced by the inclusion of C in S.

Solution: (a) 设 S 为粘合正方形 $[0, 1] \times [0, 1]$ 左右两边后得到的 Möbius 带, 则 C 为拼接正方形上下两边得到的边界圆. 取基点 x_0 为 $(0, 0)$, 则 $\pi_1(C, x_0)$ 的生成元可取为 $\langle \alpha \rangle$, 其中

$$\alpha : [0, 1] \to C, \quad \alpha(t) = \begin{cases} (2t, 0), & 0 \le t \le \frac{1}{2} \\ (2t - 1, 1), & \frac{1}{2} \le t \le 1 \end{cases}.$$

对于 S, 我们有 $\pi_1(S, x_0) \simeq \mathbb{Z}$, 其生成元可取为 $\langle \beta \rangle$, 其中

$$\beta : [0, 1] \to S, \quad \beta(t) = \begin{cases} (2t, 0), & 0 \le t \le \frac{1}{2} \\ (1, 2t - 1), & \frac{1}{2} \le t \le 1 \end{cases}.$$

注意上述 β 相当于绕边界圆 C 的一半 (底边) 后沿粘合边走回基点, 等价于沿 $\left(0, \frac{1}{2}\right)$ 从左走到右的环路 γ (β 和 γ 只是基点不同). 而 α 则为在 C 中先走底边, 再走上边回到基点, 相当于走了两圈 γ, 于是嵌入映射 $i : C \to S$ 诱导的同态为

$$i_*(\langle \alpha \rangle) = \langle \beta \rangle^2.$$

(b) 取 $x_0 = (0, 0) \in C$ 为基点, 则 $\pi_1(C, x_0)$ 的生成元可取为 $\langle \alpha \rangle$, 其中

$$\alpha : [0, 1] \to C, \quad \alpha(t) = (e^{2\pi i t}, e^{2\pi i t})$$

对于环面 $S = S^1 \times S^1$, 我们有 $\pi^1(S) \simeq \pi_1(S^1) \times \pi_1(S^1) \simeq \mathbb{Z}^2$, 其生成元可取为 $\langle \beta \rangle, \langle \gamma \rangle$, 其中

$$\beta : [0,1] \to S, \quad \beta(t) = (e^{2\pi i t}, 1),$$
$$\gamma : [0,1] \to S, \quad \gamma(t) = (1, e^{2\pi i t})$$

我们知道 $\pi_1(S) \simeq \pi_1(S^1) \times \pi_1(S^1)$ 是通过投影 $\phi(\langle\delta\rangle) = (\langle p_1 \circ \delta\rangle, \langle p_2 \circ \delta\rangle)$ 实现的, 注意

$$\phi(\langle\beta\cdot\gamma\rangle) = (\langle p_1 \circ (\beta\cdot\gamma)\rangle, \langle p_2 \circ (\beta\cdot\gamma)\rangle) = (\langle p_1 \circ \beta\rangle, \langle p_2 \cdot \gamma\rangle) = (\langle p_1 \circ \alpha\rangle, \langle p_2 \circ \alpha\rangle) = \phi(\langle\alpha\rangle)$$

因此 $\langle\alpha\rangle = \langle\beta\cdot\gamma\rangle$, 故嵌入映射 $i : C \to S$ 诱导的同态为

$$i_*(\langle\alpha\rangle) = \langle\beta\cdot\gamma\rangle = \langle\beta\rangle \cdot \langle\gamma\rangle.$$

(c) 记 $S = S^1 \times I$, 取 $x_0 = (1,0) \in C$ 为基点, 则 $\pi_1(C, x_0)$ 的生成元可取为 $\langle\alpha\rangle$, 其中

$$\alpha : [0,1] \to C, \quad \alpha(t) = (e^{2\pi i t}, 0)$$

对于圆柱 S, 我们有 $\pi_1(S, x_0) \simeq \mathbb{Z}$, $\langle\alpha\rangle$ 也是其生成元. 因此嵌入映射 $i : C \to S$ 诱导的同态为

$$i_*(\langle\alpha\rangle) = \langle\alpha\rangle.$$

Prob 5.27

Prove that if $f, g : S^1 \to X$ are homotopic maps, then the spaces formed from X by attaching a disc using f and using g are homotopy equivalent; in other words, $X \cup_f D \simeq X \cup_g D$.

Proof: 设 $H : S^1 \times I \to X$ 是对应的同伦映射, 满足 $H(x,0) = f(x), H(x,1) = g(x)$, 考虑

$$Y = (X \times I) \cup_H (D^2 \times I)$$

这里将 $H : S^1 \times I \to X$ 视为 $H : S^1 \times I \to X \times I$, 即 $(x,t) \sim (H(x,t), t)$. 定义映射

$$R : Y \times I \to Y$$

在 $X \times I$ 上, 定义 $R((x,t),s) = (x, (1-s)t)$;

在 $D^2 \times I$ 上, 定义

$$R((z,t),s) = \begin{cases} (z, (1-s)t), & |z| < 1 \\ (H(z,(1-s)t), (1-s)t), & |z| = 1 \end{cases}$$

可以验证, R 是良定义的连续映射, 且

$$R(y,0) = y, \quad \forall y \in Y,$$
$$R(y,1) \in X \times \{0\}, \quad \forall y \in Y$$

将 $X \cup_f D^2$ 视为 $X \times \{0\} \cup_f D^2 \times \{0\}$ 为 Y 的子集, 可以验证

$$\forall (x,0) \in X \times \{0\}, R((x,0),s) = (x,0),$$
$$\forall (z,0) \in D^2 \times \{0\}, R((z,0),s) = (z,0)$$

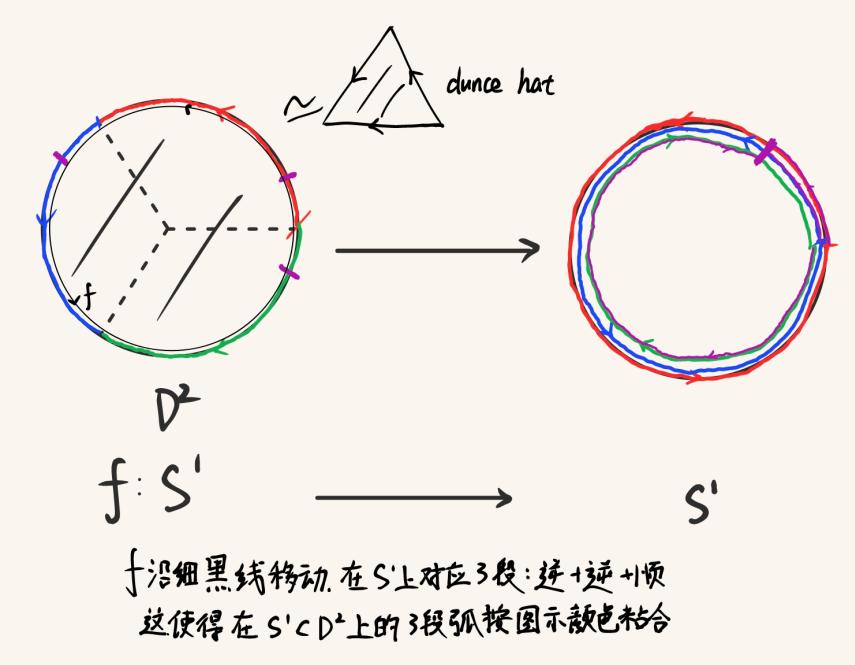
因此 R 是 Y 到 $X \cup_f D^2$ 的形变收缩, 故 $Y \simeq X \cup_f D^2$. 同理可证 $Y \simeq X \cup_g D^2$, 综上得到

$$X \cup_f D^2 \simeq Y \simeq X \cup_g D^2.$$

Prob 5.28

Use Problem 27, and the third example of a homotopy given in Section 5.1, to show that the ‘dunce hat’ has the homotopy type of a disc, and is therefore contractible.

Proof: 如图所示, 将等边三角形的粘合视作圆 $S^1 \subset D^2$ 上三段圆弧的粘合



考虑 $f : S^1 \to S^1$ 为 5.1 中第三个例子定义的环路 α 的类似物, 即

$$f\left(e^{i\theta}\right) = \begin{cases} e^{3i\theta}, & 0 \le \theta \le \frac{2\pi}{3} \\ e^{i(3\theta - 2\pi)}, & \frac{2\pi}{3} \le \theta \le \frac{4\pi}{3} \\ e^{i(6\pi - 3\theta)}, & \frac{4\pi}{3} \le \theta \le 2\pi \end{cases}.$$

由 5.1 的例子知 $f \simeq \mathrm{id}_{S^1}$, 于是根据图示以及 Prob 5.27, 可知

$$\text{dunce hat} \cong S^1 \cup_f D^2 \simeq S^1 \cup_{\mathrm{id}_{S^1}} D^2 \cong D^2$$

进而 dunce hat $\simeq D^2 \simeq \{0\}$, 故 dunce hat 可缩.

Prob 5.33

Which of the following spaces have the fixed-point property?

(a) The 2-sphere;

(b) the torus;

(c) the interior of the unit disc;

(d) the one-point union of two circles.

Solution:

(a) 取 $f : S^2 \to S^2$ 为 $f(x, y, z) = (-x, -y, -z)$, 则 f 无不动点, 故 S^2 无不动点性质.

(b) 取 $f : S^1 \times S^1 \to S^1 \times S^1$ 为 $f(x, y) = (-x, -y)$, 则 f 无不动点, 故环面无不动点性质.

(c) 定义 $f : D^\circ \to D^\circ$ 为

$$f(r\cos\theta, r\sin\theta) = \left(\frac{1}{2} + \frac{1}{2}r\cos\theta, \frac{1}{2}r\sin\theta\right), \quad 0 \le r < 1, 0 \le \theta < 2\pi$$

解不动点方程可得 $r\cos\theta = 1, r\sin\theta = 0 \Longrightarrow (r, \theta) = (1, 0) \notin D^\circ$, 故 f 无不动点, 因此 D° 无不动点性质.

(d) 设 $S = S^1 \bigvee S^1$, 将 S 视作两个单位圆周 A, B 粘合得到的空间, 其中 p 表示两个圆周的粘合点. 定义 $f : S \to S$ 为: 若 $x \in A$, 则 $f(x)$ 为 x 在 A 上逆时针旋转固定角度后得到的点; 若 $x \in B$, 则 $f(x)$ 为常值映射 $f(x) = f(p)$, 于是 f 为连续映射且无不动点, 故 S 无不动点性质.

Prob 5.34

Suppose X and Y are of the same homotopy type and X has the fixed-point property. Does Y also have it? If X retracts onto the subspace A, and A has the fixed-point property, need X also have it?

Solution: 两个都是不一定.

考虑 $Y = D^\circ$, 由于 Y 为凸集, 因此是 contractible 的, 故 $Y \simeq X = \{0\}$, 后者是单点集, 显然有不动点性质. 但由 5.33(c) 可知 Y 无不动点性质.

显然 $Y \to \{0\}$ 的常值映射使得 Y retracts onto $\{0\}$, $\{0\}$ 有不动点性质, 但 Y 无不动点性质.

Prob 5.35

Show that if X has the fixed-point property, and if X retracts onto the subspace A, then A also has the fixed-point property.

Proof: 设 $\tau : X \to A$ 为对应的收缩映射, $i : A \hookrightarrow X$ 为包含映射. 任取 $f : A \to A$ 为连续映射, 则

$$g = i \circ f \circ \tau : X \to X$$

为连续映射, 从而存在 $x \in X$ s.t. $g(x) = x$, 注意 $i(A) = A$, 因此 $x \in A$, 再由 $\tau|_A = \mathrm{id}_A$ 即知

$$x = g(x) = i \circ f \circ \tau(x) = i \circ f(x) = f(x)$$

从而 A 有不动点性质.

H-Prob 1.1.12

Show that every homomorphism $\pi_1(S^1) \to \pi_1(S^1)$ can be realized as the induced homomorphism φ_* of a map $\varphi : S^1 \to S^1$.

Proof: 设 $f : \pi_1(S^1, 1) \to \pi_1(S^1, 1)$ 为任意同态. 由于 $\pi_1(S^1, 1) \simeq \mathbb{Z}$, 取生成元为 $\langle \alpha \rangle$, 其中 $\alpha : [0, 1] \to S^1, \alpha(t) = e^{2\pi i t}$, 则

$$f\big(\langle \alpha \rangle^k\big) = f(\langle \alpha \rangle)^k$$

设 $f(\langle \alpha \rangle) = \langle \alpha \rangle^m = \langle \alpha^m \rangle$, 构造映射 $\varphi : S^1 \to S^1$ 为

$$\varphi\big(e^{i\theta}\big) = e^{im\theta}, \quad \theta \in [0, 2\pi)$$

于是

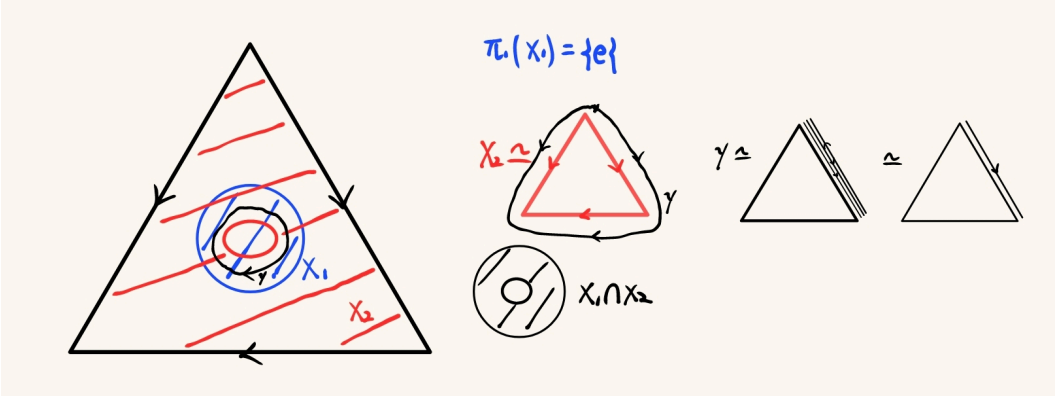
$$\varphi_*(\langle \alpha \rangle) = \langle \varphi \circ \alpha \rangle = \langle \alpha^m \rangle = f(\langle \alpha \rangle)$$

即 $\varphi_* = f$.

Prob 6.22

Prove that the 'dunce hat' is simply connected using Van Kampen's theorem.

Proof: 设 X 为 dunce hat. 考虑如下两个子集 X_1, X_2:



可见 $\pi_1(X_1)=\{e\}, \pi_1(X_2)=\mathbb{Z}, \pi_1(X_1\cap X_2)=\mathbb{Z}$, 且对于 $X_1\cap X_2$ 的生成元 γ, 对应的嵌入映射 $(i_{21})_*:\pi_1(X_1\cap X_2)\to\pi_1(X_2)$ 将 γ 也映到 $\pi_1(X_2)$ 的生成元, 故由 Van Kampen 知

$$\pi_1(X)=\langle a\mid a\rangle=\{e\}$$

即 X 单连通.

H-Prob 1.2.2

Let $X\subset\mathbb{R}^m$ be the union of convex open sets $X_1,\cdots,X_n$ such that $X_i\cap X_j\cap X_m\neq\emptyset$ for all i,j,k. Show that X is simply-connected.

Proof: 对 n 进行归纳: $n=1,2$ 时显然, $n=3$ 时, 由 $\pi_1(X_1\cup X_2)=\{e\}=\pi_1(X_3)$ 且

$$(X_1\cup X_2)\cap X_3=(X_1\cap X_3)\cup(X_2\cap X_3)$$

为道路连通的非空集可知 $\pi_1(X)=\{e\}$. 下设 $n-1$ 处成立, 则 n 时有

$$X=A\cup B,\quad A=\bigcup_{i=1}^{n-1}X_i, B=X_n$$

有归纳假设可知 $\pi_1(A)=\{e\}=\pi_1(B)$, 下考察 $A\cap B$ 的连通性: 我们有

$$A\cap B=\bigcup_{i=1}^{n-1}(X_i\cap X_n)$$

任取 $x,y\in A\cap B$, 则存在 i,j 使得

$$x\in X_i\cap X_n,\ y\in X_j\cap X_n$$

由 $X_i\cap X_j\cap X_n\neq\emptyset$, 可取 $z\in X_i\cap X_j\cap X_n$, 则存在 $X_i\cap X_n$ 中的道路 α 连接 x 与 z, 存在 $X_j\cap X_n$ 中的道路 β 连接 z 与 y, 故 $\alpha\cdot\beta$ 为 $A\cap B$ 中连接 x,y 的道路, 故 $A\cap B$ 道路连通. 因此由 Van Kampen 定理可知 $\pi_1(X)=\{e\}$. 归纳即知命题成立.

H-Prob 1.2.3

Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply-connected if $n\geq 3$.

Solution: 设 $\{x_1,\cdots,x_m\}\subset\mathbb{R}^n$, 我们要求 $X=\mathbb{R}^n\setminus\{x_1,\cdots,x_m\}$ 的基本群. 取充分小的 $\varepsilon>0$, 则易见 X 能够形变收缩到

$$Y = \mathbb{R}^n \setminus \bigcup_{i=1}^{m} B(x_i, \varepsilon)$$

于是只需求 $\pi_1(Y)$. 注意到可以在 Y 中做出一颗 "树", 即连通无环图 T, 使得每个球面上有且仅有一个点与 T 相交, 从而 Y 可以形变收缩到

$$W = T \cup \bigcup_{i=1}^{m} S_i^{n-1}$$

其中 $S_i^{n-1} = \partial B(x_i, \varepsilon)$. 由于 T 是连通无环图, 故 T 同胚于一个点, 因此 W 同胚于

$$Z = \bigvee_{i=1}^{m} S_i^{n-1}$$

使用归纳法来证明 $\pi_1(Z) = \{e\}$:

当 $m = 1$ 时, 由 $n - 1 \geq 2$ 即知 $\pi_1(Z) = \{e\}$; 设 $m - 1$ 时成立, 则 m 时有

$$Z = A \cup B, \quad A = \bigvee_{i=1}^{m-1} S_i^{n-1}, \ B = S_m^{n-1}$$

由归纳假设知 $\pi_1(A) = \{e\} = \pi_1(B)$, 且 $A \cap B = \{e\}$, 故由 Van Kampen 定理知 $\pi_1(Z) = \{e\}$. 归纳即知命题成立, 即 $\pi_1(\mathbb{R}^n \setminus \{x_1, \cdots, x_m\}) = \{e\}$.

H-Prob 1.2.4

Let $X \subset \mathbb{R}^3$ be the union of n lines through the origin. Compute $\pi_1(\mathbb{R}^3 \setminus X)$.

Solution: 记 $Y = \mathbb{R}^3 \setminus X$, 考虑映射

$$\begin{aligned} F : Y \times I &\to Y \\ (x, t) &\mapsto (1 - t)x + t\frac{x}{|x|} \end{aligned}$$

则 F 是 Y 到 $Y \cap S^2$ 的一个形变收缩, 其中 $Y \cap S^2$ 是 S^2 上去掉 $2n$ 个点的空间. 由球极投影, 可知 S^2 去掉 $2n$ 个点同胚于 $\mathbb{R}^2$ 去掉 $2n - 1$ 个点的空间, 记为 Z, 于是由上一题的方法可知

$$Z \simeq \mathbb{R}^2 \setminus \bigcup_{i=1}^{2n-1} B(y_i, \varepsilon) \simeq \bigvee_{i=1}^{2n-1} S^1$$

同理, 归纳即知 $\pi_1(Z) = *_{i=1}^{2n-1} \mathbb{Z}$, 因此

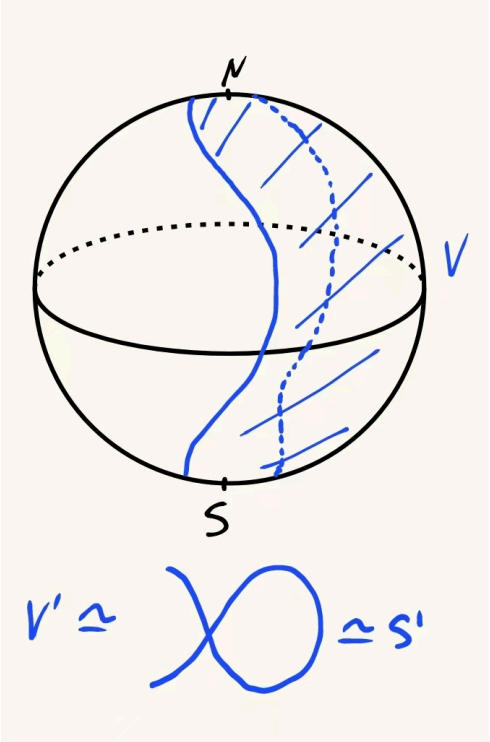
$$\pi_1(\mathbb{R}^3 \setminus X) = \mathop{*}_{i=1}^{2n-1} \mathbb{Z}.$$

H-Prob 1.2.7

Let X be the quotient space of S^2 obtained by identifying the north and south poles to a single point. Compute $\pi_1(X)$.

Proof: 记将 S^2 中的 N, S 识别为一点得到的空间为 X, 对应的粘合映射为 $q : S^2 \to X$.

取 $U = S^2 \setminus \{N, S\}$, V 为 S^2 中包含 N, S 的单连通开集, 记 $U' = q(U), V' = q(V)$, 容易验证 U', V' 都是 X 中道路连通的开集, 且 $X = U' \cup V'$ 为开覆盖, $U' \cap V' = q(U \cap V)$ 也为道路连通的开集. 故由 Van Kampen 定理可知



$$\pi_1(X) = \pi_1(U') * \pi_1(V')/N$$

其中, $\pi_1(U') = \pi_1(U) = \pi_1(S^1) = \mathbb{Z}$, V' 为将一个单连通集的两点识别为一点得到的空间, 可以想象其可以形变收缩为图中的形状, 因此有 $\pi_1(V') = \pi_1(S^1) = \mathbb{Z}$. 而 $\pi_1(U' \cap V') = \pi_1(U \cap V)$, 其中 $U \cap V \simeq S^1 \bigvee S^1$, 于是 $\pi_1(U' \cap V') = \mathbb{Z} * \mathbb{Z}$. 考察嵌入映射:

$$\begin{aligned} (i_{U'V'})_* : \pi_1(U' \cap V') &\to \pi_1(U') \\ a &\mapsto a \\ (i_{V'U'})_* : \pi_1(U' \cap V') &\to \pi_1(V') \\ a &\mapsto e \end{aligned}$$

其中第二个映射将 $\pi_1(U' \cap V')$ 中的所有元素均映成 e, 这是因为 $U' \cap V'$ 中的环路均不通过 N, S 两点, 因此在 V' 中可以收缩为一点. 而 $(i_{U'V'})_*$ 是满射, 故有 $N = \pi_1(U')$, 于是

$$\pi_1(X) = \langle a, b \mid a \rangle = \langle b \rangle = \mathbb{Z}.$$

H-Prob 1.2.8

Compute the fundamental group of the space obtained from two tori $S^1 \times S^1$ by identifying a circle $S^1 \times \{x_0\}$ in one torus with the corresponding circle $S^1 \times \{x_0\}$ in the other torus.

Solution: 不妨设 $x_0 = e^{\frac{\pi}{2}i}$, 可以想见粘合后的空间 X 为将两个环面 T^2 与 $T^{2'}$ 上下"贴起来", 相交面为一个圆. 与 $S^1 \bigvee S^1$ 的处理手法类似, 我们取如图所示的 $N, M \subset S^1 \bigvee S^1$, 并记

$$A_1 = S^1 \times M, \quad A_2 = S^1 \times N$$

则 A_1, A_2 均可形变收缩为 T^2, 从而 $\pi_1(A_1) = \pi_1(A_2) = \mathbb{Z}^2$. 并且 $A_1 \cap A_2$ 可以形变收缩为 S_1. 记 $\pi_1(A_1 \cap A_2)$ 的生成元为 w. 并分别记 $\pi_1(A_1), \pi_1(A_2)$ 的生成元为

$$\begin{aligned} \pi_1(A_1) &= \langle \{a, b\} \rangle, \quad a = S^1 \times \{x_0\}, b = \{x_0\} \times S^1 \\ \pi_1(A_2) &= \langle \{c, d\} \rangle, \quad c = S^1 \times \{x_0\}, d = \{x_0\} \times S^1 \end{aligned}$$

则 i_{12} 与 i_{21} 诱导的同态为

$$(i_{12})_*(w)=a,\quad (i_{21})_*(w)=c$$

因此 Van Kampen 定理中的正规子群 N 为

$$N=\left\langle\left\{(i_{12})_*(w)\cdot(i_{21})_*(w)^{-1}\mid w\in\pi_1(A_1\cap A_2)\right\}\right\rangle=\langle ac^{-1}\rangle$$

可知在 $\pi_1(A_1)*\pi_1(A_2)$ 中商掉 N 等价于将 a 与 c 视作同一个元素, 而 b,d 则互不影响. 因此

$$\pi_1(X)=\mathbb{Z}\times(\mathbb{Z}*\mathbb{Z}).$$

H-Prob 1.2.20

Let X be the subspace of $\mathbb{R}^2$ that is the union of the circles C_n of radius n and center $(n,0)$ for $n=1,2,\cdots$. Show that $\pi_1(X)$ is the free group $*_n\,\pi_1(C_n)$, the same as for the infinite wedge sum $\bigvee_\infty S^1$. Show that X and $\bigvee_\infty S_1$ are in fact homotopy equivalent, but not homeomorphic.

Proof: 先证明:

$$\pi_1(X)=\overset{\infty}{\underset{n=1}{*}}\pi_1(C_n)=*(n=1)^\infty\pi_1(S^1)=\overset{\infty}{\underset{n=1}{*}}\mathbb{Z}=\pi_1\left(\bigvee_\infty S^1\right).$$

易见 X 实际上就是 $\bigvee_{n=1}^\infty C_n$, 其中 0 为楔点. 在 0 处取半径小于 1 的圆盘 $B(0,\varepsilon)$, 记

$$V=B(0,\varepsilon)\cap X=\bigcup_{n=1}^\infty(C_n\cap B(0,\varepsilon))$$

于是 $V\subset X$ 为道路连通的开集. 据此, 我们取

$$A_n=V\cup C_n=V\cup(C_n\setminus\{0\})$$

注意对任意 $x\in C_n\setminus\{0\}$, 可以取到 $\mathbb{R}^2$ 中充分小的开圆盘 $B(x,\delta_x)$ s.t. 其仅与 C_n 有交集, 故

$$C_n\setminus\{0\}=X\cap\bigcup_{x\in C_n\setminus\{0\}}B(x,\delta_x)$$

从而 $C_n\setminus\{0\}$ 为 X 中的开集, 故 A_n 也为开集且道路连通. 于是 $X=\bigcup_{n=1}^\infty A_n$ 为开覆盖. 易见

$$A_i\cap A_j=V,\quad\forall i\neq j;\quad A_i\cap A_j\cap A_k=V,\quad\forall i\neq j\neq k$$

而 V 道路连通, 故 Van Kampen 定理适用, 从而

$$\pi_1(X)=\overset{\infty}{\underset{n=1}{*}}\pi_1(A_n)/N$$

根据 V 的构造, 我们可以将 V 中的每段圆弧缩到 0, 故 V 是可缩的, 进而 A_n 可形变收缩到 C_n, 因此 $\pi_1(A_n)=\pi_1(C_n)=\mathbb{Z}$. 而 $A_\alpha\cap A_\beta=V\simeq\{0\},\forall\alpha\neq\beta$, 因此每个 $(i_{\alpha\beta})_*,(i_{\beta\alpha})_*$ 均平凡, 从而 $N=\{e\}$ 平凡, 综上得到

$$\pi_1(X)=\overset{\infty}{\underset{n=1}{*}}\pi_1(A_n)=\overset{\infty}{\underset{n=1}{*}}\pi_1(C_n)=\overset{\infty}{\underset{n=1}{*}}\pi_1(S^1)=\overset{\infty}{\underset{n=1}{*}}\mathbb{Z}$$

同理可知

$$\pi_1\left(\bigvee_\infty S^1\right)=\overset{\infty}{\underset{n=1}{*}}\mathbb{Z}.$$

楔和保持同伦等价, 因此 $X \simeq \bigvee_{n=1}^{\infty} S_n^1$. 为证 X 与 $\bigvee_{n=1}^{\infty} S_n^1$ 不同胚, 我们引入第一可数的概念:

一个拓扑空间 X 称为第一可数的, 如果对任意 $x \in X$, 存在点 x 的一个可数邻域基, 即存在 x 的邻域列 $\{U_n\}$, 使得对任意 x 的邻域 V, 存在某个 n 使得 $U_n \subset V$.

关于第一可数, 我们会用到以下容易验证的基本性质:

(1) 若 X 是第一可数的, 则 $Y \subset X$ 携带子空间拓扑后也是第一可数的;

(2) 若 $f : X \to Y$ 是同胚, 则 X 是第一可数的当且仅当 Y 是第一可数的.

回到本题, 易见 $\mathbb{R}^2$ 是第一可数的, 因此 $X \subset \mathbb{R}^2$ 也是第一可数的. 下面证明 $\bigvee_{n=1}^{\infty} S^1$ 不是第一可数的, 反设 $\bigvee_{n=1}^{\infty} S^1$ 第一可数, 设 $a \in S_n^1$ 为每个圆的楔点, 则存在 a 的可数邻域基 $\{U_n\}$, 考虑

$$V_n = U_n \cap S_n^1$$

于是 V_n 为 S_n^1 中包含 a 的邻域. 由于 S_n^1 是度量空间, 故存在 $\varepsilon_n > 0$ 使得

$$W_n = B(a, \varepsilon_n) \cap S_n^1 \subsetneq V_n$$

则 W_n 为 S_n^1 中包含 a 的开邻域. 考虑

$$W = \bigcup_{n=1}^{\infty} W_n$$

易见其为 $\bigvee_{n=1}^{\infty} S_n^1$ 中包含 a 的开邻域. 由邻域基的定义, 存在某个 U_k 使得 $U_k \subset W$, 从而

$$V_k = U_k \cap S_k^1 \subset W \cap S_k^1 = W_k \subsetneq V_k$$

矛盾! 因此 $\bigvee_{n=1}^{\infty} S^1$ 不是第一可数的, 故 X 与 $\bigvee_{n=1}^{\infty} S^1$ 不同胚.

Prob 10.18

If $\tilde{X}$ is a covering space of X, and $\tilde{Y}$ a covering space of Y, show that $\tilde{X} \times \tilde{Y}$ is a covering space of $X \times Y$.

Proof: 设 $p : \tilde{X} \to X, q : \tilde{Y} \to Y$ 分别为覆盖映射, 则定义映射

$$\begin{aligned} r : \tilde{X} \times \tilde{Y} &\to X \times Y \\ (\tilde{x}, \tilde{y}) &\mapsto (p(\tilde{x}), q(\tilde{y})) \end{aligned}$$

下证 r 即为覆盖映射. 设 $X = \bigcup_\alpha U_\alpha, Y = \bigcup_{\alpha'} V_{\alpha'}$ 为对应的开覆盖, 于是

$$X \times Y = \bigcup_{\alpha, \alpha'} (U_\alpha \times V_{\alpha'})$$

任取任意 $U_\alpha \times V_{\alpha'}$, 则 $p^{-1}(U_\alpha) = \bigcup_\beta U'_{\alpha\beta}, q^{-1}(V_{\alpha'}) = \bigcup_\gamma V'_{\alpha'\gamma}$, 于是

$$r^{-1}(U_\alpha \times V_{\alpha'}) = p^{-1}(U_\alpha) \times q^{-1}(V_{\alpha'}) = \bigcup_{\beta,\gamma} \left(U'_{\alpha\beta} \times V'_{\alpha'\gamma} \right)$$

并且由于 $p|_{U'_{\alpha\beta}} : U'_{\alpha\beta} \to U_\alpha$ 与 $q|_{V'_{\alpha'\gamma}} : V'_{\alpha'\gamma} \to V_{\alpha'}$ 均为同胚, 故

$$r|_{U'_{\alpha\beta} \times V'_{\alpha'\gamma}} : U'_{\alpha\beta} \times V'_{\alpha'\gamma} \to U_\alpha \times V_{\alpha'}$$

也是同胚. 因此 $r : \tilde{X} \times \tilde{Y} \to X \times Y$ 为覆盖映射.

Prob 10.19

Is the map $f : (0, 3) \to S^1$ defined by $f(x) = e^{2\pi i x}$ a covering map?

Solution: 不是, 反设 f 为覆盖映射, 由 S^1 连通可知 $\left|p^{-1}(x)\right|$ 为常数, 但 $\left|p^{-1}(1)\right| = 2$, 对任意 $1 \neq x \in S^1, \left|p^{-1}(x)\right| = 3$, 矛盾! 因此 f 不是覆盖映射.

Prob H-1.3.1

For a covering space $p : \tilde{X} \to X$ and a subspace $A \subset X$, let $\tilde{A} = p^{-1}(A)$. Show that the restriction $p : \tilde{A} \to A$ is a covering space.

Proof: 设 $X = \bigcup_\alpha U_\alpha$ 为对应的开覆盖, 于是 $A = \bigcup_\alpha (A \cap U_\alpha)$, 为 A 的开覆盖, 记 $U'_\alpha = A \cap U_\alpha$, 则 $p^{-1}(U'_\alpha) = \tilde{A} \cap p^{-1}(U_\alpha)$, 由 p 为覆盖映射可知

$$p^{-1}(U'_\alpha) = \tilde{A} \cap \left(\bigsqcup_\beta V_{\alpha\beta} \right) = \bigsqcup_\beta \left(\tilde{A} \cap V_{\alpha\beta} \right)$$

其中每个 $\tilde{A} \cap V_{\alpha\beta}$ 为 $\tilde{A}$ 中不交的开集, 并且

$$p|_{\tilde{A} \cap V_{\alpha\beta}} = \left(p|_{V_{\alpha\beta}} \right)|_{\tilde{A} \cap V_{\alpha\beta}} : \tilde{A} \cap V_{\alpha\beta} \to U'_\alpha$$

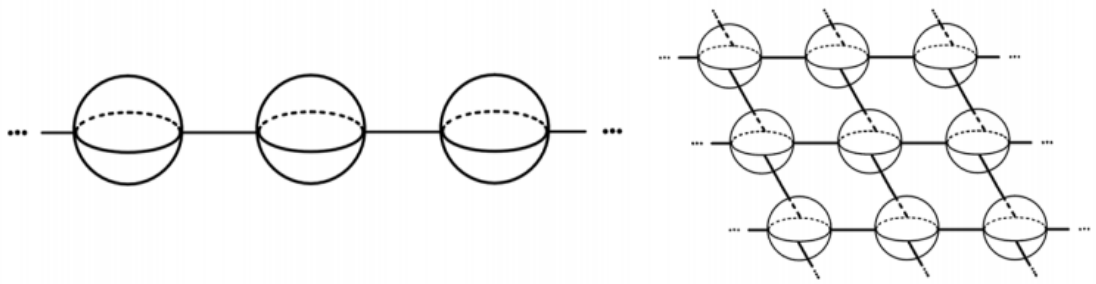
其中 $p|_{V_{\alpha\beta}} : V_{\alpha\beta} \to U_\alpha$ 为同胚, 故其限制映射 $p|_{\tilde{A} \cap V_{\alpha\beta}}$ 也是同胚. 因此 $p : \tilde{A} \to A$ 为覆盖映射.

Prob H-1.3.4

Construct a simply-connected covering space of the space $X \subset \mathbb{R}^3$ that is the union of a sphere and a diameter. Do the same when X is the union of a sphere and a circle intersecting it in two points.

Solution: 设 X 为球与直径的并, 将直径的两个端点进行移动, 可知 $X \simeq S^2 \bigvee S^1$, S^1 有覆叠空间 $\mathbb{R}$, S^2 单连通, 设楔点 $x_0 \in S^1$, 则可在每个 $p^{-1}(x_0) \in \mathbb{R}$ 处放置 S^2, 得单连通的覆叠空间 $\tilde{X}$.

同理, 当 X 为球与圆的并时, 可知 $X \simeq S^2 \bigvee S^1 \bigvee S^1$, 即可得到其单连通的覆叠空间.



Prob H-1.3.5

Let X be the subspace of $\mathbb{R}^2$ consisting of the four sides of the square $[0,1]\times[0,1]$ together with the segments of the vertical lines $x=\frac{1}{2},\frac{1}{3},\frac{1}{4},\cdots$ inside the square. Show that for every covering space $\tilde{X}\to X$ there is some neighborhood of the left edge of X that lifts homeomorphically to $\tilde{X}$. Deduce that X has no simply-connected covering space.

Proof[*]:

设 $p:\tilde{X}\to X$ 为覆盖映射, 由定义, 存在 X 的开覆盖 $\bigcup_\alpha U_\alpha$, 使得对每个 α 都有

$$p^{-1}(U_\alpha)=\bigsqcup_{i\in I_\alpha}\tilde{U}_{\alpha,i},\quad p|_{\tilde{U}_{\alpha,i}}:\tilde{U}_{\alpha,i}\to U_\alpha \text{ 为同胚.}$$

每个 U_α 都是若干 $R\cap X$ 的并, $R\subset\mathbb{R}^2$ 为开矩形, 故不妨设 $U_\alpha=R_\alpha\cap X$, 其中 R_α 为开矩形.

记左边界 $L=\{0\}\times[0,1]\subset X$. 由 L 紧, 存在有限个 U_α 覆盖 L. 通过重新编号可设它们依次为

$$\begin{aligned}U_1&=([0,\varepsilon_1)\times[0,b_1))\cap X\\U_2&=([0,\varepsilon_2)\times(a_2,b_2))\cap X\\&\cdots\\U_N&=([0,\varepsilon_N)\times(a_N,1])\cap X\end{aligned}$$

满足

$$0<a_2<b_1<a_3<b_2<\ldots<a_N<b_{N-1}<1,$$

并且对每个 n 有

$$p^{-1}(U_n)=\bigsqcup_{i\in I_n}\tilde{U}_{n,i},\quad p:\tilde{U}_{n,i}\to U_n \text{ 为同胚.}$$

接下来我们那构造某个 $c>0$, 以及提升

$$f_N:([0,c)\times[0,1])\cap X\to\tilde{X}$$

使得 $p\circ f_N=\mathrm{id}$ 且 f_N 是到其像上的同胚. 我们用归纳在 $n=1,\ldots,N$ 上一步步扩张定义域.

从 U_1 开始, 我们有 $p^{-1}(U_1)=\bigsqcup_{i\in I_1}\tilde{U}_{1,i}$, 任选其一, 记为 $\tilde{U}_{1,i_1}$. 由 p 在其上为同胚, 得到

$$f_1=\left(p|_{\tilde{U}_{1,i_1}}\right)^{-1}:U_1\to\tilde{U}_{1,i_1}.$$

于是 f_1 给出 $U_1=([0,c_1)\times[0,b_1))\cap X$ 到 $\tilde{U}_{1,i_1}$ 的同胚提升, 其中 $c_1=\varepsilon_1$.

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[*]本解答参考了 https://math.stackexchange.com/q/4650495

再考虑 $p^{-1}(U_2)=\bigsqcup_{i\in I_2}\tilde{U}_{2,i}$, 取一点 $t\in(a_2,b_1)$. 点 $(0,t)$ 在 $U_1\cap U_2$ 中, 因此 $f_1(0,t)$ 落在某个唯一的 $\tilde{U}_{2,i_2}$ 里. 由 f_1 的连续性, 存在 $0<c_2\leq\min(c_1,\varepsilon_2)$ 使得

$$f_1(([0,c_2)\times\{t\})\cap X)\subset\tilde{U}_{2,i_2}.$$

由于 $([0,c_2)\times(a_2,b_1))\cap X$ 可视为 X 中若干竖直线段的并, 且每条竖直线段连通并含有一点 (x,t), 结合 $\bigsqcup_{i\in I_2}\tilde{U}_{2,i}$ 是互不相交开集的并, 因此只能是

$$f_1(([0,c_2)\times(a_2,b_1))\cap X)\subset\tilde{U}_{2,i_2}.$$

现在利用同胚

$$g=\left(p|_{\tilde{U}_{2,i_2}}\right)^{-1}:U_2\to\tilde{U}_{2,i_2}$$

在 U_2 上定义一个提升. 在交集

$$([0,c_2)\times(a_2,b_1))\cap X=U_1\cap U_2\cap([0,c_2)\times[0,1])$$

上, 因为 f_1,g 的像都落在同一层 $\tilde{U}_{2,i_2}$ 且都是 p 的逆, 故 g 与 f_1 一致. 因此可以把

$$f_1|_{([0,c_2)\times[0,b_1))\cap X},\quad g|_{([0,c_2)\times(a_2,b_2))\cap X}$$

拼接成一个连续映射 f_2, 它同胚地提升了

$$([0,c_2)\times[0,b_2))\cap X$$

依此类推, 对 $n=1,...,N$ 递推地进行构造, 我们得到 $c_N>0$ 与提升

$$f_N:([0,c_N)\times[0,1])\cap X\to\tilde{X},$$

满足 $p\circ f_N=\mathrm{id}$ 且 f_N 为到其像的同胚, 故存在左边界的某个邻域可以在 $\tilde{X}$ 中被同胚地提升.

最后说明 $\tilde{X}$ 非单连通. 取整数 k 使得 $\frac{1}{k}<c_N$. 考虑 X 中的矩形环路

$$\gamma:(0,0)\to\left(\frac{1}{k},0\right)\to\left(\frac{1}{k},1\right)\to(0,1)\to(0,0),$$

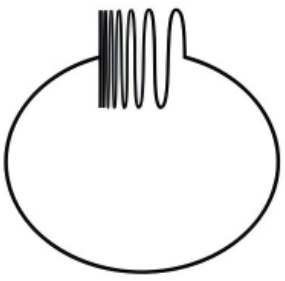
γ 完全包含在 $([0,c_N)\times[0,1])\cap X$ 中, 故可用 f_N 提升为 $\tilde{X}$ 中的环路 $\tilde{\gamma}$. 覆盖映射诱导的同态

$$p_*:\pi_1\left(\tilde{X}\right)\to\pi_1(X)$$

是单射, 而 $p_*(\tilde{\gamma})=\langle\gamma\rangle$. 而矩形环路 γ 在 $\pi_1(X)$ 中是非平凡元, 故 $\tilde{\gamma}$ 在 $\pi_1\left(\tilde{X}\right)$ 中也非平凡, 从而 $\tilde{X}$ 不是单连通空间.

H-Prob 1.3.7

Let Y be the quasi-circle shown in the figure, a closed subspace of $\mathbb{R}^2$ consisting of a portion of the graph of $y=\sin\frac{1}{x}$, the segment $[-1,1]$ in the y-axis, and an arc connecting these two pieces. Collapsing the segment of Y in the y-axis to a point gives a quotient map $f:Y\to S^1$. Show that f does not lift to the covering space $\mathbb{R}\to S^1$, even though $\pi_1(Y)=0$. Thus local path-connectedness of Y is a necessary hypothesis in the lifting criterion.



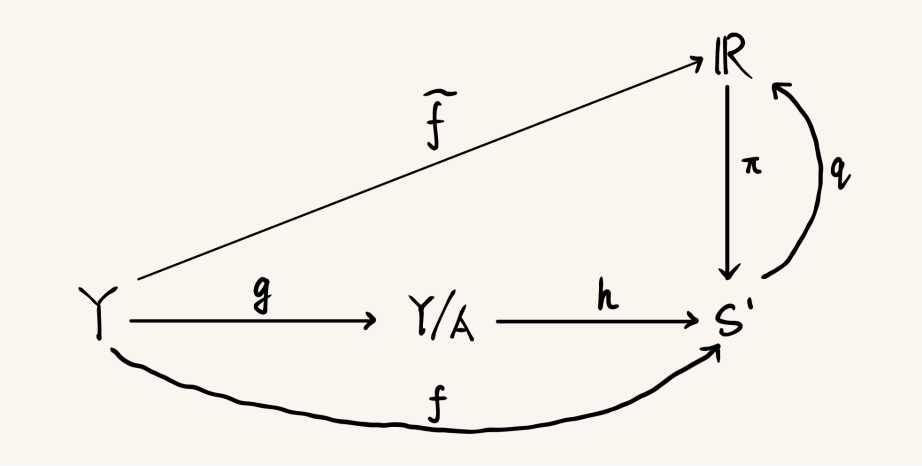
Proof: 先证 $\pi_1(Y)=\{e\}$: 将 Y 写为 $Y=A\cup B\cup C$, 其中 A 为 y 轴上的 $[-1,1]$, B 为 $y=\sin\frac{1}{x}$ 的一部分, C 为连接 A,B 的弧. 考虑 Y 中的环路 $\gamma: I\to Y$, 则 γ 为 Y 中可求长的闭合路径, 而 B 是不可求长的, 因此 γ 无法做到绕 Y 一圈, 从而 γ 可以收缩为一点, 故 $\pi_1(Y)=\{e\}$.

再证 $f: Y\to S^1$ 无法提升到 $\mathbb{R}$: 首先明确 f 的定义. 先将 Y 中 A 识别为一点 a, 给出识别映射 $g: Y\to Y/A$, 通过将 Y/A 中 B 中的点投影到 x 轴, 可以给出 $h: Y/A\to S^1$ 为同胚映射, 从而 $f=h\circ g$, 并且 f 可视为识别映射 $Y\to S^1$. 我们不妨设 $h(a)=1$, 于是 $f(A)=1\in S^1$.

记 $\pi:\mathbb{R}\to S^1$ 为识别映射. 反设存在提升 $\tilde{f}: Y\to\mathbb{R}$ 满足 $\pi\circ\tilde{f}=f$, 于是 $\pi\circ\tilde{f}(A)=f(A)=1$, 故 $\tilde{f}(A)\subset\mathbb{Z}$, 而 A 连通, 故 $\tilde{f}(A)=\{n\}$ 为单点集. 从而根据商映射的泛性质, 存在 $q: S^1\to\mathbb{R}$ 使得 $\tilde{f}=q\circ f$. 于是

$$\pi\circ q\circ f=f$$

由 f 为满射知 $\pi\circ q=\mathrm{id}_{S^1}$, 从而 $\pi_*\circ q_*=\mathrm{id}_{\pi_1(S^1)}$, 但 $\pi_*:\{e\}\to\mathbb{Z}$ 是平凡映射, 故 $\pi_*\circ q_*=\{e\}$ 也为平凡映射, 矛盾! 因此 f 无法提升到 $\mathbb{R}$.



H-Prob 1.3.9

Show that if a path-connected, locally path-connected space X has $\pi_1(X)$ finite, then every map from X to S^1 is nullhomotopic.

Proof: 设 $f: X\to S^1$ 为连续映射, 于是其给出群同态

$$f_*:\pi_1(X)\to\pi_1(S^1)=\mathbb{Z}$$

由 $\pi_1(X)$ 为有限群, 而 $\mathbb{Z}$ 的有限子群只能是 $\{0\}$, 故 $\pi_1(X)=\{e\}$ 平凡.

设 $p:\mathbb{R}\to S^1$ 为覆盖映射, 于是

$$f_*(\pi_1(X))=\{0\},\quad p_*(\pi_1(\mathbb{R}))=\{0\}\Longrightarrow f_*(\pi_1(X))\subset p_*(\pi_1(\mathbb{R}))$$

于是 lifting criterion 适用, 故存在提升 $\tilde{f}: X\to\mathbb{R}$ 使得 $p\circ\tilde{f}=f$. 由于 $\mathbb{R}$ 可缩, 故存在

$$H:\mathbb{R}\times I\to\mathbb{R}\ \text{ s.t. }\ H(x,0)=x,\quad H(x,1)=x_0$$

于是考虑

$$\begin{aligned}G: X\times I&\to S^1\\(x,t)&\mapsto p\Big(H\Big(\tilde{f}(x),t\Big)\Big)\end{aligned}$$

其满足 $G(x,0)=p\circ\tilde{f}(x)=f(x), G(x,1)=p(x_0)\in S^1$, 故 $f\underset{G}{\simeq}c_{p(x_0)}$, 即 f 零同伦.

Prob 10.20, 10.21

Describe all the covering spaces of the torus, projective plane, Möbius strip, and cylinder. Find the group of covering transformations for each of the coverings.

Solution: 设 $H \subset \pi_1(X)$ 为子群, 我们要找对应的覆叠空间 X_H 和甲板变换群 $\mathrm{Deck}(X_H)$. 注意, 容易验证题中涉及到的覆叠空间均正规, 因此 $\mathrm{Deck}(X_H) = \pi_1(X)/H$.

(1) Torus: 设 $T = S^1 \times S^1$, 则 $\pi_1(T) = \mathbb{Z}^2$. T 可由 $\mathbb{R}^2$ 商去 Z^2 通过平移在其上的群作用得到, 万有覆叠空间是 $\mathbb{R}^2$, 覆叠映射 p_H 为 $\mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2 = T$ 的商映射.

- $H = \{e\}$, 则 $X_H = \mathbb{R}^2$, 覆叠映射为上面的商映射, $\mathrm{Deck}(X_H) = \mathbb{Z}^2$.
- $H = \mathbb{Z}$, 则 $\mathbb{R}^2$ 商去 $\mathbb{Z}$ 的平移群作用后得到 $S^1 \times \mathbb{R}$, 取 $X_H = S^1 \times \mathbb{R}$, 覆叠映射为

$$p_H : S^1 \times \mathbb{R} \to S^1 \times S^1 \quad (e^{2\pi i x}, y) \mapsto (e^{2\pi i x}, e^{2\pi i y})$$

  对应 $\mathrm{Deck}(X_H) = \mathbb{Z}$.

- $H = m\mathbb{Z} \times n\mathbb{Z}$, 则 $\mathbb{R}^2/(m\mathbb{Z} \times n\mathbb{Z}) = S^1 \times S^1$, 取 $X_H = S^1 \times S^1$, 覆叠映射为

$$p_H : S^1 \times S^1 \to S^1 \times S^1 \quad (e^{2\pi i x}, e^{2\pi i y}) \mapsto (e^{2\pi i m x}, e^{2\pi i n y}).$$

  对应 $\mathrm{Deck}(X_H) = \mathbb{Z}_m \times \mathbb{Z}_n$.

(2) $\mathbb{R}\mathrm{P}^2$: 设 $\mathbb{R}\mathrm{P}^2$ 由 S^2 商去 $\mathbb{Z}/2\mathbb{Z}$ 的群作用得到, 即 $\mathbb{R}\mathrm{P}^2 = S^2/\mathbb{Z}_2$. 我们计算过 $\pi_1(\mathbb{R}\mathrm{P}^2) = \mathbb{Z}_2$, 上述 S^2 即为 $\mathbb{R}\mathrm{P}^2$ 的万有覆叠空间, 而商映射给出了对应的覆叠映射 $p : S^2 \to \mathbb{R}\mathrm{P}^2$.

- $H = \{e\}$, 则 $X_H = S^2$, 覆叠映射为上面的商映射, $\mathrm{Deck}(X_H) = \mathbb{Z}_2$.
- $H = \mathbb{Z}_2$, 则 $X_H = \mathbb{R}\mathrm{P}^2$, 覆叠映射为恒等映射, $\mathrm{Deck}(X_H) = \{e\}$.

(3) Möbius strip: 设 M 为 Möbius strip, $\pi_1(M) = \mathbb{Z}$. M 由 $\mathbb{R} \times [-1, 1]$ 商去 $\mathbb{Z}$ 的群作用得到, 即

$$\mathbb{Z} \curvearrowright \mathbb{R} \times [-1, 1], \quad (n, (x, y)) \mapsto (x + n, (-1)^n y)$$

因此万有覆叠空间为 $\mathbb{R} \times [-1, 1]$, 覆叠映射为商映射 $p : \mathbb{R} \times [-1, 1] \to M$.

- $H = \{e\}$, 则 $X_H = \mathbb{R} \times [-1, 1]$, 覆叠映射为上面的商映射, $\mathrm{Deck}(X_H) = \mathbb{Z}$.
- $H = n\mathbb{Z}$
  - n 为奇数, 此时 $\mathbb{R} \times [-1, 1]/n\mathbb{Z}$ 对应的群作用为

    $$n\mathbb{Z} \curvearrowright \mathbb{R} \times [-1, 1], \quad (kn, (x, y)) \mapsto \big(x + kn, (-1)^{kn} y\big) = (x + kn, -y)$$

    因此 X_H 仍为 Möbius strip, 对应 $\mathrm{Deck}(X_H) = \mathbb{Z}_n$.
  - n 为偶数, 此时 $\mathbb{R} \times [-1, 1]/n\mathbb{Z}$ 对应的群作用为

    $$n\mathbb{Z} \curvearrowright \mathbb{R} \times [-1, 1], \quad (kn, (x, y)) \mapsto \big(x + kn, (-1)^{kn} y\big) = (x + kn, y)$$

    因此 X_H 为 cylinder, 对应 $\mathrm{Deck}(X_H) = \mathbb{Z}_n$.

(4) Cylinder: 设 $C = S^1 \times \mathbb{R}$ 为 cylinder, 则 $\pi_1(C) = \mathbb{Z}$, 万有覆叠空间为 $\mathbb{R} \times \mathbb{R}$, 覆叠映射为

$$p : \mathbb{R} \times \mathbb{R} \to S^1 \times \mathbb{R}, \quad (x, y) \mapsto (e^{2\pi i x}, y)$$

- $H = \{e\}$, 则 $X_H = \mathbb{R} \times \mathbb{R}$, 覆叠映射为上面的映射, $\mathrm{Deck}(X_H) = \mathbb{Z}$.
- $H = n\mathbb{Z}$, 则 $X_H = S^1 \times \mathbb{R}$, 覆叠映射为 $p_H(e^{2\pi i x}, y) = (e^{2\pi i n x}, y)$, 对应 $\mathrm{Deck}(X_H) = \mathbb{Z}_n$.

Prob 10.22

The space shown in Fig.6.19 is a covering space of the one-point union of two circles. What is the corresponding subgroup of $\mathbb{Z} * \mathbb{Z}$.

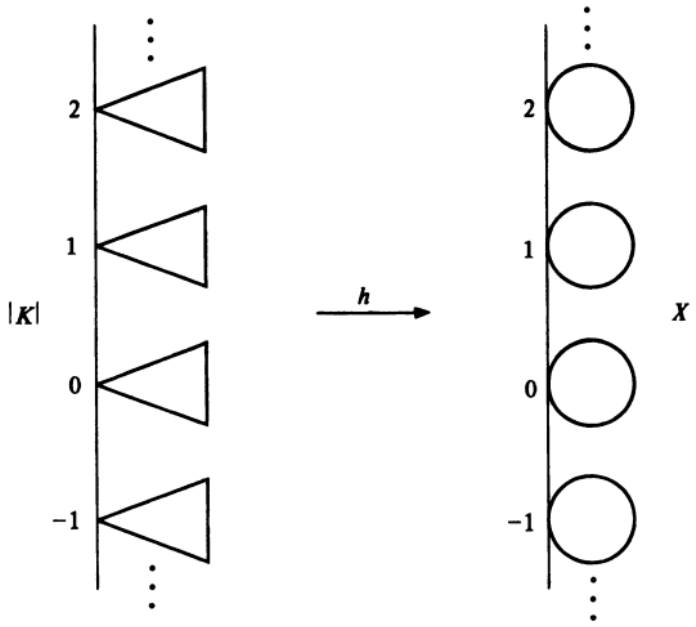
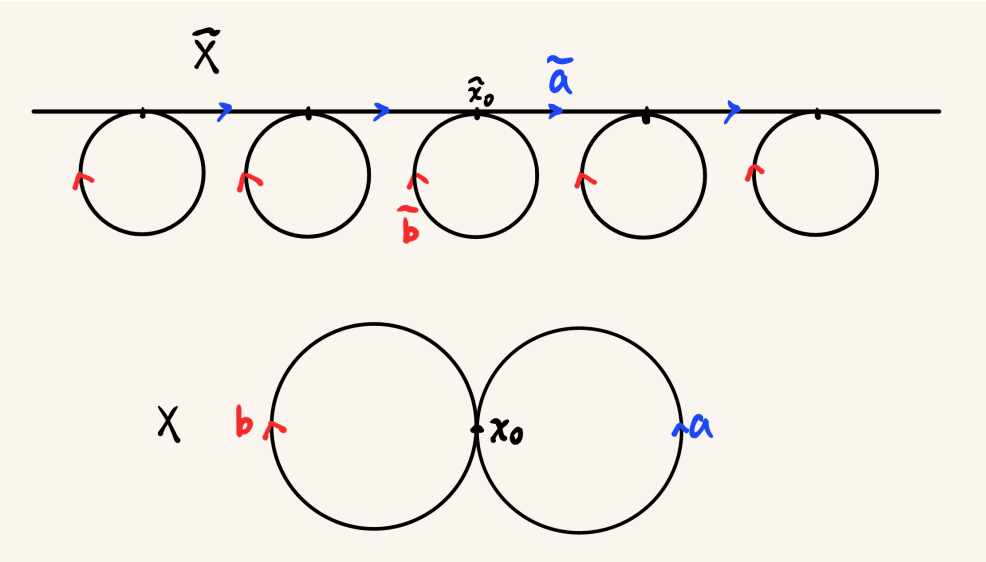


Figure 6.19

Solution: 如图, $X = S^1 \vee S^1$, $\tilde{X}$ 表示图中的覆叠空间.



构造 $p : \tilde{X} \to X$ 使得 X 中红色环路 a 对应 $\tilde{X}$ 中的圆上环路 $\tilde{a}$, 蓝色环路 b 对应连接圆环的线段 $\tilde{b}$, 于是 p 为覆叠映射. 考察 $p_*(\pi_1(\tilde{X}))$ 在 $\pi_1(X)$ 中对应的子群: 观察发现 $\tilde{X}$ 中的环路在 X 中的像可表示为

$$\prod_{i=1}^{N}(a^{x_i}b^{y_i})a^{-\sum_i x_i} = \prod_{i=1}^{N}\left(a^{\sum_{j=1}^{i} x_j}b^{y_i}a^{-\sum_{j=1}^{i} x_j}\right) = \prod_{i=1}^{N}\left(a^{\sum_{j=1}^{i} x_j}ba^{-\sum_{j=1}^{i} x_j}\right)^{y_i}$$

因此

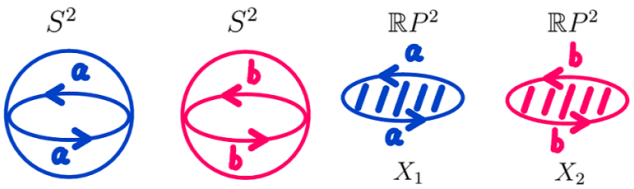
$$p_*(\pi_1(\tilde{X})) = \langle\{a^k b a^{-k} \mid k \in \mathbb{Z}\}\rangle \subset \pi_1(X) = \mathbb{Z} * \mathbb{Z}.$$

H-Prob 1.3.14

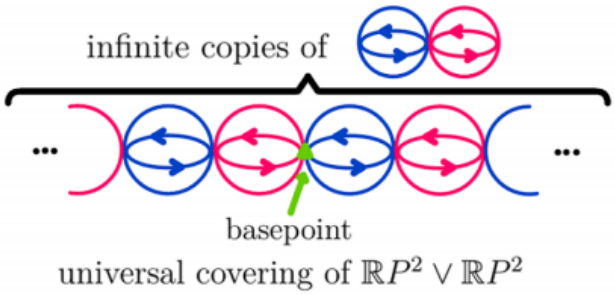
Find all the connected covering spaces of $\mathbb{R}\mathrm{P}^2 \vee \mathbb{R}\mathrm{P}^2$.

Solution[†]: 分别记 X_1, X_2 为第一个与第二个 $\mathbb{R}\mathrm{P}^2$, $\pi_1(X_1) = \mathbb{Z}_2 = \langle a \rangle, \pi_1(X_2) = \mathbb{Z}_2 = \langle b \rangle$.

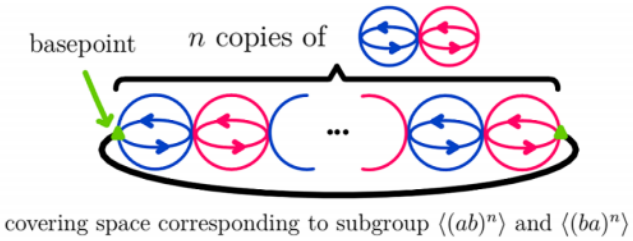
考察 $\mathbb{R}\mathrm{P}^2$ 的形成过程: 商映射将蓝色的 S^2 中的对称点识别为同一点, 从而得到 X_1; 再将红色的 S^2 中的对称点识别为同一点, 从而得到 X_2. 考虑 $\pi_1(X_1 \vee X_2) = \mathbb{Z}_2 * \mathbb{Z}_2 = \langle a \rangle * \langle b \rangle$ 的子群:



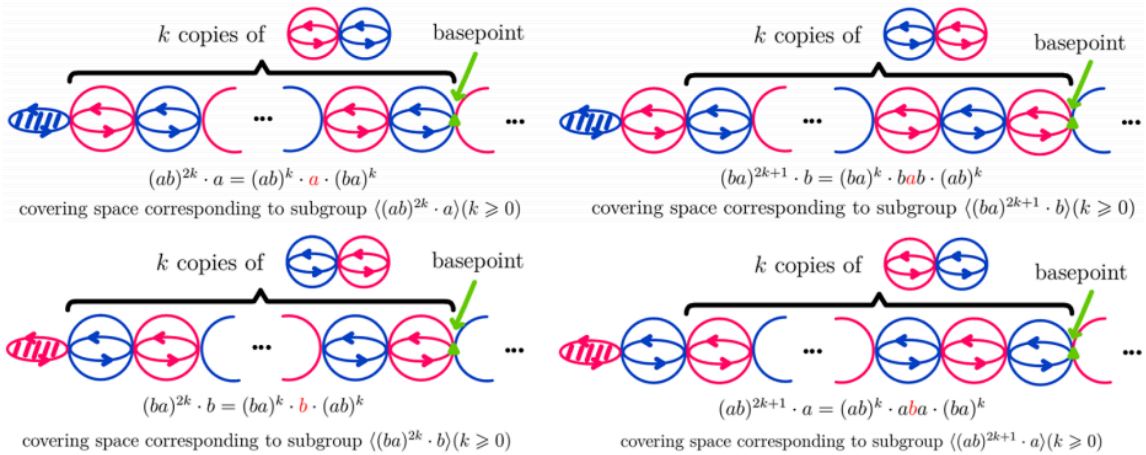
(1) 对平凡子群 $H = \{e\}$, 对应覆叠空间为万有覆叠空间, 即无穷个 S^2 形成的链.



(2) 对 $H = \mathbb{Z}$, 其由形如 ab 或 ba 的元素生成, 此时对应的覆叠空间是偶数个 S^2 形成的闭环.

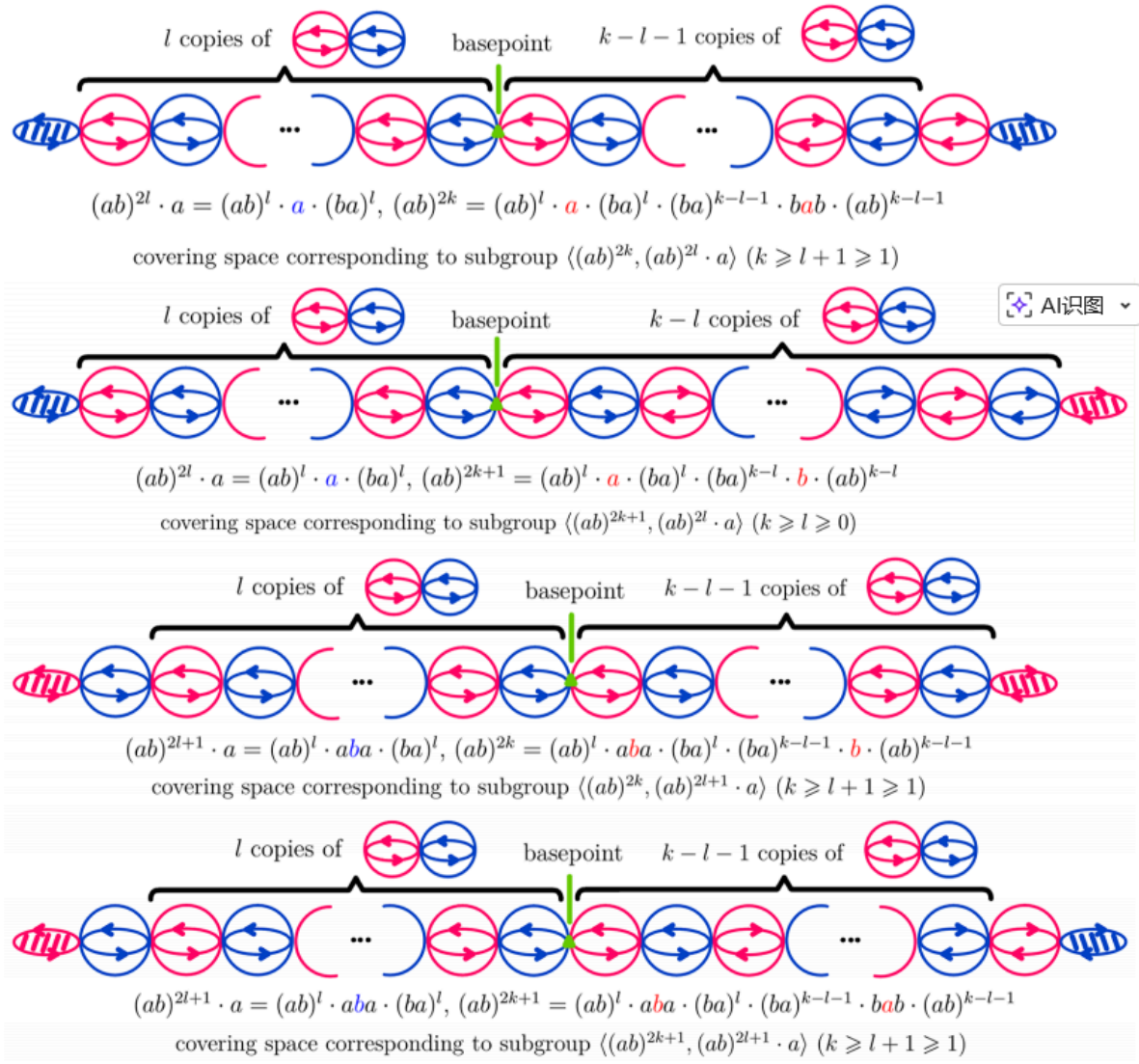


(3) 对 $H = \mathbb{Z}_2$, 其由 $(ab)^m \cdot a$ 或 $(ba)^m \cdot b$ 生成, $m \geq 0$. 对应的覆叠空间为 $\mathbb{R}\mathrm{P}^2$ 粘合上无穷个 S^2 形成的链.



(4) 对 $H = \mathbb{Z}_2 * \mathbb{Z}_2$, 其由 $(ab)^n$ 和 $(ab)^m \cdot a$ $(m \leq n)$ 生成. 对应的覆叠空间为有限个 S^2 形成的链, 两端分别粘合上 $\mathbb{R}\mathrm{P}^2$.

[†]本解答参考了 https://math.stackexchange.com/questions/3472796



H-Prob 1.3.15

Let $p : \tilde{X} \to X$ be a simply-connected covering space of X and let $A \subset X$ be a path-connected, locally path-connected subspace, with $\tilde{A} \subset \tilde{X}$ a path-component of $p^{-1}(A)$. Show that $p : \tilde{A} \to A$ is the covering space corresponding to the kernel of the map $\pi_1(A) \to \pi_1(X)$.

Proof: 之前的作业已经证明了 $p|_{\tilde{A}} : \tilde{A} \to A$ 确为覆叠映射. 设

$$i : A \hookrightarrow X, \quad j : \tilde{A} \hookrightarrow \tilde{X}$$

为嵌入映射, 我们需要证明

$$(p|_{\tilde{A}})^*\left(\pi_1\left(\tilde{A}\right)\right) = \operatorname{Ker} i_*$$

首先, 由于 $\tilde{X}$ 单连通, 因此 $\pi_1\left(\tilde{X}\right) = \{e\}$, $p_* : \pi_1\left(\tilde{X}\right) \to X$ 是平凡的. 注意到

$$i \circ (p|_{\tilde{A}}) = p \circ j \Longrightarrow i_* \circ (p|_{\tilde{A}})_* = p_* \circ j_* = e$$

因此 $(p|_{\tilde{A}})_*\left(\pi_1\left(\tilde{A}\right)\right) \subset \operatorname{Ker} i_*$. 反过来, 设 $\langle\alpha\rangle \in \operatorname{Ker} i_*$, 则 $\alpha : I \to A$ 为环, 且在 X 中可收缩. 由道路提升引理, α 可以被提升为 $\tilde{\alpha} : I \to \tilde{A}$ 为环路, 且

$$\alpha = p \circ \tilde{\alpha} = (p|_{\tilde{A}}) \circ \tilde{\alpha} \Longrightarrow \langle\alpha\rangle = (p|_{\tilde{A}})_*(\langle\tilde{\alpha}\rangle) \in (p|_{\tilde{A}})_*\left(\pi_1\left(\tilde{A}\right)\right)$$

因此 $\operatorname{Ker} i_* \subset (p|_{\tilde{A}})_*\left(\pi_1\left(\tilde{A}\right)\right)$. 综上 $(p|_{\tilde{A}})_*\left(\pi_1\left(\tilde{A}\right)\right) = \operatorname{Ker} i_*$.

H-Prob 1.3.16

Given maps $X \to Y \to Z$ such that both $Y \to Z$ and the composition $X \to Z$ are covering spaces, show that $X \to Y$ is a covering space if Z is locally path-connected, and show that this covering space is normal if $X \to Z$ is a normal covering space.

Proof: 设 $X \xrightarrow{f} Y \xrightarrow{g} Z$, 并记 $X \xrightarrow{h} Z, h \coloneqq g \circ f$. 若 h, g 均为覆叠映射, 任取 $y \in Y$, 记 $z = g(y)$, 通过对开邻域取交集, 可知存在 $z \in Z$ 的开邻域 U 使得

$$g^{-1}(U) = \bigsqcup_{\alpha} V_\alpha, \quad g|_{V_\alpha} : V_\alpha \to U \text{ 为同胚.}$$

$$h^{-1}(U) = \bigsqcup_{\beta} W_\beta, \quad h|_{W_\beta} : W_\beta \to U \text{ 为同胚.}$$

注意 $h = g \circ f$, 因此

$$\bigsqcup_{\beta} W_\beta = h^{-1}(U) = f^{-1} \circ g^{-1}(U) = f^{-1}\left(\bigsqcup_{\alpha} V_\alpha\right) = \bigsqcup_{\alpha} f^{-1}(V_\alpha)$$

选取 V 使得 $y \in V_0$, 进而 $f^{-1}(V_0)$ 可以写成若干 W_β 的不交并, 即

$$f^{-1}(V_0) = \bigsqcup_{\beta' \in B} W_{\beta'}$$

且 $f|_{W_{\beta'}} : W_{\beta'} \to V_0$ 为同胚. 由 $y \in Y$ 的任意性即知 $f : X \to Y$ 为覆叠映射.

下设 $h : X \to Z$ 为正规覆叠空间, 选取基点 x_0, y_0, z_0 s.t. $y_0 = f(x_0), z_0 = g(y_0)$. 根据定义, 我们有 $h_*(\pi_1(X, x_0))$ 为 $\pi_1(Z, z_0)$ 的正规子群. 注意到有如下包含关系

$$h_*(\pi_1(X, x_0)) = g_* \circ f_*(\pi_1(X, x_0)) \subset g_*(\pi_1(Y, y_0)) \subset \pi_1(Z, z_0)$$

故 $g_*(f_*(\pi_1(X, x_0)))$ 为 $g_*(\pi_1(Y, y_0))$ 的正规子群. 由同伦提升引理知 $g_* : \pi_1(Y, y_0) \to \pi_1(Z, z_0)$ 为单射, 故 g_* 可视作 $\pi_1(Y, y_0) \to g_*(\pi_1(Y, y_0))$ 的同构, 因此 $f_*(\pi_1(X, x_0))$ 为 $\pi_1(Y, y_0)$ 的正规子群, 故 $f : X \to Y$ 为正规覆叠空间.

Prob 5.15

Use theorem (5.13) to show that the Möbius strip and the cylinder both have fundamental group $\mathbb{Z}$.

Theorem (5.13). If G acts as a group of homeomorphisms on a simply connected space X, and if each point $x \in X$ has a neighbourhood U which satisfies $U \cap g(U) = \emptyset$ for all $g \in G \setminus \{e\}$ then $\pi_1(X/G)$ is isomorphic to G.

Solution: 设 M 为 Möbius strip, 则 M 可由 $\mathbb{R} \times [-1,1]$ 商去 $\mathbb{Z}$ 的群作用得到, 其中

$$\mathbb{Z} \curvearrowright \mathbb{R} \times [-1,1], \quad (n,(x,y)) \mapsto (x+n, (-1)^n y)$$

注意到 $\mathbb{R} \times [-1,1]$ 为单连通空间, 且对任意 $0 \neq n \in \mathbb{Z}$, 其在 $\mathbb{R} \times [-1,1]$ 上的作用是同胚的. 对任意 $(x,y) \in \mathbb{R} \times [-1,1]$, 取其邻域 $U = \left(x - \frac{1}{2}, x + \frac{1}{2}\right) \times (y - \varepsilon, y + \varepsilon)$, 这里 ε 充分小使得 y 方向包含在 $[-1,1]$ 内 (当 $y = -1$ 或 1 时取单侧区间即可) , 则对任意 $0 \neq n \in \mathbb{Z}$, $U \cap n(U) = \emptyset$. 故由定理 (5.13) 可知

$$\pi_1(M) = \pi_1(\mathbb{R} \times [-1,1]/\mathbb{Z}) = \mathbb{Z}.$$

类似地, 设 C 为 cylinder, 则 C 可由 $\mathbb{R}^2$ 商去 $\mathbb{Z}$ 的群作用得到, 其中

$$\mathbb{Z} \curvearrowright \mathbb{R}^2, \quad (n,(x,y)) \mapsto (x+n, y)$$

同理可证此时定理 (5.13) 适用, 故

$$\pi_1(C) = \pi_1(\mathbb{R}^2/\mathbb{Z}) = \mathbb{Z}.$$

Prob 10.23

If X is a connected, locally path-connected, Hausdorff space, and if a finite group G of order n acts freely on X, show that X is an n-sheeted covering of X/G.

Proof: 设 $p : X \to X/G$ 为商映射, 我们证明过当 X/G 是群作用诱导的商空间时, p 为开映射, 且对每个开集 $U \subset X$, 有

$$p^{-1}(p(U)) = \bigcup_{g \in G} gU$$

下证 p 为开映射. Claim: 任取 $x \in X$, 存在其邻域 U 使得 $U \cap gU = \emptyset, \forall g \in G$.

这是因为, 固定 $x \in X$, 由 X 为 Hausdorff 空间, 对每个 $g \in G \setminus \{e\}$, 存在 x 的邻域 U_g 与 $g \cdot x$ 的邻域 V_g 使得 $U_g \cap V_g = \emptyset$. 注意对固定的 $g \in G \setminus \{e\}$, 左乘算子 L_g 为同胚, 于是取

$$U = \bigcap_{g \in G} \left(U_g \cap g^{-1}(V_g)\right)$$

则 U 为 x 的邻域, 且对 $g \in G \setminus \{e\}$, 有

$$gU = g\left(\bigcap_{h \in G} (U_h \cap h^{-1}(V_h))\right) = \bigcap_{h \in G} (gU_h \cap gh^{-1}(V_h)) \subset gU_g \cap V_g$$

而 $U \subset U_g$, 故由 $U_g \cap V_g = \emptyset$ 即知 $U \cap gU = \emptyset$. 因此 Claim 成立. 由此, 对任意 $[x] \in X/G$, 取 x 在 X 中的邻域 U s.t. $U \cap gU = \emptyset, \forall g \in G$, 则 $p(U)$ 为 X/G 中 $[x]$ 的邻域, 并且

$$p^{-1}(p(U)) = \bigcup_{g \in G} gU$$

若 $g_1U\cap g_2U\neq\emptyset$, 则存在 $x,y\in U$ s.t. $g_1x=g_2y\Longrightarrow x=g_1^{-1}g_2y\Longrightarrow U\cap(g_1^{-1}g_2)U\neq\emptyset$ 矛盾! 因此上述并为不交并.

固定 $g\in G$, 考察 $p|_{gU}:gU\to p(U)$. 对 $\forall[u]\in p(U), p(gu)=[gu]=[u]$, 于是 $p|_{gU}$ 为满射. 若 gu_1,gu_2 满足 $p(gu_1)=p(gu_2)$, 则

$$[gu_1]=[gu_2]\Longrightarrow\exists g'\in G\ \text{s.t.}\ gu_1=g'gu_2\Longrightarrow u_1=g^{-1}g'gu_2$$

而 $U\cap gU=\emptyset,\forall g\in G\setminus\{e\}$, 因此只能是 $g^{-1}g'g=e\Longrightarrow g'g=g\Longrightarrow gu_1=gu_2$, 故 $p|_{gU}$ 为单射. 再由 p 的连续性以及其为开映射可知 $p|_{gU}$ 为同胚.

综上, 对每个 $[x]\in X/G$, 存在其邻域 $p(U)$ 使得

$$p^{-1}(p(U))=\bigsqcup_{g\in G}gU,\quad p|_{gU}:gU\to p(U)\ \text{为同胚},\forall g\in G$$

因此 $p:X\to X/G$ 为覆叠映射.

由于 $|G|=n$ 且 G 作用自由, 因此

$$p^{-1}([x])=\{y\in X\mid[y]=[x]\}=\{gx\mid g\in G\}$$

若 $g_1x=g_2x$, 则 $g_2^{-1}g_1x=x\Longrightarrow g_2^{-1}g_2=e\Longrightarrow g_1=g_2$, 故 $\left|p^{-1}(x)\right|=|G|=n$, 因此 X 是 X/G 的层数为 n 的覆叠空间.

H-Prob 2.1.11

Show that if A is a retract of X then the map $H_n(A)\to H_n(X)$ induced by the inclusion $A\subset X$ is injective.

Proof: 设 $r:X\to A$ 满足 $r|_A=\mathrm{id}_A$, 考虑嵌入 $A\overset{i}{\hookrightarrow}X$, 于是 $r\circ i=\mathrm{id}_A$, 进而

$$r_*\circ i_*=(r\circ i)_*=(\mathrm{id}_A)_*=\mathrm{id}_{H_n(A)}$$

故 i_* 有左逆, 故为单射.

H-Prob 2.1.13

Verify that $f\simeq g$ implies $f_*=g_*$ for induced homomorphisms of reduced homology groups.

Proof: 课上已经证明, 对 $g_*,f_*:H_n(X)\to H_n(Y)$, 有 $f_*=g_*$. 这里需要对 reduced homology group $\tilde{H}_n(X)$ 进行证明. 当 $n>1$ 时, $\tilde{H}_n(X)=H_n(X)$, 故结论成立. 当 $n=0$ 时, 由定义可知

$$\cdots\overset{\partial_1}{\to}C_0(X)\overset{\varepsilon_X}{\to}\mathbb{Z}\to0,\quad\tilde{H}_0(X)=\operatorname{Ker}\varepsilon_X/\operatorname{Im}\partial_1$$

课上已证 $\operatorname{Im}\partial_1<\operatorname{Ker}\varepsilon_X<C_0(X)$, 这里 $<$ 表示子群. 于是

$$\tilde{H}_0(X)=\operatorname{Ker}\varepsilon_X/\operatorname{Im}\partial_1<C_0(X)/\operatorname{Im}\partial_1=H_0(X)$$

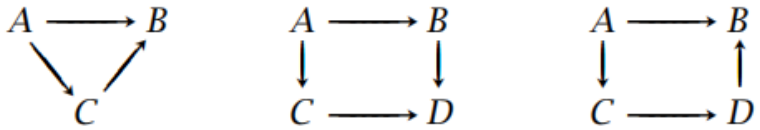
即 $\tilde{H}_0(X)$ 可视作 $H_0(X)$ 的子群. 用 $\tilde{f}_*,\tilde{g}_*$ 表示 reduced homology group 上的诱导同态, 于是

$$\tilde{f}_*=f_*|_{\tilde{H}_0(X)},\quad\tilde{g}_*=g_*|_{\tilde{H}_0(X)}$$

而 $f_*=g_*$, 故 $\tilde{f}_*=\tilde{g}_*$.

H-Prob 2.1.30

In each of the following commutative diagrams assume that all maps but one are isomorphisms. Show that the remaining map must be an isomorphism as well.



Proof: 第一张图: 设 $A \overset{f}{\to} B, A \overset{g}{\to} C \overset{h}{\to} B$, 于是 $f = h \circ g$, 其中任意两个为通过, 则可通过逆映射得到第三个映射为练个同构的复合映射, 故第三个映射也是同构映射.

第二张图: 设 $A \overset{f}{\to} B \overset{g}{\to} D, A \overset{h}{\to} C \overset{w}{\to} D$, 于是 $g \circ f = w \circ h$, 其中任意三个为同构映射, 则可通过逆映射得到第四个映射为三个同构的复合映射, 故第四个映射也是同构映射.

第三张图: 设 $A \overset{f}{\to} B, A \overset{g}{\to} C \overset{h}{\to} D \overset{w}{\to} B$, 于是 $w \circ h \circ g = f$, 其中任意三个为同构映射, 则可通过逆映射得到第四个映射为三个同构的复合映射, 故第四个映射也是同构映射.

H-Prob 2.1.12

Show that chain homotopy of chain maps is an equivalence relation.

Proof: 设 $C_\cdot, D_\cdot$ 为链复形.

(1) 正定性: 设 $f_\cdot : C_\cdot \to D_\cdot$ 为链映射, 考虑 $P_n : C_n \to D_{n+1}$ 为零映射, 于是

$$\partial_{n+1} \circ P_n + P_{n-1} \circ \partial_n = 0 = f_n - f_n$$

故 $f_\cdot$ 与其自身链同伦.

(2) 对称性: 设 $f_\cdot, g_\cdot : C_\cdot \to D_\cdot$ 为链映射, 且 $P_n : C_n \to D_{n+1}$ 满足

$$\partial_{n+1} \circ P_n + P_{n-1} \circ \partial_n = f_n - g_n, \forall n$$

于是考虑 $-P_n$, 则有

$$\partial_{n+1} \circ (-P_n) + (-P_{n-1}) \circ \partial_n = g_n - f_n, \forall n$$

故 $g_\cdot$ 与 $f_\cdot$ 链同伦.

(3) 传递性: 设 $f_\cdot, g_\cdot, h_\cdot : C_\cdot \to D_\cdot$ 为链映射, 且 $P_n, Q_n : C_n \to D_{n+1}$ 分别为 $f_\cdot$ 与 $g_\cdot$ 以及 $g_\cdot$ 与 $h_\cdot$ 之间的链同伦映射, 即

$$\partial_{n+1} \circ P_n + P_{n-1} \circ \partial_n = f_n - g_n, \forall n$$
$$\partial_{n+1} \circ Q_n + Q_{n-1} \circ \partial_n = g_n - h_n, \forall n$$

于是考虑 $P_n + Q_n : C_n \to D_{n+1}$, 有

$$\partial_{n+1} \circ (P_n + Q_n) + (P_{n-1} + Q_{n-1}) \circ \partial_n = f_n - h_n, \forall n$$

故 $f_\cdot$ 与 $h_\cdot$ 链同伦.

综上, 链同伦为链映射之间的等价关系.

H-Prob 2.1.15

For an exact sequence $A \to B \to C \to D \to E$ show that $C = 0$ iff the map $A \to B$ is surjective and $D \to E$ is injective. Hence for a pair of spaces (X, A), the inclusion $A \hookrightarrow X$ induces isomorphisms on all homology groups iff $H_n(X, A) = 0$ for all n.

Proof: 设 $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D \xrightarrow{w} E$, 先证明 $C = 0$ 当且仅当 f 为满射且 w 为单射.

$\Longrightarrow$: 当 $C = 0$ 时, 有 $\operatorname{Ker} g = B, \operatorname{Im} h = 0$, 由正合列的定义即知 $\operatorname{Im} f = \operatorname{Ker} g = B$, 故 f 为满射. 同理, 由 $\operatorname{Im} h = 0 = \operatorname{Ker} w$ 可知 w 为单射.

$\Longleftarrow$: 当 f 为满射且 w 为单射时, 有 $\operatorname{Im} f = B = \operatorname{Ker} g$, 故 $\operatorname{Im} g = 0 = \operatorname{Ker} h$. 而 w 为单射则给出 $\operatorname{Ker} w = 0 = \operatorname{Im} h$, 故 $\operatorname{Ker} h = \operatorname{Im} h = 0$, 即 $C = 0$.

考虑 (X, A), 其中 $A \subset X$, 则有短正合列 $0 \to C_\cdot(A) \xrightarrow{i_\cdot} C_\cdot(X) \xrightarrow{j_\cdot} C_\cdot(X, A) \to 0$, 其引出长正合列

$$\cdots \xrightarrow{\partial} H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X) \xrightarrow{j_*} H_{n-1}(X, A) \xrightarrow{\partial} \cdots$$

若每个 $H_n(X, A) = 0$, 则由上面证明的结论可知 $i_* : H_n(A) \to H_n(X)$ 为满射, 且 $i_* : H_{n-1}(A) \to H_{n-1}(X)$ 为单射, 对每个 n 应用结论即知 i_* 为同构. 反之, 若 $i_* : H_n(A) \to H_n(X)$ 为同构映射, 则其为满射且单射, 由上面证明的结论可知 $H_n(X, A) = 0$.

H-Prob 2.1.16

(a) Show that $H_0(X, A) = 0$ iff A meets each path-component of X.

(b) Show that $H_1(X, A) = 0$ iff $H_1(A) \to H_1(X)$ is surjective and each path-component of X contains at most one path-component of A.

Proof: (a) 考虑长正合列

$$\cdots \xrightarrow{j_*} H_1(X, A) \xrightarrow{\partial} H_0(A) \xrightarrow{i_*} H_0(X) \xrightarrow{j_*} H_0(X, A) \to 0$$

类似上一题可知 $H_0(X, A) = 0 \iff H_0(A) \xrightarrow{i_*} H_0(X)$ 为满射. 设 X 的路径分支为 $\{X_\alpha \mid \alpha \in J\}$, 下证 i_* 为满射当且仅当 A 与每个 X_α 相交.

$\Longrightarrow$: 设 i_* 为满射, 我们熟知

$$H_0(X) = \bigoplus_{\alpha \in J} \mathbb{Z}$$

考虑 X_α, 设其在 $H_0(X)$ 中的生成元为 $[a]$, 其中 $a : \Delta^0 \to X_\alpha$ 为点映射, 可视作 $a \in X_\alpha$. 由于 i_* 为满射, 因此存在 $[b] \in H_0(A)$ 使得 $i_*([b]) = [a]$, 这里 $b \in A$. 由 $i_*([b]) = [a]$ 可知 a, b 在 $H_0(X)$ 中只相差一个 $\mathrm{Im}(\partial_1)$ 的元素, 即 a, b 可被 X 中的一条道路连接, 故 $b \in X_\alpha$, 因此 $A \cap X_\alpha \neq \emptyset$.

$\Longleftarrow$: 设 A 与每个 X_α 相交. 对于 $H_0(X)$ 中的任意 $\mathbb{Z}$ 的生成元 $[a]$, 其中 $a : \Delta^0 \to X_\alpha$ 为点映射, 可视作 $a \in X_\alpha$. 由于 $A \cap X_\alpha \neq \emptyset$, 取 $\beta \in A \cap X_\alpha$, 由于 α, β 均在 X_α 中, 因此它们在 X 中可被一条道路连接, 故 $i_*([\beta]) = [\alpha]$. 因此 $i_*(H_0(A))$ 包含了 $H_0(X)$ 中所有生成元, 故 i_* 为满射.

(b) 考虑长正合列

$$\cdots \xrightarrow{\partial} H_1(A) \xrightarrow{i_*} H_1(X) \xrightarrow{j_*} H_1(X, A) \xrightarrow{\partial} H_0(A) \xrightarrow{i_*} H_0(X) \xrightarrow{j_*} H_0(X, A) \to 0$$

由上一题可知, $H_1(X, A) = 0 \iff i_* : H_1(A) \to H_1(X)$ 为满射且 $i_* : H_0(A) \to H_0(X)$ 为单射. 故只需证 $i_* : H_0(A) \to H_0(X)$ 为单射当且仅当 X 的每个路径分支最多含 A 的一个路径分支.

$\Longrightarrow$: 设 i_* 为单射, 反设存在某个 X_α 包含 A 的两个不同路径分支 A_1, A_2. 取 $a_1 \in A_1, a_2 \in A_2$, 则 $[a_1], [a_2] \in H_0(A)$, 且在 $H_0(A)$ 中 $[a_1] \neq [a_2]$. 而 a_1, a_2 在 X 中位于同一道路连通分支, 故 $i_*([a_1]) = i_*([a_2])$, 矛盾!

$\Longleftarrow$: 设 X 的每个路径分支最多包含 A 的一个路径分支. 设 $[a_1], [a_2] \in H_0(A)$ 满足 $i_*([a_1]) = i_*([a_2])$, $a_1, a_2 : \Delta^0 \to A$. 由 $i_*([a_1]) = i_*([a_2])$ 可知 a_1, a_2 在 X 中可被一条道路连接, 故二者位于某个 X_α. 由假设可知 X_α 最多包含 A 的一个路径分支, 故 a_1, a_2 在 A 中也位于同一个路径分支中, 因此它们在 $H_0(A)$ 中相等, 即 $[a_1] = [a_2]$. 综上, i_* 为单射.

H-Prob 2.1.18

Show that for the subspace $\mathbb{Q} \subset \mathbb{R}$, the relative homology group $H_1(\mathbb{R}, \mathbb{Q})$ is free abelian and find a basis.

Proof: 考虑长正合列

$$\cdots \xrightarrow{\partial} H_1(\mathbb{Q}) \xrightarrow{i_*} H_1(\mathbb{R}) \xrightarrow{j_*} H_1(\mathbb{R}, \mathbb{Q}) \xrightarrow{\partial} H_0(\mathbb{Q}) \xrightarrow{i_*} H_0(\mathbb{R}) \xrightarrow{j_*} H_0(\mathbb{R}, \mathbb{Q}) \to 0$$

由于 $\mathbb{R}$ 可缩, 因此 $H_1(\mathbb{R}) = 0$, 又由上一题可知 $H_0(\mathbb{R}, \mathbb{Q}) = 0$, 因此有短正合列

$$0 \to H_1(\mathbb{R}, \mathbb{Q}) \xrightarrow{\partial} H_0(\mathbb{Q}) \xrightarrow{i_*} H_0(\mathbb{R}) \to 0$$

其中 ∂ 为单射, i_* 为满射. 由此可知

$$H_1(\mathbb{R},\mathbb{Q}) \cong \operatorname{Ker} i_* = \operatorname{Ker}\left(H_0(\mathbb{Q}) \xrightarrow{i_*} H_0(\mathbb{R})\right)$$

我们有

$$H_0(\mathbb{Q}) = \bigoplus_{q\in\mathbb{Q}} \mathbb{Z}, \quad H_0(\mathbb{R}) = \mathbb{Z}$$

设 $q_0 \in \mathbb{Q}$, 将 q_0 视作 $\Delta^0 \to q_0$ 的点映射, 于是 $[q_0] \in H_0(\mathbb{Q})$ 为某个 $\mathbb{Z}$ 的生成元, 则 $i_*([q_0])$ 为将 q_0 看做 $\mathbb{R}$ 中的点映射, 故对应 $H_0(\mathbb{R})$ 中的生成元. 故 $\forall q \in \mathbb{Q}, i_*([q]) = 1$. 因此 i_* 形如

$$i_*\left(\sum_{i=1}^n n_i[q_i]\right) = \sum_{i=1}^n n_i$$

于是

$$\operatorname{Ker} i_* = \left\{ \sum_{i=1}^n a_i[q_i] \mid \sum_{i=1}^n a_i = 0 \right\}$$

下给出其的一组基: 固定 $q_0 \in \mathbb{Q}$, 考虑所有形如 $[q]-[q_0], \forall q \in \mathbb{Q}\setminus\{q_0\}$ 的元素, 断言:

$$B = \{[q]-[q_0] \mid q \in \mathbb{Q}\setminus\{q_0\}\}$$

为 $\operatorname{Ker} i_* = H_0(\mathbb{R},\mathbb{Q})$ 的一组基. 任取 $\sum_{i=1}^n a_i[q_i] \in \operatorname{Ker} i_*$, 则有 $\sum_{i=1}^n a_i = 0$, 故

$$\sum_{i=1}^n a_i[q_i] = \sum_{i=1}^n a_i([q_i]-[q_0]).$$

因此 $\operatorname{span}\{B\} = \operatorname{Ker} i_*$. 下证线性无关性: 设

$$\sum_{i=1}^n b_i([q_i]-[q_0]) = 0 \Longrightarrow \sum_{i=1}^n b_i[q_i] + \left(-\sum_{i=1}^n b_i\right)[q_0] = 0$$

其中每个 q_i 互不相同, 因此 $b_i = 0, \forall i$.

综上, B 为 $H_1(\mathbb{R},\mathbb{Q})$ 的一组基, 故 $H_1(\mathbb{R},\mathbb{Q})$ 为自由阿贝尔群.

H-Prob 2.1.27

Let $f : (X, A) \to (Y, B)$ be a map such that both $f : X \to Y$ and the restriction $f : A \to B$ are homotopy equivalences.

(a) Show that $f_* : H_n(X, A) \to H_n(Y, B)$ is an isomorphism for all n.

(b) For the case of the inclusion $f : (D^n, S^{n-1}) \hookrightarrow (D^n, D^n - \{0\})$, show that f is not a homotopy equivalence of pairs, i.e. there is no $g : (D^n, D^n - \{0\}) \to (D^n, S^{n-1})$ such that fg and gf are homotopic to the identity through maps of pairs.

[Observe that a homotopy equivalence of pairs $(X, A) \to (Y, B)$ is also a homotopy equivalence for the pairs obtained by replacing A and B by their closures.]

Proof: (a) 由于 $f : X \to Y$ 及其限制 $f : A \to B$ 均为同伦等价映射, 因此二者诱导的同态 $f_* : H_n(X) \to H_n(Y), f_* : H_n(A) \to H_n(B)$ 均为同构映射. 考虑如下交换图:

$$\begin{pmatrix} \cdots \to & H_n(A) & \xrightarrow{i_*} & H_n(X) & \xrightarrow{j_*} & H_n(X,A) & \xrightarrow{\partial} & H_{n-1}(A) & \xrightarrow{i_*} & H_{n-1}(X) & \to \cdots \\ & \downarrow f_* & \circlearrowright & \downarrow f_* & \circlearrowright & \downarrow f_* & \circlearrowright & \downarrow f_* & \circlearrowright & \downarrow f_* & \\ \cdots \to & H_n(B) & \xrightarrow{i_*} & H_n(Y) & \xrightarrow{j_*} & H_n(Y,B) & \xrightarrow{\partial} & H_{n-1}(B) & \xrightarrow{i_*} & H_{n-1}(Y) & \to \cdots \end{pmatrix}$$

注意上图中除了中间的 $f_*: H_n(X,A) \to H_n(Y,B)$ 之外, 其余同态均为同构映射, 因此由 *Five Lemma* 可知 $f_*: H_n(X,A) \to H_n(Y,B)$ 亦为同构映射.

(b) 先证明: 若 $f: (X,A) \to (Y,B)$ 为 pairs 之间的同伦等价映射, 则 $f: \left(X, \overline{A}\right) \to \left(Y, \overline{B}\right)$ 也是 pairs 之间的同伦等价映射: 设 $g: (Y,B) \to (X,A)$ 为 f 的同伦逆映射, 由定义, 存在同伦

$$H: X \times I \to X \text{ s.t. } H(x,0) = g \circ f(x), H(x,1) = \mathrm{id}_X(x), \forall x \in X$$

$$H_t \coloneqq H|_{X \times \{t\}}: X \to Y \text{ 满足 } H_t(A) \subset A, \forall t \in I$$

并且也存在 $K: Y \times I \to Y$ 满足类似条件. 我们证明过任意连续映射具有性质 $h\left(\overline{A}\right) \subset \overline{h(A)}$, 因此 f, g 可分别视作 $f: \left(X, \overline{A}\right) \to \left(Y, \overline{B}\right), g: \left(Y, \overline{B}\right) \to \left(X, \overline{A}\right)$. 进一步, 对 H 有

$$H_t\left(\overline{A}\right) \subset \overline{H_t(A)} \subset \overline{A}, \forall t \in I$$

同理, $K_t\left(\overline{B}\right) \subset \overline{B}$, 因此若将 f, g 分别视作 $f: \left(X, \overline{A}\right) \to \left(Y, \overline{B}\right), g: \left(Y, \overline{B}\right) \to \left(X, \overline{A}\right)$, 则 f, g 仍然互为同伦逆映射, 故 $f: \left(X, \overline{A}\right) \to \left(Y, \overline{B}\right)$ 为 pairs 之间的同伦等价映射.

回到本题, 反设 $f: (D^n, S^{n-1}) \hookrightarrow (D^n, D^n - \{0\})$ 为 pairs 之间的同伦等价映射, 则由上面的结论可知 $f: (D^n, S^{n-1}) \hookrightarrow (D^n, D^n)$ 也是 pairs 之间的同伦等价映射. 从而由 (a) 可知

$$f_*: H_n(D^n, S^{n-1}) \to H_n(D^n, D^n)$$

为同构映射. 但是我们熟知

$$H_n(D^n, S^{n-1}) \cong \mathbb{Z}, \quad H_n(D^n, D^n) = 0$$

因此得到矛盾, 故 $f: (D^n, S^{n-1}) \hookrightarrow (D^n, D^n - \{0\})$ 不是 pairs 之间的同伦等价映射.

H-Prob 2.1.26

Show that $H_1(X, A)$ is not isomorphic to $\tilde{H}_1(X/A)$ if $X = [0, 1]$ and A is the sequence $1, \frac{1}{2}, \frac{1}{3}, \cdots$ together with its limit 0. [See Example 1.25.]

Proof: 我们用到以下基本事实:

对任意拓扑空间 X, 一维同调群 $H_1(X)$ 是 $\pi_1(X)$ 的阿贝尔化 *(abelianization)*.

一方面, 由 Example 1.25 可知, 存在满射

$$f : \pi_1(X/A) \to \prod_{i=1}^{\infty} \mathbb{Z}$$

由于 $\prod_{i=1}^{\infty} \mathbb{Z}$ 为阿贝尔群, 故由上述基本事实可知, f 诱导出 $H_1(X/A)$ 到 $\prod_{i=1}^{\infty} \mathbb{Z}$ 的满射, 因此

$$|H_1(X/A)| \geq \left| \prod_{i=1}^{\infty} \mathbb{Z} \right| = \left| \mathbb{Z}^{\mathbb{N}} \right| = \aleph_1$$

即 $H_1(X/A)$ 不可数.

另一方面, 由于 X 可缩, 我们有 $H_1(X) = H_0(X) = 0$, 因此长正合列

$$\cdots \xrightarrow{\partial} H_1(A) \xrightarrow{i_*} H_1(X) \xrightarrow{j_*} H_1(X, A) \xrightarrow{\partial} H_0(A) \xrightarrow{i_*} H_0(X) \xrightarrow{j_*} H_0(X, A) \to 0$$

可以化简为短正合列

$$0 \to H_1(X, A) \xrightarrow{\partial} H_0(A) \to 0$$

因此 $H_1(X, A) \cong H_0(A) = \bigoplus_{a \in A} \mathbb{Z} = \bigoplus_{i=1}^{\infty} \mathbb{Z}$. 注意直和 $\bigoplus_{i=1}^{\infty} \mathbb{Z}$ 可数, 故 $H_1(X, A)$ 可数.

综上, $H_1(X, A)$ 与 $\tilde{H}_1(X/A)$ 不同构.

H-Prob 2.2.2

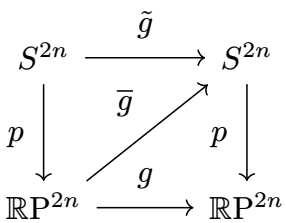
Given a map $f : S^{2n} \to S^{2n}$, show that there is some point $x \in S^{2n}$ with either $f(x) = -x$ or $f(x) = x$. Deduce that every map $\mathbb{R}\mathrm{P}^{2n} \to \mathbb{R}\mathrm{P}^{2n}$ has a fixed point. Construct maps $\mathbb{R}\mathrm{P}^{2n-1} \to \mathbb{R}\mathrm{P}^{2n-1}$ without fixed points from linear transformations $\mathbb{R}^{2n} \to \mathbb{R}^{2n}$ without eigenvectors.

Proof: 反设 $\forall x \in S^{2n}$, 均有 $f(x) \neq x$ 且 $f(x) \neq -x$, 于是 $(1-t)f(x) + tx$ 与 $(1-t)f(x) - tx$ 均不通过原点, 因此定义 $F, G : S^{2n} \times I \to S^{2n}$ 为

$$F(x,t) = \frac{(1-t)f(x) + tx}{\|(1-t)f(x) + tx\|} \quad , G(x,t) = \frac{(1-t)f(x) - tx}{\|(1-t)f(x) - tx\|}$$

可见 $f \underset{F}{\simeq} \mathrm{id}_{S^{2n}}$, $f \underset{G}{\simeq} -\mathrm{id}_{S^{2n}}$, 进而 $\mathrm{id}_{S^{2n}} \simeq -\mathrm{id}_{S^{2n}}$, 于是二者的度应相同, 但 $\deg(-\mathrm{id}_{S^{2n}}) = (-1)^{2n+1} = -1$, 矛盾! 因此存在 $x \in S^{2n}$ 使得 $f(x) = x$ 或 $f(x) = -x$.

设 $g : \mathbb{R}\mathrm{P}^{2n} \to \mathbb{R}\mathrm{P}^{2n}$. 注意到 $p : S^{2n} \to \mathbb{R}\mathrm{P}^{2n}, x \mapsto [x]$ 为商映射, 也是覆叠映射, 因此存在 g 的提升 $\overline{g} : \mathbb{R}\mathrm{P}^{2n} \to S^{2n}$ 使得 $p \circ \overline{g} = g$, 进而定义 $\tilde{g} : S^{2n} \to S^{2n}$ 为 $\tilde{g} = \overline{g} \circ p$, 于是有交换图



即 $p\circ\tilde{g}=g\circ p$. 由上面结论可知, 存在 $x\in S^{2n}$ 使得 $\tilde{g}(x)=x$ 或 $\tilde{g}(x)=-x$, 由 $p(x)=p(-x)$ 即知 $g([x])=p(\tilde{g}(x))=[x]$, 故 g 有不动点.

最后, 我们给出一个 $\mathbb{R}\mathrm{P}^{2n-1}\to\mathbb{R}\mathrm{P}^{2n-1}$ 的没有不动点的映射. 设 $T:\mathbb{R}^{2n}\to\mathbb{R}^{2n}$ 为

$$T(x_1,x_2,\cdots,x_{2n})=(-x_{2n},x_1,x_2,\cdots,x_{2n-1})$$

于是 $T^{2n}=-\mathrm{id}$, 进而 T 的特征值满足 $\lambda^{2n}+1=0$, 故 T 无实特征值, 因此无实特征向量. 考虑将 $\mathbb{R}^{2n}\setminus\{0\}$ 中同一条直线上的点识别为一点得到的 $\mathbb{R}\mathrm{P}^{2n-1}$. 由于 T 线性, 因此 T 可以诱导出 $\tilde{T}:\mathbb{R}\mathrm{P}^{2n-1}\to\mathbb{R}\mathrm{P}^{2n-1}$. 若存在 $[x]\in\mathbb{R}\mathrm{P}^{2n-1}$ 使得 $\tilde{T}([x])=[x]$, 则 $[T(x)]=[x]$, 即 $T(x)=\lambda x$ 对某个 $\lambda\in\mathbb{R}$ 成立, 矛盾! 因此 $\tilde{T}$ 无不动点.

H-Prob 2.2.5

Show that any two reflections of S^n across different n-dimensional hyperplanes are homotopic, in fact homotopic through reflections. [The linear algebra formula for a reflection in terms of inner products may be helpful.]

Proof: 设 P 为某个超平面, 对应的法向量为 $k\in S^n$, 记 S^n 上关于 P 的反射为 $f_k:S^n\to S^n$, 则

$$f_k(x)=x-2\langle x,k\rangle k,\forall x\in S^n$$

设 f_a,f_b 为任意两个反设. 考虑 $\gamma:[0,1]\to S^n$ 为连接 a,b 的道路, 定义 $H:S^n\times I\to S^n$ 为

$$H(x,t)=f_{\gamma(t)}(x)$$

于是 $H(x,t)$ 给出了 f_a 与 f_b 间的同伦, 且对固定的 $t\in I$, $H_t=f_{\gamma(t)}$ 为关于某个超平面的反射.

H-Prob 2.2.6

Show that every map $S^n\to S^n$ can be homotoped to a map which has a fixed point if $n>0$.

Proof: 先证明一个简单的引理:

lemma: 当 n 为奇数时, S^n 上的恒等映射与对径点映射同伦.

proof: 记旋转矩阵为

$$R(\theta)=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$$

于是定义 $n+1$ 阶矩阵为

$$A_t:=\begin{pmatrix}R(\pi t) & & & \\ & R(\pi t) & & \\ & & \ddots & \\ & & & R(\pi t)\end{pmatrix},\quad t\in[0,1]$$

定义 $H:S^n\times I\to S^n$ 为

$$H(x,t)=\frac{A_tx}{\|A_tx\|}$$

则 $H(x,0)=x,H(x,1)=-x$, H 给出了恒等映射与对径点映射之间的同伦.

回到本题: 设 $f:S^n\to S^n$, 不妨设 f 没有不动点, 类似之前构造可知此时 f 与对径点映射同伦. 当 n 为奇数时, 由引理即知 $f\simeq-\mathrm{id}_{S^n}\simeq\mathrm{id}_{S^n}$, 且 id_{S^n} 有不动点.

当 n 为偶数时, 设 $n=2m$, 考虑 $H: S^{2m}\times I\to S^{2m}$ 为

$$\begin{aligned}H((x_1,\cdots,x_{2m+1}),t)=(&x_1\cos\pi t-x_2\sin\pi t,x_1\sin\pi t+x_2\cos\pi t,\cdots,\\&x_{2m-1}\cos\pi t-x_{2m}\sin\pi t,x_{2m-1}\sin\pi t+x_{2m}\cos\pi t,-x_{2m+1})\end{aligned}$$

于是 $H(x,0)=g$, $H(x,1)=-x$, 其中 g 为对 S^{2m} 上最后一个坐标的反射. 由此可知

$$f\simeq -\mathrm{id}_{S^{2m}}\simeq g$$

并且 g 有不动点 $(1,0,\cdots,0)$, 因此 f 可同伦为有不动点的映射.

Prob 9.3

If the degree of $f: S^n\to S^n$ is not $+1$, show that f must map some point to its antipode.

Proof: 反设 $\forall x\in S^n$, $f(x)\neq -x$, 则 $(1-t)f(x)+tx$ 不通过原点. 定义 $H: S^n\times I\to S^n$ 为

$$H(x,t)=\frac{(1-t)f(x)+tx}{\|(1-t)f(x)+tx\|}$$

则 $f\underset{H}{\simeq}\mathrm{id}_{S^n}$, 进而 $\deg(f)=1$, 矛盾!

Prob 9.4

Show that the antipodal map of the circle is homotopic to the identity.

Proof: 由 Hatcher Prob 2.2.6 中使用的引理即知.

Prob 9.10

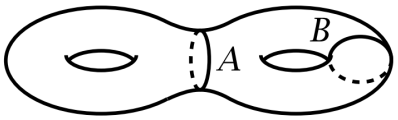
If $f: S^n\to S^n$ is a map, and if n is even, show that f^2 must have a fixed point. Even better, prove that either f has a fixed point, or it sends some point to its antipode.

Proof: 若 f^2 没有不动点, 则 $H(x,t)=((1-t)f^2(x)-tx)/\|(1-t)f^2(x)-tx\|$ 给出了 f^2 与 $-\mathrm{id}_{S^n}$ 之间的同伦, 进而 $\deg(f^2)=\deg(-\mathrm{id}_{S^n})=(-1)^{n+1}=-1$. 而 $\deg(f^2)=(\deg(f))^2\geq 0$, 因此矛盾. 后一条结论在 Hatcher Prob 2.2.2 中已证明.

H-Prob 2.1.17

(a) Compute the homology groups $H_n(X, A)$ when X is S^2 or $S^1 \times S^1$ and A is a finite set of points in X.

(b) Compute the groups $H_n(X, A)$ and $H_n(X, B)$ for X a closed orientable surface of genus two with A and B the circles shown. [What are X/A and X/B ?]



Solution: (a) 当 $X = S^2$ 时, 设 $A = \{x_1, x_2, \cdots, x_k\}$, 易知 (X, A) 是好对, 因此 $H_n(X, A) = \tilde{H}_n(X/A)$. 由 2.2.31, 我们有 $X/A \cong S^2 \vee S^1_1 \vee \cdots \vee S^1_{k-1}$ ($k = 1$ 时即 $X/A = X$), 因此

$$\begin{aligned} H_n(X/A) \cong \tilde{H}_n(S^2 \vee S^1_1 \vee \cdots \vee S^1_{k-1}) &= \tilde{H}_n(S^2) \oplus \tilde{H}_n(S^1) \oplus \cdots \oplus \tilde{H}_n(S^1) \\ &= \begin{cases} \mathbb{Z}, & n = 2 \\ \mathbb{Z}^{k-1}, & n = 1 \\ 0, & \text{else} \end{cases} \end{aligned}$$

当 $X = S^1 \times S^1 = T^2$ 时, (X, A) 仍为好对. 我们熟知

$$\tilde{H}_n(X) = \begin{cases} \mathbb{Z}, & n = 2 \\ \mathbb{Z}^2, & n = 1 \\ 0, & \text{else} \end{cases}$$

因此由 $X/A \cong T^2 \vee S^1_1 \vee \cdots \vee S^1_{k-1}$ 可知

$$\begin{aligned} H_n(X, A) \cong \tilde{H}_n(X/A) &= \tilde{H}_n(T^2) \oplus \tilde{H}_n(S^1) \oplus \cdots \oplus \tilde{H}_n(S^1) \\ &= \begin{cases} \mathbb{Z}, & n = 2 \\ \mathbb{Z}^{k+1}, & n = 1 \\ 0, & \text{else} \end{cases} \end{aligned}$$

(b) 由几何结构可知 $X/A = T^2 \vee T^2$, 且 (X, A) 为好对, 因此

$$H_n(X, A) = \tilde{H}_n(X/A) = \tilde{H}_n(T^2) \oplus \tilde{H}_n(T^2) = \begin{cases} \mathbb{Z}^2, & n = 2 \\ \mathbb{Z}^4, & n = 1 \\ 0, & \text{else} \end{cases}$$

由几何结构可知 $X/B = T^2 \vee S^1$, 且 (X, B) 为好对, 因此

$$H_n(X, B) = \tilde{H}_n(X/B) = \tilde{H}_n(T^2) \oplus \tilde{H}_n(S^1) = \begin{cases} \mathbb{Z}, & n = 2 \\ \mathbb{Z}^3, & n = 1 \\ 0, & \text{else} \end{cases}$$

H-Prob 2.1.29

Show that $S^1 \times S^1$ and $S^1 \vee S^1 \vee S^2$ have isomorphic homology groups in all dimensions, but their universal covering spaces do not.

Proof: 我们有

$$H_n(S_1 \times S_1) = \begin{cases} \mathbb{Z}, & n = 0, 2 \\ \mathbb{Z}^2, & n = 1 \\ 0, & \text{else} \end{cases}$$

当 $n=0$ 时, 显然 $H_0(S^1 \vee S^1 \vee S^2)=\mathbb{Z}$. 当 $n \geq 1$ 时, 有

$$H_n(S^1 \vee S^1 \vee S^2)=H_n(S^1) \oplus H_n(S^1) \oplus H_n(S^2)=\begin{cases}\mathbb{Z}, & n=2 \\ \mathbb{Z}^2, & n=1 \\ 0, & \text{else}\end{cases}$$

可见 $S^1 \times S^1$ 与 $S^1 \vee S^1 \vee S^2$ 在各维度上具有相同的同调群.

$S^1 \times S^1$ 的万有覆叠空间为 $\mathbb{R}^2$, $H_n(\mathbb{R}^2)=0, \forall n \geq 1$.

$S^1 \vee S^1 \vee S^2$ 的万有覆叠空间是: 在 $S^1 \vee S^1$ 的万有覆叠空间的每个顶点处 attach 一个 S^2 得到的空间, 记这个空间为 X, 于是 $X=X^2$ 为 2 为胞腔复形, 其一维骨架 X^1 即为 $S^1 \vee S^1$ 的万有覆叠空间. 我们有正合列

$$0 \to H_2(X^2) \to H_2(X^2, X^1) \to H_1(X^1) \to 0$$

注意 X^1 可缩, 因此 $H_1(X^1)=0$, 从而上述正合列给出 $H_2(X^2) \cong H_2(X^2, X^1)$, 进而

$$H_2(X^2)=H_2(X^2, X^1)=\tilde{H}_2(X^2/X^1)=\tilde{H}_2\left(\bigvee_\alpha S^1_\alpha\right) \neq 0$$

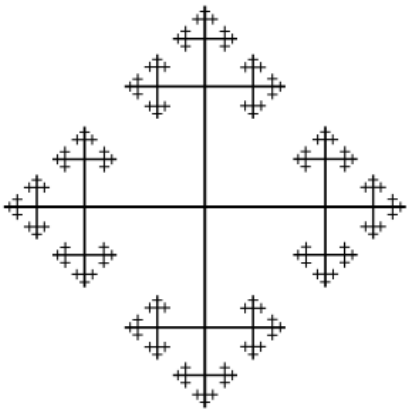


Figure 15: $S^1 \vee S^1$ 的万有覆叠

综上, $S^1 \times S^1$ 与 $S^1 \vee S^1 \vee S^2$ 的万有覆叠空间的同调群不同.

H-Prob 2.2.9

Compute the homology groups of the following 2-complexes:

(a) The quotient of S^2 obtained by identifying north and south poles to a point.

(b) $S^1 \times (S^1 \vee S^1)$.

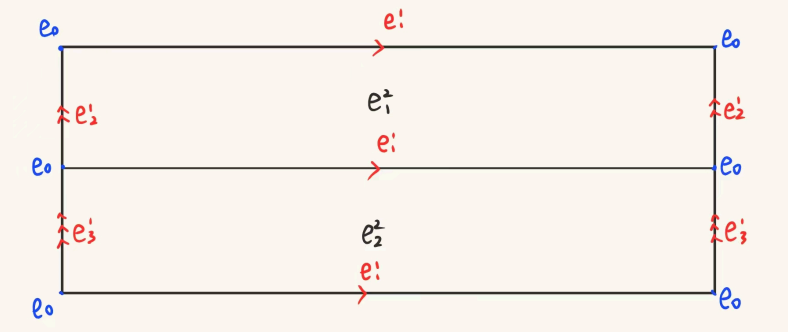
(c) The space obtained from D^2 by first deleting the interiors of two disjoint subdisks in the interior of D^2 and then identifying all three resulting boundary circles together via homeomorphisms preserving clockwise orientations of these circles.

(d) The quotient space of $S^1 \times S^1$ obtained by identifying points in the circle $S^1 \times \{x_0\}$ that differ by $2\pi/m$ rotation and identifying points in the circle $\{x_0\} \times S^1$ that differ by $2\pi/n$ rotation.

Solution: (a) 题中空间 X 同伦等价于 $S^2 \vee S^1$, 因此

$$H_n(X)=H_n(S^2 \vee S^1)=H_n(S^2) \oplus H_n(S^1)\ (n \geq 1)=\begin{cases}\mathbb{Z}, n=0,1,2 \\ 0, \text{ else}\end{cases}.$$

(b) $X=S^1 \times (S^1 \vee S^2)$ 可以视作如下胞腔复形: $X=e^0 \cup e_1^1 \cup e_2^1 \cup e_3^1 \cup e_1^2 \cup e_2^2$.



我们有 $d_1(e_i^1)=e^0-e^0=0, i=1,2,3$, 因此 $d_1=0$.

$d_2(e_i^2)=d_{i1}e_1^1+d_{i2}e_2^1+d_{i3}e_3^1, i=1,2$. 其中 $d_{ij}=\deg\Delta_{ij}$,

$$\Delta_{ij}:S^1\to S^1$$

从图中的粘合方式, 可见所有 d_{ij} 均为 0, 因此 $d_2=0$. 从而我们有胞腔同调链复形

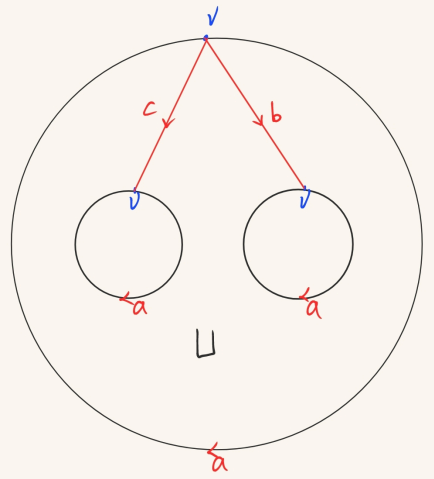
$$0\to\mathbb{Z}e_1^2\oplus\mathbb{Z}e_2^2\xrightarrow{d_2=0}\mathbb{Z}e_1^1\oplus\mathbb{Z}e_2^1\oplus\mathbb{Z}e_3^1\xrightarrow{d_1=0}\mathbb{Z}e^0\to 0$$

因此可得 X 的(胞腔)同调群为

$$H_0(X)=\mathbb{Z},\quad H_1(X)=\mathbb{Z}^3,\quad H_2(X)=\mathbb{Z}^2,\quad H_n(X)=0,\forall n\geq 3.$$

(c) 为使 X 成为胞腔复形, 在原先被粘合的三个圆 (a) 的基础上连出 b,c 作为 1-cell, 于是

$$X=e_v^0\cup e_a^1\cup e_b^1\cup e_c^1\cup e_U^2$$



于是 $d_1(e_i^1)=e_v^0-e_v^0=0, i=a,b,c$, 因此 $d_1=0$.

对于 d_2, 注意到沿 U 的边界走一圈, 对应在 e_a^1,e_b^1,e_c^1 上的走法为 $ca^{-1}c^{-1}ba^{-1}b^{-1}a$, 因此设 $d_2(e_U^2)=d_ae_a^1+d_be_b^1+d_ce_c^1$, 则

$$d_a=-1,\quad d_b=0,\quad d_c=0$$

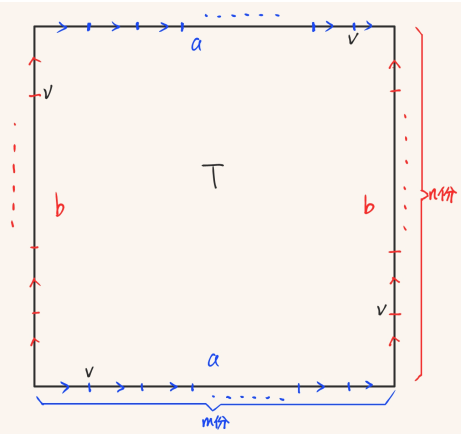
即 $d_2(e_U^2)=-e_a^1$. 于是我们有胞腔同调链复形

$$0\to\mathbb{Z}e_U^2\xrightarrow{d_2}\mathbb{Z}e_a^1\oplus\mathbb{Z}e_b^1\oplus\mathbb{Z}e_c^1\xrightarrow{d_1=0}\mathbb{Z}e_v^0\to 0$$

因此可得 X 的同调群为

$$H_0(X)=\mathbb{Z},\quad H_1(X)=\mathbb{Z}^2,\quad H_n(X)=0,\forall n\geq 2.$$

(d) 考虑 X 的矩形粘合, 题中操作等价于对 X 的上下两边进行 m 等分, 左右两边进行 n 等分, 并按照原先的对应的关系进行粘合, 于是 X 可视作图中所示的胞腔复形.



X 具有 1 个 0-cell e_v^0, 2 个 1-cell e_a^1,e_b^1 和 1 个 2-cell e_T^2. 因此

$$X=e_v^0\cup e_a^1\cup e_b^1\cup e_T^2$$

易见 $d_1=0$. 对于 d_2, 易见沿 T 的边界走一圈, 对应在 e_a^1,e_b^1 上的走法为 $a^mb^{-n}a^{-m}b^n$, 因此设 $d_2(e_T^2)=d_ae_a^1+d_be_b^1$, 则可知 $d_a=0,d_b=0$, 即 $d_2(e_T^2)=0$. 于是我们有胞腔同调链复形

$$0\to\mathbb{Z}\xrightarrow{d_2=0}\mathbb{Z}^2\xrightarrow{d_1=0}\mathbb{Z}\to 0$$

因此可得 X 的同调群为

$$H_0(X)=\mathbb{Z},\quad H_1(X)=\mathbb{Z}^2,H_2(X)=\mathbb{Z},\quad H_n(X)=0,\forall n\geq 3.$$

H-Prob 2.2.10

Let X be the quotient space of S^2 under the identifications $x\sim -x$ for x in the equator S^1. Compute $H_i(X)$. Do the same for S^3 with antipodal points of the equatorial $S^2\subset S^3$ identified.

Solution: (a) 在 S^2 中的 S^1 进行对径点粘合, 于是这个 S^1 变为 $\mathbb{R}\mathrm{P}^1$, 而 S^2 的上下球面分别作为 e_1^2, e_2^2 粘合到 $\mathbb{R}\mathrm{P}^1 = e^0 \cup e^1$ 上, 并且这两个半球面的粘合方式与 $\mathbb{R}\mathrm{P}^2 = \mathbb{R}\mathrm{P}^1 \cup e^2$ 的粘合方式一致, 因此我们得到

$$X = e^0 \cup e^1 \cup e_1^2 \cup e_2^2$$

并且对 $\mathbb{R}\mathrm{P}^n$ 的计算给出:

$$d_k(e_i^k) = (1 + (-1)^k)e^{k-1}$$

因此 $d_1 = 0, d_2(e_i^2) = 2e^1, i = 1, 2$. 于是 X 的胞腔同调链复形为

$$0 \to \mathbb{Z}e_1^2 \oplus \mathbb{Z}e_2^2 \xrightarrow{d_2=2} \mathbb{Z}e^1 \xrightarrow{d_1=0} \mathbb{Z}e^0 \to 0, \quad d_2(a+b) = 2(a+b), \forall a, b \in \mathbb{Z}$$

因此可得 X 的同调群为

$$H_0(X) = \mathbb{Z}, \quad H_1(X) = \mathbb{Z}/\operatorname{Im} d_2 = \mathbb{Z}/2\mathbb{Z} = \mathbb{Z}_2,$$
$$H_2(X) = \operatorname{Ker} d_2 = \{(a, -a) |\ a \in \mathbb{Z}\} \cong \mathbb{Z}, \quad H_n(X) = 0, \forall n \geq 3$$

(b) 对于 S^3 中将 S^2 进行对径点粘合, 类似地, 我们得到

$$X = e^0 \cup e^1 \cup e^2 \cup e_1^3 \cup e_2^3$$

其中 $d_1 = 0, d_2(e^2) = 2e^1, d_3(e_i^3) = 0$, 因此 X 的胞腔同调链复形为

$$0 \to \mathbb{Z}e_1^3 \oplus \mathbb{Z}e_2^3 \xrightarrow{d_3=0} \mathbb{Z}e^2 \xrightarrow{d_2=2} \mathbb{Z}e^1 \xrightarrow{d_1=0} \mathbb{Z}e^0 \to 0$$

因此可得 X 的同调群为

$$H_0(X) = \mathbb{Z}, \quad H_1(X) = \mathbb{Z}_2, \quad H_2(X) = 0, \quad H_3(X) = \mathbb{Z}^2, \quad H_n(X) = 0, \forall n \geq 4$$

H-Prob 2.2.12

Show that the quotient map $S^1 \times S^1 \to S^2$ collapsing the subspace $S^1 \vee S^1$ to a point is not null-homotopic by showing that it induces an isomorphism on H_2. On the other hand, show via covering spaces that any map $S^2 \to S^1 \times S^1$ is nullhomotopic.

Proof: 注意 $(S^1 \times S^1, S^1 \vee S^1)$ 为好对, 故由相对同调群的长正合列以及

$$H_2(S^1 \times S^1, S^1 \vee S^1) = \tilde{H}_2((S^1 \times S^1)/(S^1 \vee S^1)) = \tilde{H}_2(S^2) = \mathbb{Z}$$
$$H_1(S^1 \times S^1, S^1 \vee S^1) = \tilde{H}_1((S^1 \times S^1)/(S^1 \vee S^1)) = \tilde{H}_1(S^2) = 0$$

可得

$$\cdots \to H_2(S^1 \vee S^1) \to H_2(S^1 \times S^1) \xrightarrow{\pi_*} \tilde{H}_2(S^2) \to H_1(S^1 \vee S^1) \to H_1(S^1 \times S^1) \to 0$$

其中 π_* 是我们关心的映射. 利用熟知的 $S^1 \times S^1$ 与 $S^1 \vee S^1$ 的同调群, 上述正合列化为

$$0 \to 0 \to \mathbb{Z} \xrightarrow{\pi_*} \mathbb{Z} \to \mathbb{Z}^2 \to \mathbb{Z}^2 \to 0$$

于是 π_* 为单射, 注意 π_* 为群同态, 故有 $\operatorname{Im} \pi_* = \mathbb{Z}/\operatorname{Ker} \pi_* = \mathbb{Z}$, 因此 π_* 为同构映射. 由此可知, 若 π 零同伦, 即 π 与某个常数映射 e_p 同伦, 则有 $\pi_* = e_{p*}$, 而后者根据定义只能是零映射, 矛盾! 因此 π 不是零同伦映射.

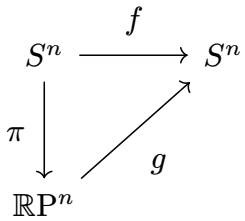
设 $f: S^2 \to S^1 \times S^1$, 由 $S^1 \times S^1$ 的万有覆叠空间为 $\mathbb{R}^2$ 可知, 存在 f 的提升 $\overline{f}: S^2 \to \mathbb{R}^2$. 由于 $\mathbb{R}^2$ 可缩, 故 $\overline{f}$ 零同伦, 进而 $f = p \circ \overline{f}$ 也是零同伦映射.

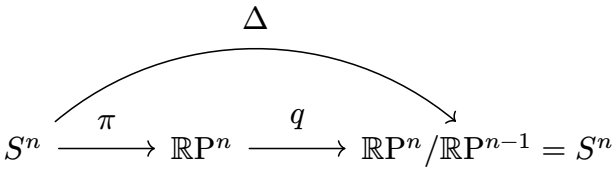
H-Prob 2.2.14

A map $f: S^n \to S^n$ satisfying $f(x) = f(-x)$ for all x is called an *even map*. Show that an even map $S^n \to S^n$ must have even degree, and that the degree must in fact be zero when n is even. When n is odd, show there exist even maps of any given even degree.

[Hints: If f is even, it factors as a composition $S^n \to \mathbb{R}\mathrm{P}^n \to S^n$. Using the calculation of $H_n(\mathbb{R}\mathrm{P}^n)$ in the text, show that the induced map $H_n(S^n) \to H_n(\mathbb{R}\mathrm{P}^n)$ sends a generator to twice a generator when n is odd. It may be helpful to show that the quotient map $\mathbb{R}\mathrm{P}^n \to \mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^{n-1}$ induces an isomorphism on H_n when n is odd.]

Proof: 注意 $f = f \circ (-\mathrm{id}_{S^n})$, 因此 $\deg f = (-1)^{n+1} \deg f$. 若 n 为偶数, 则 $\deg f = 0$. 下设 n 为奇数. 设 $\pi: S^n \to \mathbb{R}\mathrm{P}^n$ 为商映射, 定义 $g: \mathbb{R}\mathrm{P}^n \to S^n$ 为 $g(\{-x, x\}) = f(x)$, 于是有交换图





即 $f = g \circ \pi$. 回顾计算 $\mathbb{R}\mathrm{P}^n$ 的同调群时使用的映射 $\Delta = q \circ \pi$, 其中 $q: \mathbb{R}\mathrm{P}^n \to \mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^{n-1}$ 为商映射. 注意 $(\mathbb{R}\mathrm{P}^n, \mathbb{R}\mathrm{P}^{n-1})$ 为好对, 故由相对同调群的长正合列以及 $\mathbb{R}\mathrm{P}^n$ 同调群的计算结果得

$$\cdots \to H_n(\mathbb{R}\mathrm{P}^{n-1}) = 0 \to H_n(\mathbb{R}\mathrm{P}^n) = \mathbb{Z} \xrightarrow{q_*} \tilde{H}_n(\mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^{n-1}) = \mathbb{Z} \to H_{n-1}(\mathbb{R}\mathrm{P}^{n-1}) = 0 \to \cdots$$

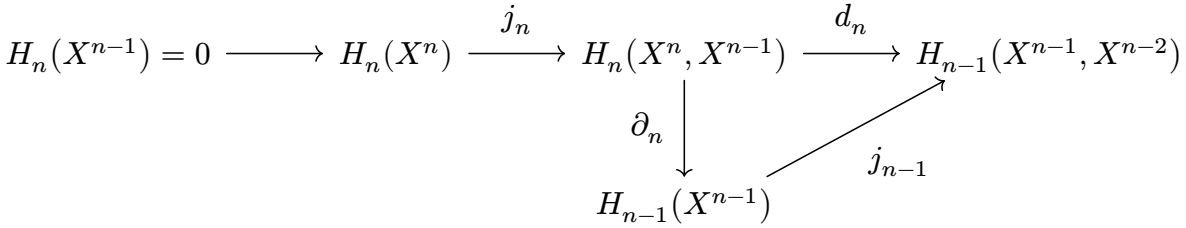
即 $0 \to \mathbb{Z} \xrightarrow{q_*} \mathbb{Z} \to 0$, 因此 q_* 为同构, 故 $\deg q = \pm 1$. 另一方面, 有 $\deg \Delta = 1 + (-1)^{n+1} = 2$, 因此 $2 = \deg q \deg \pi$, 进而 $\deg \pi = \pm 2$. 于是 $\deg f = \deg g \deg \pi = \pm 2 \deg g$ 为偶数.

由书中例子 2.31, 可知对任意 k, 存在 $f: S^n \to S^n, \deg f = k$, 定义 $g = f \circ \Delta$, 由 Δ 为偶映射可知 g 也是偶映射, 且 $\deg g = 2k$. 因此当 n 为奇数时, 存在任意给定的偶数度的偶映射.

H-Prob 2.2.15

Show that if X is a CW complex then $H_n(X^n)$ is free by identifying it with the kernel of the cellular boundary map $H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$.

Proof: 我们有交换图



其中 j_n, j_{n-1} 均为单射, 因此由正合列的性质, 有 $H_n(X^n) \cong \mathrm{Im}\, j_n = \mathrm{Ker}\, \partial_n$. 由 j_{n-1} 为单射知 $\mathrm{Ker}\, \partial_n = \mathrm{Ker}\, d_n$, 于是 $H_n(X^n) = \mathrm{Ker}\, d_n$ 为自由阿贝尔群 $H_n(X^n, X^{n-1})$ 的子群, 故 $H_n(X^n)$ 也是自由阿贝尔群.

H-Prob 2.2.17

Show the isomorphism between cellular and singular homology is natural in the following sense: A map $f : X \to Y$ that is *cellular* – satisfying $f(X^n) \subset Y^n$ for all n – induces a chain map f_* between the cellular chain complexes of X and Y, and the map $f_* : H_n^{\mathrm{CW}}(X) \to H_n^{\mathrm{CW}}(Y)$ induced by this chain map corresponds to $f_* : H_n(X) \to H_n(Y)$ under the isomorphism $H_n^{\mathrm{CW}} \cong H_n$.

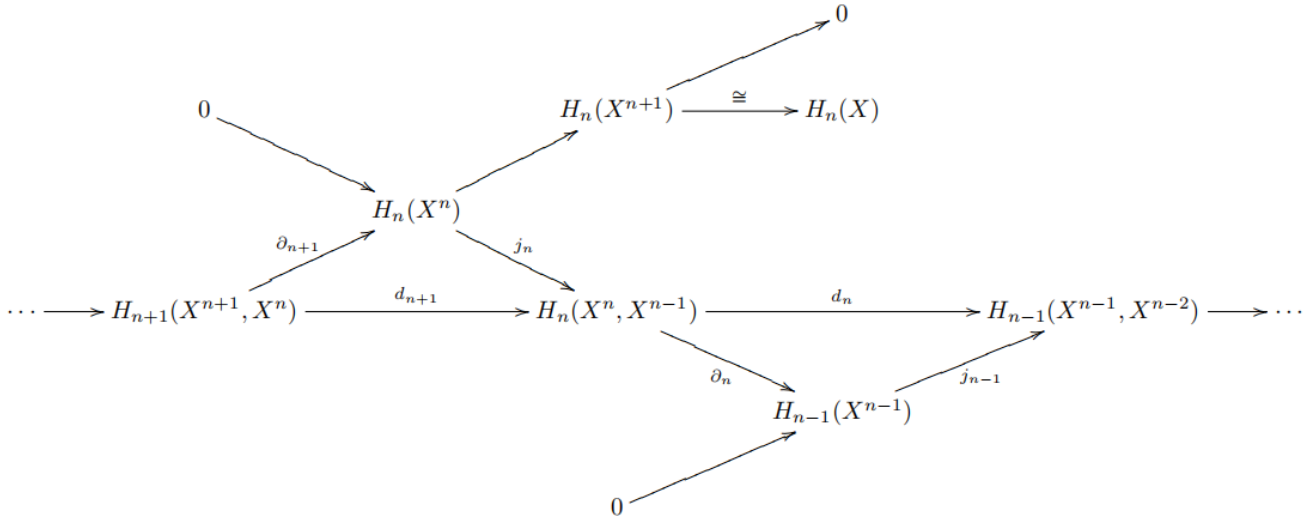
Proof: 设 X, Y 为两个胞腔复形, $f : X \to Y$ 为胞腔映射, 由胞腔同调的自然性, 我们有如下交换图, 其中 f_* 表示 $f.$ 诱导的映射.

$$\begin{array}{ccccc}
 & & \overset{d_n}{\frown} & & \\
H_n(X^n, X^{n-1}) & \xrightarrow{\partial_n} & H_{n-1}(X^{n-1}) & \xrightarrow{j_{n-1}} & H_{n-1}(X^{n-1}, X^{n-2}) \\
\downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} \\
H_n(Y^n, Y^{n-1}) & \xrightarrow{\partial_n} & H_{n-1}(Y^{n-1},) & \xrightarrow{j_{n-1}} & H_{n-1}(Y^{n-1}, Y^{n-2}) \\
 & & \underset{d_n}{\smile} & &
\end{array}$$

于是 f 在 X, Y 的胞腔链复形之间诱导了 $f_{\#}$, 即如下交换图

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & H_{n+1}(X^{n+1}, X^n) & \xrightarrow{d_{n+1}} & H_n(X^n, X^{n-1}) & \xrightarrow{d_n} & H_{n-1}(X^{n-1}, X^{n-2}) & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f_{\#}} & & \downarrow{\scriptstyle f_{\#}} & & \downarrow{\scriptstyle f_{\#}} & & \\
\cdots & \longrightarrow & H_{n+1}(Y^{n+1}, Y^n) & \xrightarrow{d_{n+1}} & H_n(Y^n, Y^{n-1}) & \xrightarrow{d_n} & H_{n-1}(Y^{n-1}, Y^{n-2}) & \longrightarrow & \cdots
\end{array}$$

于是 $f_{\#}$ 诱导了映射 $f_*^{\mathrm{CW}} : H_n^{\mathrm{CW}}(X) \to H_n^{\mathrm{CW}}(Y)$.



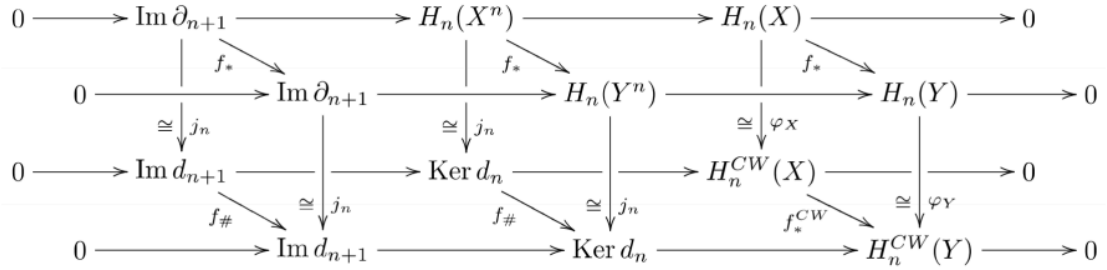
其中 j_n 是单射, 且有

$$\operatorname{Im}\partial_{n+1} \cong j_n(\operatorname{Im}\partial_{n+1}) = \operatorname{Im} d_{n+1}, \quad H_n(X^n) \cong j_n(H_n(X^n)) = \operatorname{Im} j_n = \operatorname{Ker}\partial_n = \operatorname{Ker} d_n$$

因此有交换图

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \operatorname{Im}\partial_{n+1} & \longrightarrow & H_n(X^n) & \longrightarrow & H_n(X) & \longrightarrow & 0 \\ & & \cong\downarrow & & \cong\downarrow j_n & & \cong\downarrow \varphi_X & & \\ 0 & \longrightarrow & \operatorname{Im} d_{n+1} & \longrightarrow & \operatorname{Ker} d_n & \longrightarrow & H_n^{\mathrm{CW}}(X) & \longrightarrow & 0 \end{array}$$

其中同构 $\varphi_X : H_n(X) \to H_n^{\mathrm{CW}}(X)$ 由 j_n 诱导



在上图中, 前后上下左的方块均交换, 因此右侧方块也交换, 即奇异同调和胞腔同调之间的同构是自然的.

H-Prob 2.2.19

Compute $H_i(\mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^m)$ for $m < n$ by cellular homology, using the standard CW structure on $\mathbb{R}\mathrm{P}^n$ with $\mathbb{R}\mathrm{P}^m$ as its m-skeleton.

Solution: 将 $\mathbb{R}\mathrm{P}^m = e^0 \cup e^1 \cup \cdots \cup e^m$ 视作 $\mathbb{R}\mathrm{P}^n$ 的 m-骨架, 则

$$\mathbb{R}\mathrm{P}^n = \mathbb{R}\mathrm{P}^m \cup e^{m+1} \cup \cdots \cup e^n$$

于是 $\mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^m$ 的胞腔复形结构为

$$\mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^m = e^0 \cup e^{m+1} \cup \cdots \cup e^n$$

于是 $d_k = 0, 1 \le k \le m+1$. 对 $m+2 \le k \le n$, 有 $d_k(e^k) = (1+(-1)^k)e^{k-1}$. 因此 $\mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^m$ 的胞腔同调链复形为

$$0 \to \mathbb{Z}e^n \xrightarrow{d_n} \cdots \xrightarrow{d_{m+2}} \mathbb{Z}e^{m+1} \xrightarrow{d_{m+1}=0} 0 \to \cdots \to 0 \to \mathbb{Z}e^0 \to 0$$

可见当 $m+1 \le i \le n$, i 为奇数时, $H_i(\mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^m) = \mathbb{Z}_2$; $i = 0$ 或 $i = n$ (n 为奇数) 或 $i = m+1$ (m 为奇数) 时, $H_i(\mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^m) = \mathbb{Z}$; 其余情况下 $H_i(\mathbb{R}\mathrm{P}^n/\mathbb{R}\mathrm{P}^m) = 0$.

H-Prob 2.2.21

If a finite CW complex X is the union of subcomplexes A and B, show that

$$\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B).$$

Proof: 设 X 为 n 为胞腔复形, $a_i(X)$ 表示 X 中 n-cell 的数量. 由 Euler-Poincaré 公式, 我们有

$$\chi(X) = \sum_{i=0}^{n} (-1)^i a_i(X), \quad \chi(A) = \sum_{i=0}^{n} (-1)^i a_i(A),$$

$$\chi(B) = \sum_{i=0}^{n} (-1)^i a_i(B), \quad \chi(A \cap B) = \sum_{i=0}^{n} (-1)^i a_i(A \cap B)$$

由于 $X = A \cup B$, 因此由容斥原理可知

$$a_i(X) = a_i(A) + a_i(B) - a_i(A \cap B), \forall i = 0, 1, \cdots, n$$

由此即得 $\chi(X)=\chi(A)+\chi(B)-\chi(A\cap B)$.

H-Prob 2.2.28

(a) Use the Mayer-Vietoris sequence to compute the homology groups of the space obtained from a torus $S^1\times S^1$ by attaching a Möbius band via a homeomorphism from the boundary circle of the Möbius band to the circle $S^1\times\{x_0\}$ in the torus.

(b) Do the same for the space obtained by attaching a Möbius band to $\mathbb{R}\mathrm{P}^2$ via a homeomorphism of its boundary circle to the standard $\mathbb{R}\mathrm{P}^1\subset\mathbb{R}\mathrm{P}^2$.

Solution: (a) 设题中空间为 X, X 中 Möbius 带的部分记为 A, 环面的部分记为 B, 取 U 为 X 中 Möbius 带的与环面的粘合圆的邻域, 记 $A'=A\cup U, B'=B\cup U$, 则 $X=A'\cup B'$, 且 A', B' 均为开集且 $A'\cap B'=U\simeq S^1, A'\simeq A, B'\simeq B$, 故 Mayer-Vietoris 序列为

$$0\to\tilde{H}_2(A)\oplus\tilde{H}_2(B)\xrightarrow{\Psi}\tilde{H}_2(X)\xrightarrow{\partial}\mathbb{Z}\xrightarrow{\Phi}\tilde{H}_1(A)\oplus\tilde{H}_1(B)\xrightarrow{\Psi}\tilde{H}_1(X)\to 0$$

其中 $A\simeq S^1$, 因此 $\tilde{H}_1(A)=\mathbb{Z}, \tilde{H}_2(A)=0$. 由环面的同调群可知 $\tilde{H}_1(B)=\mathbb{Z}^2, \tilde{H}_2(B)=\mathbb{Z}$. 于是上式化为

$$0\to\mathbb{Z}\xrightarrow{\Psi}\tilde{H}_2(X)\xrightarrow{\partial}\mathbb{Z}\xrightarrow{\Phi}\mathbb{Z}\oplus\mathbb{Z}^2\xrightarrow{\Psi}\tilde{H}_1(X)\to 0$$

考虑 $\Phi:\tilde{H}_1(S^1)\to\tilde{H}_1(A)\oplus\tilde{H}_1(B)$ 的具体形式, 取 $[a]\in\tilde{H}_1(S^1)$ 为生成元, 由于 $U\simeq S^1$ 对应 Möbius 带 A 中的边界圆, 沿边界圆走一圈相当于沿 A 的中心圆 (对应 $\tilde{H}_1(A)$ 的生成元) 走两圈. 设 $\tilde{H}_1(A)=\mathbb{Z}[b]$, $\tilde{H}_1(B)=\mathbb{Z}[c]\oplus\mathbb{Z}[d]$, 其中 $[c]$ 对应环面中与 Möbius 带粘合的圆 $S^1\times\{x_0\}$, $[d]$ 对应环面中 $\{y_0\}\times S^1$ 的另一个圆. 于是

$$\Phi([a])=(2[b],[c],0)\Longrightarrow\Phi(1)=(2,1,0)$$

故由 Ψ 为满射, 可得

$$\tilde{H}_1(X)=\operatorname{Im}\Psi=(\mathbb{Z}\oplus\mathbb{Z}^2)/\operatorname{Ker}\Psi=(\mathbb{Z}\oplus\mathbb{Z}^2)/\operatorname{Im}\Phi=(\mathbb{Z}\oplus\mathbb{Z}^2)/\langle(2,1,0)\rangle=\mathbb{Z}^2$$

并且 $\operatorname{Ker}\Phi=0\Longrightarrow\operatorname{Im}\partial=0\Longrightarrow\operatorname{Ker}\partial=\tilde{H}_2(X)\Longrightarrow\operatorname{Im}\Psi=\tilde{H}_2(X)$, Ψ 为单射, 故 $\tilde{H}_2(X)=\mathbb{Z}$. 综上, 我们有

$$H_0(X)=\mathbb{Z},\quad H_1(X)=\mathbb{Z}^2,\quad H_2(X)=\mathbb{Z},\quad H_n(X)=0,\forall n\geq 3.$$

(b) 记题中空间为 X, $\mathbb{R}\mathrm{P}^2$ 的部分记为 A, Möbius 带的部分记为 B, 二者通过 $\mathbb{R}\mathrm{P}^1\subset\mathbb{R}\mathrm{P}^2$ 粘合, 记粘合圆的邻域为 U, 则 $X=A'\cup B'$, 其中 $A'=A\cup U, B'=B\cup U$, $A'\cap B'=U\simeq S^1$, 并且 $A'\simeq A, B'\simeq B$. 由 $\mathbb{R}\mathrm{P}^2$ 的同调群可知 $\tilde{H}_1(A)=\mathbb{Z}_2, \tilde{H}_2(A)=0$, 由 Möbius 带的同调群可知 $\tilde{H}_1(B)=\mathbb{Z}, \tilde{H}_2(B)=0$. 于是 Mayer-Vietoris 序列化为

$$0\to\tilde{H}_2(X)\xrightarrow{\partial}\mathbb{Z}\xrightarrow{\Phi}\mathbb{Z}_2\oplus\mathbb{Z}\xrightarrow{\Psi}\tilde{H}_1(X)\to 0$$

考察 $\Phi:\tilde{H}_1(S^1)\to\tilde{H}_1(A)\oplus\tilde{H}_1(B)$ 的具体形式: 设 $\tilde{H}_1(S^1)=\mathbb{Z}[a], \tilde{H}_1(A)=\mathbb{Z}_2[b], \tilde{H}_1(B)=\mathbb{Z}[c]$, 则 $[a]$ 对应 Möbius 带中的边界圆以及 $\mathbb{R}\mathrm{P}^1$, $[c]$ 对应 Möbius 带的中心圆, 沿边界圆走一圈相当于沿中心圆走两圈, 再由 $\mathbb{R}\mathrm{P}^1$ 在 $\mathbb{R}\mathrm{P}^2$ 中的同调性质知, $[a]$ 在 $\tilde{H}_1(A)$ 中对应生成元 $[b]$, 故

$$\Phi([a])=([b],2[c])\Longrightarrow\Phi(1)=(1,2)\Longrightarrow\operatorname{Ker}\Phi=0,\operatorname{Im}\Phi=\langle(1,2)\rangle$$

于是

$$\tilde{H}_1(X)=\operatorname{Im}\Psi=(\mathbb{Z}_2\oplus\mathbb{Z})/\operatorname{Ker}\Psi=(\mathbb{Z}_2\oplus\mathbb{Z})/\operatorname{Im}\Phi=(\mathbb{Z}_2\oplus\mathbb{Z})/\langle(1,2)\rangle=\mathbb{Z}_4$$

$$\tilde{H}_2(X)=\operatorname{Im}\partial=\operatorname{Ker}\Phi=0$$

综上, 我们有

$$H_0(X)=\mathbb{Z},\quad H_1(X)=\mathbb{Z}_4,\quad H_n(X)=0,\forall n\geq 2.$$

H-Prob 2.2.29

The surface M_g of genus g, embedded in $\mathbb{R}^3$ in the standard way, bounds a compact region R. Two copies of R, glued together by the identity map between their boundary surfaces M_g, form a closed 3-manifold X. Compute the homology groups of X via the Mayer-Vietoris sequence for decomposition of X into two copies of R. Also compute the relative groups $H_i\big(R,M_g\big)$.

Solution: (a) 设 X 中的两份 R 为 R_1,R_2, 取 A,B 为 X 中包含 R_1,R_2 的开集且 $A\simeq R_1,B\simeq R_2$, 则 $X=A\cup B$, 且 $A\cap B\simeq M_g$. 故 Mayer-Vietoris 序列为

$$\cdots\to\tilde{H}_3\big(M_g\big)\to\tilde{H}_3(R)\oplus\tilde{H}_3(R)\to\tilde{H}_3(X)\to\tilde{H}_2\big(M_g\big)\to\tilde{H}_2(R)\oplus\tilde{H}_2(R)\to\tilde{H}_2(X)$$
$$\to\tilde{H}_1\big(M_g\big)\to\tilde{H}_1(R)\oplus\tilde{H}_1(R)\to\tilde{H}_1(X)\to 0$$

注意到 $R\simeq\bigvee_g S^1$, 于是 $\tilde{H}_3(R)=\tilde{H}_2(R)=0,\tilde{H}_1(R)=\mathbb{Z}^g=\mathbb{Z}\langle c_1,\cdots,c_g\rangle$. 对 M_g, 有

$$\tilde{H}_3\big(M_g\big)=0,\quad \tilde{H}_2\big(M_g\big)=\mathbb{Z},\quad \tilde{H}_1\big(M_g\big)=\mathbb{Z}^{2g}=\mathbb{Z}\langle a_1,b_1,\cdots,a_g,b_g\rangle$$

于是 Mayer-Vietoris 序列变为

$$0\to\tilde{H}_3(X)\to\mathbb{Z}\to 0\to\tilde{H}_2(X)\xrightarrow{\partial}\mathbb{Z}^{2g}\xrightarrow{\Psi}\mathbb{Z}^g\oplus\mathbb{Z}^g\xrightarrow{\Phi}\tilde{H}_1(X)\to 0$$

于是 $\tilde{H}_3(X)=\mathbb{Z}$. 考虑 $\Psi:\mathbb{Z}\langle a_1,b_1,\cdots,a_g,b_g\rangle\to\mathbb{Z}\langle c_1,\cdots,c_g\rangle\oplus\mathbb{Z}\langle c_1',\cdots,c_g'\rangle$, 有

$$\Psi(a_i)=c_i+c_i',\Psi(b_i)=0,i=1,\cdots,g$$

于是

$$\tilde{H}_1(X)=\operatorname{Im}\Phi=(\mathbb{Z}^g\oplus\mathbb{Z}^g)/\operatorname{Ker}\Phi=(\mathbb{Z}^g\oplus\mathbb{Z}^g)/\operatorname{Im}\Psi=(\mathbb{Z}^g\oplus\mathbb{Z}^g)/\langle c_1+c_1',\cdots,c_g+c_g'\rangle=\mathbb{Z}^g$$

$$\tilde{H}_2(X)=\operatorname{Im}\partial=\operatorname{Ker}\Psi=\mathbb{Z}\langle b_1,\cdots,b_g\rangle=\mathbb{Z}^g$$

综上我们得到

$$H_0(X)=\mathbb{Z},\quad H_1(X)=\mathbb{Z}^g,\quad H_2(X)=\mathbb{Z}^g,\quad H_3(X)=\mathbb{Z},\quad H_n(X)=0,\forall n\geq 4.$$

(b) 考虑好对 $\big(R,M_g\big)$, 我们有长正合列

$$\cdots\to\tilde{H}_3\big(M_g\big)=0\to\tilde{H}_3(R)\to H_3\big(R,M_g\big)\to\tilde{H}_2\big(M_g\big)\to\tilde{H}_2(R)\to H_2\big(R,M_g\big)$$
$$\to\tilde{H}_1\big(M_g\big)\to\tilde{H}_1(R)\to H_1\big(R,M_g\big)\to 0$$

利用 $R\simeq\bigvee_g S^1$, 以上正合列化为

$$0\to 0\to H_3\big(R,M_g\big)\to\mathbb{Z}\to 0\to H_2\big(R,M_g\big)\xrightarrow{\partial'}\mathbb{Z}^{2g}\xrightarrow{\Psi'}\mathbb{Z}^g\xrightarrow{\Phi'}H_1\big(R,M_g\big)\to 0$$

于是 $H_3\big(R,M_g\big)=\mathbb{Z}$. 考察 $\Psi':\mathbb{Z}\langle a_1,b_1,\cdots,a_g,b_g\rangle\to\mathbb{Z}\langle c_1,\cdots,c_g\rangle$, 有

$$\Psi'(a_i)=c_i,\Psi'(b_i)=0,i=1,\cdots,g$$

因此

$$H_1(R, M_g) = \operatorname{Im}\Phi' = (\mathbb{Z}^g)/\operatorname{Ker}\Phi' = (\mathbb{Z}^g)/\operatorname{Im}\Psi' = (\mathbb{Z}^g)/\langle c_1, \cdots, c_g\rangle = 0$$

$$H_2(R, M_g) = \operatorname{Im}\partial' = \operatorname{Ker}\Psi' = \mathbb{Z}\langle b_1, \cdots, b_g\rangle = \mathbb{Z}^g$$

以及 $H_0(R, M_g) = \tilde{H}_0(R/M_g) = 0$. 综上我们得到

$$H_0(R, M_g) = H_1(R, M_g) = 0, \quad H_2(R, M_g) = \mathbb{Z}^g, \quad H_3(R, M_g) = \mathbb{Z}, \quad H_n(R, M_g) = 0, \forall n \geq 4.$$

H-Prob 2.2.31

Use the Mayer-Vietoris sequence to show there are isomorphisms $\tilde{H}_n(X \vee Y) \cong \tilde{H}_n(X) \oplus \tilde{H}_n(Y)$ if the basepoints of X and Y that are identified in $X \vee Y$ are deformation retracts of neighborhoods $U \subset X$ and $V \subset Y$.

Proof: 在 $X \vee Y$ 中, 设楔和的基点为 $a \in U, V$, 考虑 $X' = X \cup V, Y' = Y \cup U$, 则 $X \vee Y = X' \cup Y'$, 且 $X' \cap Y' = U \cup V$ 可缩为点 a. 由 reduced Mayer-Vietoris 序列, 我们有长正合列

$$\cdots \to \tilde{H}_n(U \cup V) = 0 \to \tilde{H}_n(X') \oplus \tilde{H}_n(Y') \to \tilde{H}_n(X \vee Y) \to \tilde{H}_{n-1}(U \cup V) = 0 \to \cdots$$

故 $\tilde{H}_n(X \vee Y) \cong \tilde{H}_n(X') \oplus \tilde{H}_n(Y')$. 由 U, V 可缩知 $X' \simeq X, Y' \simeq Y$, 于是 $\tilde{H}_n(X') \cong \tilde{H}_n(X)$, $\tilde{H}_n(Y') \cong \tilde{H}_n(Y)$, 故 $\tilde{H}_n(X \vee Y) \cong \tilde{H}_n(X) \oplus \tilde{H}_n(Y)$.

H-Prob 2.2.32

For SX the suspension of X, show by a Mayer-Vietoris sequence that there are isomorphisms $\tilde{H}_n(SX) \cong \tilde{H}_{n-1}(X)$ for all n.

Proof: 根据定义, $SX = (CX)_1 \cup_X (CX)_2$, 其中 $CX = X \times I/(X \times \{0\})$ 为 X 的锥. 设 U 为 SX 中包含 attaching 面 X 且可以形变收缩到 X 的开邻域. 考虑 SX 中的两个子空间

$$A = (CX)_1 \cup U, B = (CX)_2 \cup U$$

则 $SX = A \cup B$, $A \cap B \simeq X$, $A \simeq (CX)_1, B \simeq (CX)_2$. 因此 Mayer-Vietoris 序列为

$$\cdots \to \tilde{H}_n(CX) \oplus \tilde{H}_n(CX) \to \tilde{H}_n(SX) \to \tilde{H}_{n-1}(X) \to \tilde{H}_{n-1}(CX) \oplus \tilde{H}_{n-1}(CX) \to \cdots$$

注意 CX 是可缩的, 因此 $\tilde{H}_n(CX) = 0, \forall n \geq 0$, 于是上式化为

$$0 \to \tilde{H}_n(SX) \to \tilde{H}_{n-1}(X) \to 0$$

即 $\tilde{H}_n(SX) \cong \tilde{H}_{n-1}(X)$.

Prob 9.25

If X is a compact triangulable space, $f : X \to X$ is null homotopic, show that f have a fixed point.

Proof: 由 X 三角可剖知此时 Lefschetz 数可定义. 设 $f \simeq c_x$, 于是 $f_* = (c_x)_*$, c_x 可视为

$$X \xrightarrow{p} \{x\} \xhookrightarrow{i} X, \quad c_x = i \circ p$$

其中 p 为常值映射, i 为包含映射. 因此 $f_* = i_* \circ p_*$. 注意当 $i \geq 1$ 时

$$p_* : H_i(X; \mathbb{Q}) \to H_i(\{x\}; \mathbb{Q}) = 0$$

当 $i = 1$ 时, 设 X 有 k 个连通分支, 于是 $H_0(X; \mathbb{Q}) = \mathbb{Q}^k$, 设每个分支对应的生成元为 $e_1, \cdots, e_k$, 不妨设 x 在第一个分支上, 则

$$(c_x)_*(e_j) = e_1, \forall j = 1, \cdots, k \Longrightarrow \operatorname{tr}(c_x : H_0(X; \mathbb{Q}) \to H_0(X; \mathbb{Q})) = 1$$

因此 Lefschetz 数为

$$\Lambda_f = \sum_{i=0}^{n} (-1)^i \operatorname{tr}(f_* : H_i(X; \mathbb{Q}) \to H_i(X; \mathbb{Q})) = 1 + \sum_{i=1}^{n} (-1)^i 0 = 1 \neq 0$$

由 Lefschetz 不动点定理可知 f 有不动点.

Prob 9.26

Let G be a path-connected topological group. Show that left translation $L_g : G \to G$ by an element $g \in G$ is homotopic to the identity.

Proof: 设 $\gamma : [0, 1] \to G$ 为连接 g 与 e 的路径, 定义

$$H : G \times [0, 1] \to G, \quad H(x, t) = L_{\gamma(t)}(x) = \gamma(t) \cdot x$$

于是 $H(x, 0) = x$, $H(x, 1) = gx$, 故 $L_g \underset{H}{\cong} \mathrm{id}_G$.

Prob 9.27

Show that the Euler characteristic of a compact, connected, triangulable topological group is zero.

Proof: 取 $g \neq e$, 于是左乘算子 L_g 无不动点, 故其 Lefschetz 数为零. 由上题可知 L_g 与恒等映射同伦, 因此 $\Lambda_{\mathrm{id}_G} = \Lambda_{L_g} = 0$. 于是

$$0 = \Lambda_{\mathrm{id}_G} = \sum_{i=0}^{n} (-1)^i \operatorname{tr}\Big((\mathrm{id}_G)_* : H_i(G; \mathbb{Q}) \to H_i(G; \mathbb{Q})\Big) = \sum_{i=0}^{n} (-1)^i \operatorname{rank}(H_i(G; \mathbb{Q})) = \chi(G)$$

Prob 9.29

Prove that an even-dimensional sphere cannot be a topological group. (In fact, S^1 and S^3 are the only sphere which are topological groups, though this is much harder to prove.)

Proof: 反证, 设 S^{2n} 存在拓扑群结构, 则由上题可知 $\chi(S^{2n}) = 0$. 但我们可在 S^{2n} 上任取一点 e_0, 于是 $S^{2n} = e_0 \cup e_{2n}$ 构成 CW 复形, 于是由 Euler-Poincaré 公式可知

$$\chi(S^{2n}) = \sum_{i=0}^{2n} (-1)^i a_i(S^{2n}) = 1 + (-1)^{2n} = 2 \neq 0$$

矛盾!